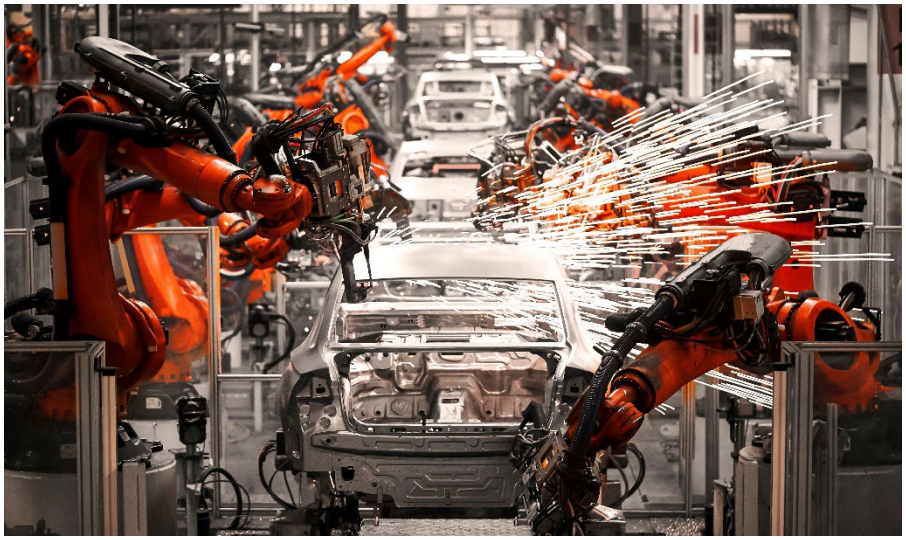
STUDY

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Industrial overcapacities, with a focus on China



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ABSTRACT

Overcapacities, particularly from China, are perceived as a threat to European manufacturers, distorting competition and straining bilateral trade. On the basis of a dynamic definition of 'industrial overcapacity', this study identifies overcapacities for most industrial sectors in China. Absorption occurs primarily through inventory accumulation, while six sectors with rising export-to-revenue ratios account for the large increase in China's exports to the EU. At the same time, case studies of semiconductors, electric vehicles and batteries, hydrogen, and robotics show that concrete risks from overcapacity for manufacturing in the EU are sector-specific and partly mitigated by value-chain dynamics and technological leadership. In China, overcapacities — discussed as 'involution' — are perceived as an unwelcome outcome of industrial policies and an obstacle to the high-priority objectives of technological upgrading and self-sufficiency. As the effectiveness of Chinese policies to reduce overcapacities remains uncertain, competitive pressures upon European companies are likely to persist or intensify. This highlights the need for context-specific EU policy responses, including the full use of traditional trade defence as well as other EU instruments. Policy space provided by WTO law should be tested, renewed efforts for a targeted reform of WTO rules be made, while cooperative efforts with China should be maintained.

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List of abbreviations and acronyms

ADA	Anti-Dumping Agreement
AEM	Anion Exchange Membrane
Art	Article
ASCM	Agreement on Subsidies and Countervailing Measures
ASEAN	Association of Southeast Asian Nations
AWE	Alkaline Water Electrolysers
BEV	Battery Electric Vehicle
bn	Billion
CAGR	compound annual growth rate
CCP	Chinese Communist Party
CEG	capacity expansion gap
CIECC	China International Engineering Consulting Corporation
CMIF	China Machinery Industry Federation
CNY	Chinese Renminbi Yuan
CUR	capacity utilisation rate
CVD	countervailing duty
DR	domestic revenues
EC	European Commission
ECB	European Central Bank
EU	European Union
EUR	Euro
EV	electric vehicle
FCEV	fuel cell electric vehicle
FDI	foreign direct investment
FSR	Foreign Subsidy Regulation
GAC	General Administration of Customs of the People's Republic of China
GATS	General Agreement on Trade in Services
GATT	General Agreement on Tariffs and Trade
GDP	gross domestic product
GGII	Gaogong Industry Research Institute
GVCs	global value chains
GW	gigawatt
GWh	gigawatt hours
ICE	internal combustion engine

IFA	investment in fixed assets
IFR	International Federation of Robotics
Inv	inventories-to-revenues
IPI	International Procurement Instrument
IPO	Initial Public Offering
ISIC	International Standard Industrial Classification
ISO	International Organization for Standardization
kW	kilowatt
kWh	kilowatt hours
LCR	local content requirement
LFP	lithium-iron-phosphate
LIB	lithium-ion battery
M&A	mergers and acquisitions
MCA	Ministry of Civil Affairs of the People's Republic of China
MEA	membrane electrode assemblies
MEE	Ministry of Ecology and Environment of the People's Republic of China
MEP	Member of European Parliament
MFN	most favoured nation
MIC	Made in China 2025 policy
MIIT	Ministry of Industry and Information Technology of the People's Republic of China
MNR	Ministry of Natural Resources of the People's Republic of China
MOC	Ministry of Commerce of the People's Republic of China
MOE	Ministry of Education of the People's Republic of China
MOF	Ministry of Finance of the People's Republic of China
MPS	Ministry of Public Security of the People's Republic of China
NBS	National Bureau of Statistics of China
NCA	non-current assets
NDRC	National Development and Reform Commission
NEV	new-energy vehicle
NFRA	National Financial Regulatory Administration (China)
nm	nanometer
OECD	Organisation for Economic Co-operation and Development
OEM	original equipment manufacturer
PEM	proton exchange membrane

PPI	producer price index
PRC	People's Republic of China
PV	photovoltaic
R&D	research and development
ROW	rest of the world
RR	real revenues
SA	Safeguards Agreement
SAMR	State Administration of Market Regulation (China)
Shanghai DRC	Shanghai Municipal Development and Reform Commission
Shanghai EIC	Shanghai Municipal Economy and Industrialisation Commission
Shanghai STC	Shanghai Municipal Science and Technology Commission
SNA	System of National Accounts
SOE	state-owned enterprise
SOEC	solid oxide electrolysers
STA	State Taxation Administration (China)
TDI	trade defence instrument
TiVA	Trade in Value Added
TRIMs	Agreement on Trade-Related Investment Measures
TRIPS	Agreement on Trade-Related Aspects of Intellectual Property
UK	United Kingdom
US	United States of America
USD	United States Dollar
VAT	Value Added Tax
WTO	World Trade Organization
yr	year

Executive summary

Global markets are currently experiencing a strong surge in China's exports, resulting in a record trade surplus of EUR 1.05 trillion in 2025. Between 2015 and 2025, the Chinese trade surplus doubled. Over the same period, the EU's trade deficit with China also doubled in value (EUR 309 billion in 2024), and expanded approximately fourfold in physical terms. Consequently, a debate has emerged around the drivers behind China's export surge, with overcapacities in industrial sectors identified as a potential factor.

The EU has repeatedly raised concerns that state-supported industries in China generate overcapacity particularly through high levels of investment in production despite low levels of domestic consumption, which leads to exports at artificially low prices in order to dispose of surplus output. This distorts competition and threatens existing manufacturing including employment and the development of new industrial capacities in emerging technologies within the EU. In response, the European Commission has launched a record number of trade defence investigations, and increasingly imposed trade defence instruments (TDIs), such as the countervailing tariffs on Chinese EVs imposed in 2024.

This report examines the scale and dynamics of overcapacity in China's industrial sectors, as well as potential risks and effects for the EU based on case studies. It further discusses potential countermeasures and offers policy recommendations for the EU to remedy the resulting problems. We propose a dynamic definition of industrial overcapacity in a high-growth environment. Accordingly, industrial overcapacity is a condition in which productive capacity expands faster than economically viable demand over more than five years, leading to sustained overproduction pressure that is absorbed through inventory accumulation, price and profit margin declines, and exports rather than through capacity exit. Policy-induced incentives and institutional arrangements, as typically implemented in non-market economies like China, shape investment behaviour and influence whether imbalances are resolved through market adjustment or persist through price pressure and external spill-overs.

Applying this definition empirically across 14 industrial sectors in China reveals widespread overcapacity. Since the mid-2010s, productive capacity indicators have grown persistently faster than domestic absorption across most sectors and ten sectors (Chemical Fibres, Electronics, Electrical Machinery, Ferrous Metals, General Machinery, Measuring Instruments, Medicines, Non-metallic Minerals, Rubber & Plastics, Special Machinery) clearly meet the criteria for overcapacity under the proposed definition. This capacity expansion pressure led to absorption of production primarily through inventory accumulation, while price adjustment and margin compression play a minor role. The six sectors Automobiles, Electric Machinery, General Machinery, Rubber & Plastics, Special Machinery, and Medicines saw rising export shares and accounted disproportionately for China's export growth to the EU. While our empirical analysis does not allow for a precise assessment of the role of active state involvement in industrial overcapacity, the findings point to indicative patterns including state-owned enterprises (SOEs)' role in adjustment mechanisms. Establishing a robust link requires better databases on the comprehensive effects of policy interventions in China and across other countries.

The concrete effects and risks associated with these indications of overcapacity for current and future manufacturing in the EU warrant more detailed assessment. We examine four cases in depth: (i) semiconductors, and (ii) EVs and battery production as fast-growing sectors with relatively developed production systems and potentially emerging overcapacity, while (iii) hydrogen and (iv) robotics represent high-growth sectors where Chinese capacity expansion despite significant uncertainty over future demand could undermine the technological capabilities of EU companies. The analysis reveals that in sectors where Europe retains a technological lead (semiconductors,

hydrogen and industrial robots), overcapacities concentrate mostly on low- and mid-tier segments or, in the case of electrolysers, technology-ready segments, which limits direct impact on European firms to some extent. However, as China aims to ascend to more technologically complex segments, this might change. In the EV/batteries sector, restructuring and localisation of production networks reduce overcapacity risks for EV producers in the EU, but increase technological dependence on Chinese battery suppliers. Thus, China's push toward technological upgrading and self-sufficiency poses an additional challenge, reinforcing the need to safeguard technological leadership and support scale-up in strategically important industries in the EU.

The internal economic policy debate in China has addressed the issue of overcapacity under the term of 'involution', explaining it as the unwelcome outcome of local government investment activities leading to excessive competition ('vicious competition') and low-price pressure. Current policy responses to address 'involution' range from lower export support and legislation against price competition to tighter control over local government investment. The primary motivation, however, is not overcapacity but concerns that involution is hindering technological upgrading and self-sufficiency.

Against this background, the EU increasingly relies on TDIs to counter unfair competition. Yet, apart from TDIs' lengthy procedures and high evidentiary thresholds, none of the existing instruments are designed to target the structural root causes of overcapacity but only apply once injury of EU industry has occurred or is imminent. Thus, traditional TDIs are only partially suited to counteract overcapacity issues. Where suitable, they should be complemented by the strategic use of additional instruments available at WTO and EU level, including by more intense cooperation with China and other trading partners. Cooperative solutions should be sector-specific given distinct dynamics and strategic interests. On a multilateral level, cooperation is needed to share data and align policy approaches, including on subsidization.

Overall, the EU should pursue an 'inclusive regionalism' approach, combining targeted protective measures and strategic support for key EU industries with continued commitment to a rules-based trading regime and economic cooperation. With respect to specific policy recommendations, we suggest:

(1) **Exploring potential for common ground on subsidy issues under WTO law,** particularly with respect to reviving (and potentially extending) the category of 'non-actionable' subsidies in the Agreement on Subsidies and Countervailing Measures (ASCM).

(2) **Fully utilizing and continuing to test the legal policy space provided by WTO law:** The EU should continue to apply existing rules assertively and, where necessary, put forward legislation strategically testing the limits of policy space under WTO law. Also, depending on the specific context, the modification of schedules (Art. XXVIII GATT) as well as novel instruments such as the EU Foreign Subsidies Regulation, the International Procurement Instrument and FDI Screening may be used more actively. The recently proposed steel measure could also serve as a model for similar instruments which could strategically provide sector-specific protection where overcapacity issues pose a serious threat to EU industries.

(3) **Expanding monitoring and trade intelligence capacities:** TDIs face a risk of circumvention as they rely on limited insight into complex value chains. Sufficient analytical capacity is needed in dedicated EU units to track developments in China, but also in other countries given the global competition for technology leadership.

(4) **Combining trade defence with industrial upgrading:** to avoid rent-seeking in strategic sectors and increase competitiveness, TDIs need to be complemented by active industrial policy measures that foster technological upgrading of EU companies.

(5) **Tackling adverse exchange-rate effects:** Import surges from China have been exacerbated by distorted real exchange-rates and currency management policies. In the absence of adjustment through currency movements, compensatory EU policies should be considered, either through cooperative stabilisation measures, ECB exchange rate management, or non-violation complaints for the impairment of trade benefits.

1 Introduction

Global markets are currently experiencing a strong surge in exports from China, with a record trade surplus of EUR 1.05 trillion in 2025 (NBS China, 2026). Between 2015 and 2025, the Chinese trade surplus doubled. EU-China trade dynamics have followed a similar trend. Consequently, a debate around industrial overcapacities as a potential driver behind this export surge has emerged. This phenomenon has been well-documented in Chinese sectors such as steel and solar photovoltaic (PV) modules (Jin, 2018). More recently, concerns have extended to electric vehicles (EVs), batteries, chemicals, and other green technologies (Boullenois et al., 2024; Gunter et al., 2025).

The EU has repeatedly raised concerns that state-supported industrial overcapacity in China is leading to artificially low export prices, undermining fair competition and threatening EU manufacturing sectors, including in emerging technologies. In an escalating response, the European Commission (EC) has launched a record number of trade defence investigations in 2024, more than two third of which concerned China (EC, 2025a).

Despite official denials of industrial overcapacities (Ministry of Foreign Affairs, People's Republic of China, 2025), internal policy debates in China frame overcapacity under the rubric of 'involution', characterized by fierce competition ('disorderly competition') that threatens upgrading strategies. Policies to address these issues have been introduced and announced within the framework of the upcoming 15th five-year plan (2026-2030). Against this background, this report examines the scale and dynamics of overcapacity in China's industrial sectors, assessing where overcapacities have emerged in recent years and what role they have in the development of new technologies in which the EU aims to defend or establish own production capacities. Further, we discuss potential countermeasures and offer policy recommendations for the EU to remedy the resulting problems.

In methodological terms the study uses a mixed methods approach, combining (i) analysis of the academic and policy-oriented literature including official policy documents of the EU and of China; (ii) data analysis using official statistics from China's National Bureau of Statistics (NBS), UN Comtrade, UNIDO and Eurostat, and sector-specific sources such as industry reports, allowing us to produce quantitative indicators of overcapacity; (iii) legal analysis of available trade defence instruments (TDIs), as well as relevant Chinese trade and industrial regulations; and (iv) semi-structured expert interviews as part of the sector case studies, including with EU industry representatives and company executives (see Annex I). These will provide first-hand insights into sector dynamics, validate data findings, and assess perceived risks from Chinese overcapacity.

The report is structured as follows: Chapter 2 first takes up the conceptual discussion and proposes a dynamic definition of overcapacity. Next, it provides an empirical assessment of overcapacity in China's industrial sector at the aggregate and industry-specific level, before focusing on case studies of sectors with currently emerging overcapacity issues (semiconductors, EVs and batteries). Two further case studies on hydrogen and robotics focus on sectors with potential future overcapacity. Chapter 3 discusses solutions to tackle overcapacities. First, we provide an overview of China's internal policies, with a focus on the most recent ones. Second, unilateral measures at the EU's disposal are discussed. Third, we address cooperative bilateral and multilateral approaches. Chapter 4 concludes by outlining the opportunities and limits of EU strategic responses and by offering specific policy recommendations for the EU.

2 Definitions and 'diagnoses'

The first part of the report introduces a concept and definition of 'industrial overcapacity' that is operationalised for the case of industry sectors in China. In the following, two case studies examine the emerging risks of overcapacity in EVs/batteries and semiconductors, and two case studies investigate the issues of overcapacity in the hydrogen economy and robotics as prospective key technologies for sustainable and automated industrial production.

2.1 Conceptualisation and definition of 'industrial overcapacity'

The issue of 'overcapacities' and 'excess capacities' has appeared in economic analysis since the 1930s (e.g. Cassels, 1937). However, overcapacity remains a contested concept with multiple theoretical interpretations and without a commonly accepted definition. Three analytical perspectives on overcapacity can be identified in the economic literature. First, industrial organisation research shows that excess capacity can be a rational firm-level strategy — deterring market entry or persisting due to exit barriers such as sunk costs (Baumol, 1982; Dixit, 1980). Second, various macroeconomic approaches interpret overcapacity as cyclical, driven by investment dynamics and economic fluctuations, with market mechanisms eventually restoring balance (Goodwin, 1951; Hicks, 1950; Kalecki, 1937; Minsky, 1982). Third, heterodox approaches emphasise persistent rather than cyclical excess capacity — whether from oligopolistic structures that discourage price competition (Steindl 1952), or competitive pressures from late-industrialising economies (Brenner 2001).

Basic definitions equate overcapacity with the gap between production capacity and actual output, treating it as the inverse of the capacity utilisation rate (EU Chamber of Commerce in China, 2016). However, these utilisation-based definitions face several limitations: measured capacity utilisation varies depending on whether engineering capacity (maximum physical output) or economic capacity (based on cost functions) are used (Berndt & Hesse, 1986; Klein, 1960).[1] Further, capacity is often assessed relative to the level of demand or market absorption at prevailing prices rather than to 'actual production', distinguishing between short-term and structural variations. Basic definitions also neglect the broader economic consequences such as resource misallocation towards unproductive uses, deflationary pressure, financial fragility or trade frictions (Xu, 2025).

Taken together, these limitations indicate that utilisation-based metrics alone are inadequate for identifying structurally relevant overcapacity. Moreover, existing theoretical approaches primarily explain why overcapacities emerge — temporarily or persistently — in mature, slow-growing economies, paying limited attention to domestic and international effects or how overcapacity dynamics might differ in rapidly growing sectors.

2.1.1 Overcapacities in high-growth and state-driven contexts

The developmental state literature demonstrates that capacity expansion exceeding domestic demand can be an intentional and growth-enhancing strategy rather than a market failure. For instance, Japan's export-oriented industrialisation relied on rapid capacity expansion to achieve scale economies and accelerate productivity growth in heavy, scale-intensive industries such as steel, shipbuilding, and automobiles (Johnson, 1982; Okimoto, 1989). However, Japanese authorities actively sought to contain and resolve excessive capacity expansion and potential

[1] Berndt and Morrison (1981) showed that economic capacity measures yield utilisation rates approximately ten percentage points higher than engineering measures. This distinction matters because 'overcapacity' relative to engineering maximums may represent optimal behaviour relative to economic costs.

adverse effects of internal competition through coordinated investment, entry controls, and rationalisation measures, albeit to limited avail (Noland & Pack, 2003; Yamamura, 1986).

Korea's developmental state supported more aggressive and prolonged overcapacity in strategic sectors such as steel, shipbuilding, automobiles, and electronics.[2] The government deliberately licensed multiple firms in the same industries, intentionally pushing capacity beyond both domestic and short-term export demand (Amsden, 1992; Wade, 1990). According to Wade (1990), these capacity expansions functioned as a disciplinary mechanism, forcing firms to compete for export markets, accelerate learning, and upgrade productivity.[3] Comprehensive capital flow liberalisation in the 1990s led to overcapacity and heightened financial fragility, contributing to the 1997–1998 crisis (Wade & Veneroso, 1998).

These experiences suggest that capacity expansion alone is not sufficient to define overcapacity; Instead, the economic consequences and adjustment mechanisms are decisive. When capacity outpaced demand, idle capacity accumulated, compressing profits and heightening financial fragility. Japan contained this tension through coordinated rationalisation, but it also faced pressure from Western importing countries; Korea tolerated prolonged overcapacity until financial liberalisation heightened its vulnerability, contributing to the Asian financial crisis.

2.1.2 Literature on overcapacities in China

Overcapacity in China has become a focal topic in the academic literature since the mid-2000s,[4] reflecting the scale and persistence of low utilisation rates in several sectors in China such as coal (W. Wu & Lin, 2020), steel (OECD, 2025b) and PV (Y. Wang, Luo, & Guo, 2014). The literature centres on institutional and political-economic explanations for overcapacities.

While several authors highlight that experiences and expectations of fast-growing demand triggered repeated rounds of capacity expansion (Dong & Sun, 2022; D. Xu & Liu, 2018), most analyses show a link between low capacity utilisation and economic growth targets as well as sector-specific industrial policies (W. Chen, Gong, Jiang, & Jin, 2022; Ge, Zhu, Chen, & Huang, 2024; Lu, 2024). In particular, the institutional coordination between national and local government actors and the incentives and implementation of policies on the local level through various policy instruments (such as tax incentives, subsidies, preferential loans and land supply) are identified as the major factors for high capacity expansion (J. Chen, Chen, Hou, & Hu, 2021; Fang, Li, & Lu, 2025). Furthermore, preferential access to bank credit and government subsidies enable companies to prolong production, creating 'zombie firms' (Chao & Phillips, 2023; Shen & Chen, 2017). The latter phenomenon is also linked to the social targets these companies face in maintaining excess employment and supporting local stability objectives (Y. Zhang, Zhang, Liu, & Nie, 2017).

While the literature is rich in identifying the development and causes of investment surges and the consequent decline in utilisation rates, there is a lack of analysis of the consequences of overinvestment and low capacity-utilisation such as imbalances between production levels and effective demand, prices or profits. Only the risk of higher exports as an outlet for surplus

[2] A similar strategy was employed in Taiwan, where state-guided expansion and competition among small firms created temporary overcapacity that conditioned firms to meet international standards, illustrating that deliberate excess capacity was a characteristic feature of late-developing, export-oriented developmental states (Wade 1990).
[3] The United States and European Union frequently perceived Japanese and Korean surplus exports as excessive, prompting trade disputes, voluntary export restraints, and anti-dumping measures. The resulting trade frictions were managed by Japan through Voluntary Export Restraints on textiles, automobiles, steel, and semiconductors (Berry, Levinsohn, & Pakes, 1999; Itoh & Nagaoka, 1997).
[4] The literature on overcapacity in China is dominated by Chinese scholars, with a substantial share of studies published in Chinese-language journals.

production receives attention (Dai & Zhao, 2021; Matthes, 2024), particularly in papers of European organisations (Boullenois et al., 2024; Gunter et al., 2025).

2.1.3 Defining industrial overcapacity

Existing definitions and conceptualisations of overcapacity incorporate important structural considerations, but remain anchored in static indicators of unused capacity or firm-level profitability (Boullenois et al., 2024; Gunter et al., 2025). Building on the limitations of utilisation-based definitions and integrating insights from industrial organisation, macroeconomic, developmental state, and China-focused literature, we propose the following definition of industrial overcapacity in a national economy:

Industrial overcapacity arises when, over an extended period of 5 to 7 years, productive capacity expands faster than economically viable demand in industrial sectors, generating sustained overproduction pressure, while market adjustment occurs primarily through inventory accumulation, price declines and trade spill-overs rather than output contraction and capacity exit, allowing capacity and production to persist under depressed prices and margins. The prolonged duration of such conditions increases the likelihood that they are sustained by policy-induced incentives and institutional arrangements that encourage capacity expansion while weakening mechanisms for capacity reduction and exit.

The central features of this definition are, (i) the focus on the dynamic relationship between capacity expansion and demand as drivers of overcapacity, (ii) the channels through which excess supply is absorbed and (iii) market adjustment mechanisms and their economic effects. By assessing capacity growth relative to demand that can be sustained over time at economically viable prices and margins, the definition avoids misclassifying cyclical development and innovation dynamics as overcapacity.[5] Overcapacity is identified only when capacity expansion persistently exceeds demand growth and is not absorbed through market expansion or productivity-driven cost reductions. The five-to-seven-year period accounts for the time lag between initial investment decisions, the realisation of production capacity, and subsequent market absorption.

The definition further emphasises sustained overproduction pressure as the key transmission mechanism linking excess capacity to economic outcomes. Excess capacity often leads firms to maintain or expand production to cover fixed costs, service debt or meet strategic and policy objectives (see industrial organisation literature). In such cases, market adjustment does not primarily take the form of output reduction or firm exit. Instead, markets clear through higher inventories, exports or depressed producer prices and profits.

From a policy perspective, the criterion of persistence is particularly relevant, as it indicates a reduced likelihood of self-correction and a higher risk of spill-overs, including financial stress, resource misallocation, and trade tensions. While multi-year industrial overcapacity can arise from market-driven investment cycles, persistence beyond normal cyclical adjustment horizons is more likely in environments where institutional or policy conditions weaken market-based exit and reallocation mechanisms. Public support measures, preferential financing conditions, state ownership, and social or regional policy objectives can contribute to rapid capacity expansion and weaken the economic pressure to reduce output or exit the market and increase the probability that capacity–demand imbalances. Industrial overcapacities therefore extend beyond the duration of typical cyclical overcapacity phases.

[5] In fast-growing and technologically dynamic sectors, capacity is often built up ahead of current demand in anticipation of future market growth, learning-by-doing, and economies of scale (see developmental state literature). During such phases, prices may fall and utilisation rates may fluctuate without indicating economic distress or inefficiency.

2.2 Analysis of overcapacities in China

We operationalise the definition of industrial overcapacity for industrial sectors in China, using a set of sectoral variables primarily based on the National Bureau of Statistics of China (NBS)[6], UNIDO production and UN Comtrade trade data. The analysis of capacity expansion gaps and adjustment mechanisms is conducted in the industrial sectors 27 to 40 according to the Chinese Industrial classification for national economic activities (GB/T 4754-2017). These are part of the manufacturing sector but exclude the less capital-intensive sectors (textiles and food processing) and raw materials sectors. The selected sectors represent more than 80 % of China's exports in goods (UN Comtrade data).

The first element of the analysis is the relationship between capacity expansion and changes in demand. Since actual capacity expansion cannot be observed directly, we use changes in firms' non-current assets (NCA) as an approximation. Non-current assets reflect the value of long-term productive assets such as machinery and equipment.[7] Demand growth is represented by growth of nominal revenues from principal business, adjusted for producer prices. This yields the changes of real revenues (RR). For the case of China and its large internal market, we relate capacity expansion to domestic revenues (DR), i.e. total revenues less exports. This focuses on domestic demand and removes the effect of price changes, including price declines that may occur under overcapacity. The difference of changes in NCA and the changes in DR is defined as the Capacity Expansion Gap (CEG). The CEG indicates whether productive capacity is growing faster or slower than domestic demand.

The second element of the analysis is a set of absorption and adjustment indicators. Inventories relative to revenues and export shares capture the main absorption channels of excess supply. Rising inventories indicate that output exceeds market absorption, while increasing export shares suggest that excess production is redirected to foreign markets. The changes in producer prices reflect price-based adjustment mechanisms, while loss-making shares and profit margins capture profitability-based adjustments and indicate financial stress within the sector.

We assess capacity pressure and adjustment dynamics through rolling three-, five-, and seven-year averages from 2011 to 2024.[8] The five-year windows serve as the primary indicator of sustained overcapacity, reflecting the typical lag between investment decisions and capacity realization. The three-year windows capture more detailed and recent developments, while the seven-year windows show structural persistence beyond a single investment cycle.

2.2.1 Indications of overcapacities in China's industrial sectors

China has experienced a massive increase of investment in fixed assets (IFA) as well as revenues over the last two decades. Most IFA went to the components 'construction' (53 % of IFA on average) and 'expansion' (20 %), which are closely related to capacity expansion (NBS Statistics). Consequently, China's share in global manufacturing output increased from 27 % in 2011 to 36 % in

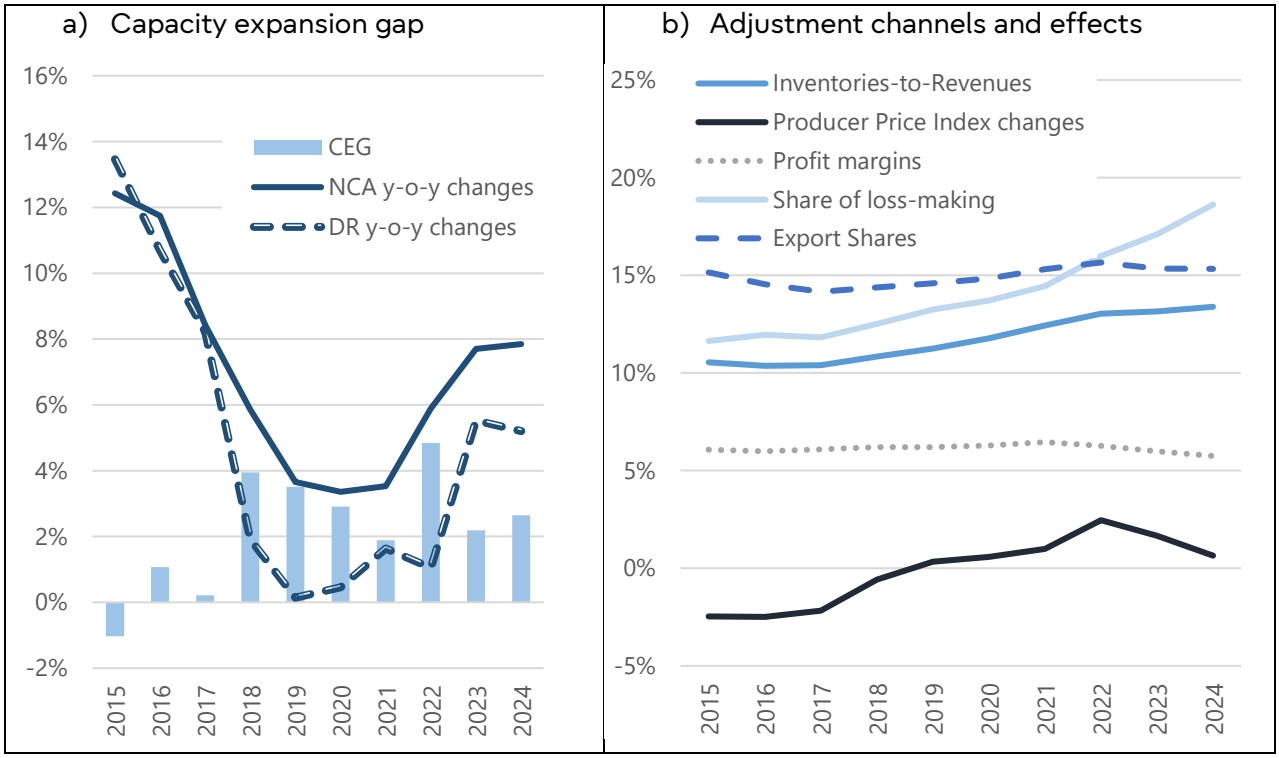
[6] The sectoral data used in this analysis are derived from China's enterprise statistical reporting system, not from the internationally standardized System of National Accounts (SNA). This limits cross-country comparability, and NBS data have been subject to critical scrutiny regarding consistency and quality (Holz, 2020). Data not available for download at the NBS Data explorer or China's Statistical Yearbook were extracted by use of an artificial intelligence tool (Claude.ai).

[7] An alternative indicator of capacity expansion is annual investment spending. However, gross investment does not directly translate into additional productive capacity, as a considerable share merely replaces depreciated or scrapped assets. To refine the analysis, the changes in NCA could be complemented by investment in new construction and expansion as components of investment in fixed-assets (IFA) relative to the NCA stock shows. This ratio indicates how intensively firms expand their productive base relative to their installed capital stock. Applying these alternative measures does not materially change the overall results.

[8] Capacity utilisation rates are however not available before 2017. Data from the GB/T 4754 2017 sectors 36 (Automobiles) and 37 (Railway, Ship, Aerospace & Other Transport) are available from 2012 onwards.

2022, while China's share in manufacturing exports increased from 15 % to 20 % (UNIDO production and UN Comtrade trade data). In several sectors, the growing dominance of China in global production is even more evident, for example, in electronics with an output share of more than 50 % and a share of more than 30 % in global exports (covering ISIC Rev 3 Sectors 30 & 31, which in sum are equivalent to sector 39 – Manufacture of Computers, Communication and Other Electronic Equipment in the Chinese sector classification). With an output surge of 36 % between 2011 and 2022, the growth in the automobile sector in China outpaced global production growth and the sector captured 34 % of global automobile output in 2024 (ISIC Rev 3 Sector 34 equivalent to C36 – Manufacture of Automobiles) (NBS, UNIDO production and UN Comtrade trade data).

Figure 1: Industry sector overcapacity indicators (5-year rolling averages)



Note: In Figure 1b Inventories-to-Revenues and Export Shares are expressed as % of total revenues and output respectively; Producer Price Index changes as % year-on-year change; Profit margins and Share of loss-making firms as % of total revenues and total firms respectively.
Source: Own elaboration based on NSB China data.

The selected sectors are aggregated to an industry sector, for which Figure 1 show the development of 5-year rolling averages of the CEG (a) and ratios and changes of absorption channels and adjustment mechanisms (b). While capacity expansion measured by the growth in NCA matched DR growth in the early 2010s, DR declined more than capacity expansion proxies thereafter, opening up a CEG since the 5-year window 2014 to 2018 (see Figure 1a).

The capacity utilisation rate (CUR), measuring the extent to which installed productive capacity is actually used in production, remained relatively stable despite a slight decline from 78 % in 2017 to 75 % in 2024 and 2025. Thus, other channels absorbed the capacity expansion pressure. Inventories and, to a smaller degree, exports both increased relative to RR from 2017 onwards. While profit margins remained largely stable and producer prices (PPI) increased slightly, the share of loss-making companies increased. Overall, this can be interpreted as an overcapacity regime with inventories functioning as the main buffer of high production levels.

Table 1 shows the CEG dynamics in the 3-year, 5-year and 7-year windows as well as the 5-year trend of the absorption channels and adjustment mechanisms at the aggregated industry sector

and for individual sub-sectors. For each indicator, coloured dots show whether the most recent observation (ending in 2024) provides a strong, moderate or weak signal of overcapacity relative to that sector's own historical distribution of each indicator. Red indicates a strong overcapacity signal, yellow an intermediate signal, and green no signal. Whether a high or a low value signals overcapacity depends on the indicator. For the variables CEG, Inventories (Inv., expressed as ratio to revenues.), share of loss-making companies (loss-making) and export shares, unusually high values show strong signals of overcapacities. For producer prices, capacity-utilisation ratios (CUR) and profit margins, unusually low values indicate overcapacity issues.

The reported ranges show the historical minimum and maximum values for each sector and window length. These ranges are sector specific and not comparable across sectors. For the CEGs the ranges are predominantly positive (see Figure 1a for the industry sector as a whole) indicating that capacity expansion proxies systematically outpace domestic sales — capacity expansion pressure is the norm, not the exception. A red dot for CEG signals that NCA growth exceeds DR growth in the current window and does so more strongly than in the sector's historical pattern.

Several distinct temporal patterns emerge. First, five sectors — Medicines, Non-metallic Minerals, Ferrous Metals, General Machinery and Measuring Instruments — exhibit persistent CEGs with red dot indicators in the 3-year and 7-year windows, showing that capacity-revenue misalignment is severe and persistent. Second, in several sectors CEGs improve in the short-run, but remain high in the 7-year window: Electronics, Chemical Fibres Electrical Machinery, and Special Machinery show moderate or improving conditions at shorter horizons but remain in the upper tercile at the 7-year window, indicating potentially large capacity expansions exceeding demand growth in the past. Notably, Metal Products, Non-ferrous Metals, Automobiles and Rail & Other Transport show moderate CEG growth in the 7-year window, but Metal Products and Non-ferrous Metals face a deterioration of the CEGs in the short-run.

The defining feature of overcapacity is that CEGs do not lead to capacity adjustment but to an elevated level of production that is instead absorbed through different channels. The analysis shows that inventory build-up serves as the dominant adjustment mechanism with eleven of fourteen sectors exhibiting elevated inventory-to-revenue ratios. Exports measured as exports relative to revenues plays a more limited and selective role as an absorption channel, with only six sectors showing elevated export shares, including medicines, rubber & plastics, general machinery, special machinery, automobiles and electric machinery.

Stimulating additional demand for higher output through lower prices has not been a widely used mechanism, yet PPIs have declined in many sectors since 2022. Rather several sectors face a downward trend in profit margins, but the historical ranges are very narrow. Strikingly, the share of loss-making enterprises has increased significantly across all fourteen sectors. Even though this indicator is highlighted as a key indicator in the literature (Boullenois et al., 2024), the general trend across sectors independent of the CEG and adjustment mechanisms points to competitive environment across all industrial sectors in China, discussed under the concept of 'involution' in China (see section 3.1.).

Overall, the analysis allows to differentiate four categories (see Table 2):

- Category 1: Five of the fourteen manufacturing sectors signal overcapacities under our definition, exhibiting red CEG indicators at the 5-year and/or 7-year horizons with CEG > 0 for most of the historical window. Adjustment occurs through inventory accumulation, rising export shares, and profit margin compression rather than production cuts as CURs remain relatively stable.

- Category 2: Five sectors also show high CEGs, but active adjustment pressure runs primarily through elevated inventories and lower profit margins, rather than through higher export shares.
- Category 3: Two sectors exhibit low or moderate CEGs, but face short-term pressure with a high 3-year average CEG value (Metal Products) or are driven by short-term surge in domestic demand (Automobiles), and show negative trends in all adjustment mechanisms.
- Category 4: Only Non-ferrous Metals and Rail & Other Transport sectors show little to no signs of overcapacity according to our definition.

Table 1: Analysis of overcapacity indicators Chinese manufacturing sectors

Sectors	CEG 3yr	CEG 5yr	CEG 7yr	CUR %	Inv.	PPI	Profit margin	Loss-making	Export shares
Industry (agg)	🔴 (-1%–7%)	🔴 (-1%–5%)	🔴 (-1%–4%)	🔴 (75–78)	🔴 (10%–13%)	🟡 (-2%–2%)	🔴 (6%–6%)	🔴 (12%–19%)	🔴 (14%–16%)
27 Medicines	🔴 (3%–15%)	🔴 (1%–12%)	🔴 (3%–10%)	🔴 (75–79)	🔴 (11%–19%)	🔴 (0%–1%)	🟢 (11%–16%)	🔴 (10%–21%)	🔴 (5%–9%)
28 Chemical fibres	🟡 (-5%–17%)	🔴 (-5%–10%)	🔴 (-4%–6%)	🟢 (81–85)	🔴 (10%–11%)	🟢 (-7%–2%)	🔴 (3.6%–4.9%)	🔴 (16%–25%)	🟢 (6%–7%)
29 Rubber & plastics	🔴 (-3%–7%)	🟡 (1%–5%)	🟡 (1%–5%)	n.a.	🔴 (8%–13%)	🟡 (-2%–1%)	🔴 (6.0%–6.5%)	🔴 (10%–16%)	🔴 (12%–15%)
30 Non-metallic minerals	🔴 (0%–17%)	🔴 (-1%–10%)	🔴 (-1%–9%)	🔴 (62–71)	🔴 (7%–12%)	🔴 (-1%–4%)	🔴 (7%–9%)	🔴 (10%–19%)	🟢 (3%–4%)
31 Ferrous metals	🔴 (-9%–15%)	🟡 (-4%–7%)	🔴 (-3%–5%)	🟡 (76–80)	🟢 (9%–11%)	🟡 (-7%–12%)	🔴 (2%–5%)	🔴 (18%–26%)	🟢 (2%–4%)
32 Non-ferrous metals	🔴 (-9%–7%)	🟢 (-3%–4%)	🟡 (-2%–4%)	🔴 (79–80)	🟢 (10%–11%)	🟢 (-5%–8%)	🟢 (3%–4%)	🔴 (17%–21%)	🟢 (2%–3%)
33 Metal products	🔴 (-5%–14%)	🟢 (-1%–10%)	🟡 (-3%–7%)	n.a.	🔴 (9%–13%)	🟡 (-2%–4%)	🔴 (5%–6%)	🔴 (10%–17%)	🟢 (10%–11%)
34 General Machinery	🔴 (-4%–11%)	🔴 (0%–5%)	🔴 (0%–5%)	🟡 (77–81)	🔴 (13%–20%)	🟢 (-1%–1%)	🟢 (7%–7%)	🔴 (10%–15%)	🔴 (11%–14%)
35 Special machinery	🟡 (-10%–24%)	🔴 (-4%–9%)	🔴 (0%–7%)	🔴 (76–80)	🔴 (15%–25%)	🟡 (0%–1%)	🟢 (7%–8%)	🔴 (10%–18%)	🔴 (9%–14%)
36 Auto-mobiles	🟢* (-2%–5%)	🟢* (-1%–4%)	🟢* (0%–3%)	🔴 (72–82)	🔴 (8%–10%)	🟡 (-1%–0%)	🔴 (6%–9%)	🔴 (12%–22%)	🔴 (4%–7%)

Sectors	CEG 3yr	CEG 5yr	CEG 7yr	CUR %	Inv.	PPI	Profit margin	Loss-making	Export shares
37 Rail & other transport	🟡 (-3%–5%)	🟢 (-1%–4%)	🟢 (1%–1%)	n.a.	🟡 (20%–30%)	🟢 (-20%–1%)	🔴 (5%–6%)	🔴 (13%–18%)	🟢 (13%–20%)
38 Electrical machinery	🟡 (-2%–7%)	🟡 (0%–7%)	🔴 (1%–5%)	🔴 (75–81)	🔴 (10%–13%)	🟢 (-2%–1%)	🔴 (6%–7%)	🔴 (12%–18%)	🔴 (14%–18%)
39 Electronics	🟢 (-7%–10%)	🟡 (-9%–10%)	🔴 (-6%–7%)	🔴 (76–81)	🔴 (9%–13%)	🟢 (-2%–-1%)	🟡 (4.8%–5.4%)	🔴 (16%–23%)	🟢 (44%–56%)
40 Measuring instruments	🔴 (-1%–10%)	🟡 (-1%–6%)	🔴 (-1%–6%)	🟢 (77–86)	🔴 (13%–22%)	🟢 (0%–1%)	🟢 (9%–10%)	🔴 (10%–15%)	🟢 (14%–18%)

Note: *The negative CEGs with domestic revenue growth exceeding NCA growth in the recent windows in Automobiles are exceptional; earlier periods show positive CEGs reflecting prior capacity expansion pressure.

Source: Own elaboration based on NBS China data.

Table 2: Sectoral overcapacity assessment results

Category	Count	Sectors	Key feature
1. High CEGs & broad adjustment	5	Medicines, Rubber & plastics, General machinery, Special machinery, Electrical machinery	CEG red at 5yr AND/OR 7yr +inventory and export adjustment mechanisms are active
2. High CEGs & national adjustment	5	Chemical fibres, Non-metallic minerals, Ferrous metals, Measuring instruments, Electronics	CEG red at 5yr AND/OR 7yr + inventory adjustment mechanisms active
3. CEG-risks	2	Metal products, Automobiles	CEG green/yellow at 5yr AND/OR 7yr but elevated risk to turn red
4. No long-term CEGs nor major adjustments	2	Non-ferrous metals, Rail & other transport	Short-term CEG spike, efficient adjustment, no absorption

Source: Own elaboration.

The aggregated industry sector confirms the widespread overcapacity issue in China's industry. However, it should be noted that the industry sector's ranges as a whole are very narrow. Thus, a red dot does not indicate a massive CEG pressure and substantial shifts in the aggregated sector. Related to this, some dominant sectors with regard to their share in total revenues and exports such as Electronics (20 % of revenues and 50 % of exports of the industry sector) and Non-ferrous Metals (10 % / 2 %) show currently no or moderate CEG and adjustment indicators.

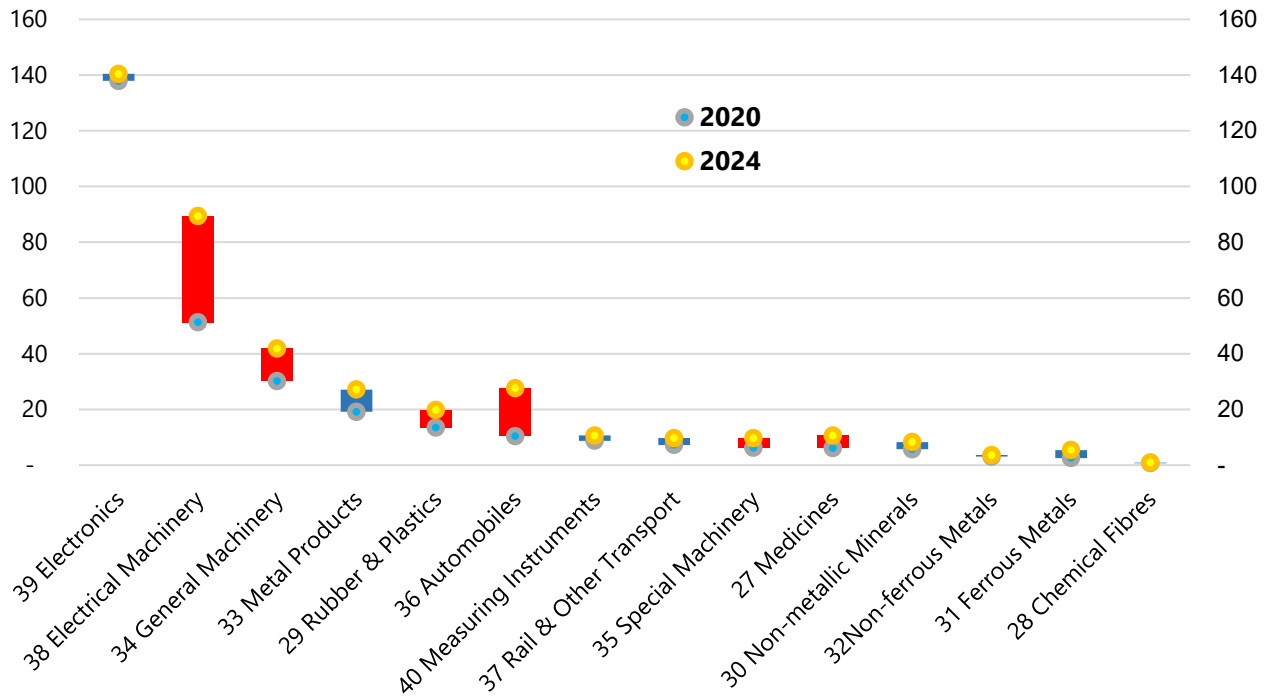
2.2.2 Trade effects of overcapacity risk

According to our analysis, exports have not emerged as the major outlet for overcapacity risks as only six of the fourteen sectors show an increased export share compared to the previous 5-year average (Figure 2). However, even unchanged export shares can lead to higher exports, as production could potentially be accelerated by overcapacity pressure. The growth of China's export of industrial goods as a whole has generally followed revenue trends. From 2020 to 2024, total revenues in the aggregated industry sector increased by 29.4 % despite notable variations due to the COVID-19 crisis (NBS data). In parallel, total exports of industrial goods to the world grew by

31 %, and to the EU by 34 %. However, six sectors with high capacity expansion show growth rates of exports to the world and to the EU above the average, in particular from 2020 onwards.[9]

The export surge between 2020 and 2024 was driven by specific sectors, including Electrical Machinery (+74 %), with lithium-ion batteries and static inverters, General Machinery (+39 %) with industrial components such as valves, transmission parts, and Automobiles with EVs (+165 %). The other sectors with higher export rates are Rubber & Plastics (+46 %), Special Machinery (+56 %) and Medicine (+72 %). Electronics as the largest export sector has seen only relatively small export growth (+2 %). Nevertheless, the size and importance of Chinese industry sectors relative to the world economy can be seen when relating additional revenues to export changes. For instance, total exports from China to the EU in the automobile sector in 2024 are equivalent to 1.9 % of total sector revenues, indicating that given the sector's total capacity, even small changes in exports could have strong effects on import markets.

Figure 2: China's exports to the EU in industrial sectors (in billion USD)



Notes: Sectors according to the Chinese classification GB/T 4754 2017.
Dots indicate export values in 2020 and 2024, respectively. The bars indicate the increase between 2020 and 2024. The red bars highlight the sectors with an increasing export share.
Source: Own elaboration based on UN Comtrade.

2.2.3 Non-market elements and overcapacities

The debate about overcapacities in China has focused on the effects of non-market elements that expand capacities and continue production despite excess capacities, including subsidies and tax incentives, direct R&D grants, and below-market loan interest rates. Furthermore, the effects of the institutional arrangements in China, in particular the lack of coordination between local and national government units are discussed (see literature overview in section 3.1). The debate, however, suffers without of a consistent and comprehensive database on these policy incentives in China (Chimits, 2023).

[9] Based on an analysis of EUROSTAT HS6 import data (weighting by base-year trade value) we cannot confirm that the changes in trade values between 2000 and 2024 in these sectors are driven by quantity expansion, with would imply declining unit prices in Chinese exports to the EU.

Recent research efforts created and analysed subsidy data. The OECD MAGIC database (OECD, 2025a), which tracks subsidies to large manufacturing firms globally from 2006 to 2022, shows that China-based firms receive larger, more persistent, and more finance-based support than firms elsewhere, primarily through below-market loans from state-controlled banks, complemented by grants and tax concessions. While average subsidy levels in China appear modest (around 0.7 % of firm revenue), support is concentrated in heavy industry and strategic sectors such as semiconductors, shipbuilding, and solar PV. Complementary analysis of China's WTO subsidy notifications by Xiang, Yin and Zi, (2025) shows a rising number of subsidy programmes since 2001, particularly at the local level, with direct fiscal outlays stabilising at around 0.8 % of GDP (gross domestic product) and a policy shift toward industry- and innovation-oriented support. However, these datasets exclude non-budgetary and implicit support and exhibit limited sectoral comparability, thereby limiting the direct applicability to this report.

We therefore rely on NBS data as the sources for IFA and for SOE-specific statistics. Annual investments in fixed assets in industry sectors that are potentially linked to capacity expansion, are, with an average share of 92 %, mainly self-financed (cash-flows, equity, bonds), followed by loans (5 %) and foreign and other sources. The state budget is a minor source of annual IFA accounting for 0.6 % of actual funds available in 2024. However, selected sectors, such as Electronics, Rail & Other Transport and Automobile fund more than 1.2 % of IFA directly from state budgets.

The role of SOEs in the production process of several industrial sectors is nevertheless significant in several industry sectors, in particular in Automobiles (SOEs generated 35 % of revenue in the sector in 2023), Rail & other Transport (58 %) and Ferrous and Non-ferrous Metals (34 % each). The CEG of SOEs at the 5-year window ending in 2023 exceeds that of private firms in several sectors, indicating that SOEs face greater pressure to expand capacities (see [OBJ][OBJ][10]The data also reveal that SOEs in many sectors have inventory-to-revenue ratios that are higher than their revenue share in the sector. Thus, SOEs disproportionately contribute to the use of inventories as a quantity buffer. SOEs across several sectors have also experienced larger declines in profit margins since 2017 than private firms. Finally, SOEs in the sectors Automobiles and Rail & other Transport sectors report negative financial expenses relative to revenues. More generally, SOEs in these sectors show lower financial expenses than SOEs in China overall. Even though financial expenses also include interest income from financial assets, the lower financial expenses could be an indicator for below-market borrowing, a major channel of subsidization (OECD, 2025a).

Table 3: Role of SOEs in China's industry sectors

	CEG (5 yr 2023)		Share of total revenues	Share of total inventories	Profit margins	Financial expenses/rev
	SOEs	All companies			Change vs 2017 in %-points	
All sectors	1 %		28 %	25 %	- 0.3	1.3 %
27 Medicines	5 %	9 %	9 %	13 %	4.3	0.0 %
28 Chemical fibres	9 %	7 %	12 %	17 %	- 4.9	1.7 %
29 Rubber & plastics	5 %	3 %	4 %	6 %	3.6	0.6 %
30 Non-metallic minerals	7 %	7 %	9 %	15 %	- 3.8	1.0 %
31 Ferrous metals	-4 %	3 %	34 %	33 %	- 3.8	0.8 %
32 Non-ferrous metals	-4 %	1 %	35 %	36 %	3.2	0.7 %
33 Metal products	-1 %	3 %	7 %	15 %	0.5	0.5 %

[10] Data on state-holding companies are mainly available in the Statistical Year Books published by the NBS (Table 13-4).

	CEG (5 yr 2023)		Share of total revenues	Share of total inventories	Profit margins	Financial expenses/rev
	SOEs	All companies			Change vs 2017 in %-points	
34 General machinery	4 %	3 %	10 %	19 %	- 0.0	0.0 %
35 Special machinery	0 %	6 %	12 %	21 %	2.2	1.0 %
36 Automobiles	3 %	1 %	44 %	29 %	- 4.2	-0.3 %
37 Rail & other transport	3 %	0 %	42 %	76 %	0.3	-0.1 %
38 Electrical machinery	4 %	0 %	9 %	10 %	- 1.7	0.4 %
39 Electronics	3 %	3 %	9 %	18 %	- 4.9	0.7 %
40 Measuring instruments	1%	1 %	8 %	12 %	3.9	0.5 %

Source: Own elaboration based on NBS data.

Overall, SOEs in Chemical Fibres, General Machinery, Electric Machinery, Electronics sectors and particularly in those in the Automobile sector may contribute to overcapacities through higher capacity expenditures, above-average inventory build-ups and low profit rates. Further, all SOEs can potentially benefit from below-market loan rates loans. However, in the sectors for which we identified the lowest overcapacity indications — Non-ferrous metals and Other Transport — the importance of SOEs is high. Thus, the contribution of SOEs to overcapacities remains highly sector-specific.

In conclusion, the empirical evidence linking existing overcapacities in industrial sectors to policy-induced measures remains scattered due to limited databases. Thus, the link between China's official policies and actual flows into investments and support measures remains weak. The main obstacles are low data transparency and the variety of instruments used by various governmental actors in China, which limit the ability to capture the comprehensive effects.

2.3 Case studies: Currently emerging overcapacities

In two case studies, the detailed dynamics of overcapacity and risk for EU manufacturing are assessed for the examples of semiconductors and EVs/batteries. Both cases illustrate cases with projected high demand growth and developing production systems in the EU that are potentially threatened by Chinese overcapacity.

2.3.1 Case study: Semiconductors

Industry overview and capacity dynamics

The semiconductor value chain consists of design, manufacturing (frontend and backend), and integration into electronic systems. Frontend fabrication, the most capital-intensive part of the value chain, builds circuits on silicon wafers in fabrication facilities ('fabs'), producing hundreds or thousands of chips per wafer. Backend factories cut wafers into chips, assemble packages, test them, and ship them to integration facilities. Europe's global share is 8 % in frontend capacity but only 1 % in backend, which has largely been offshored to Asia since the 1980s (ZVEI, 2025). A single chip typically passes through three to six factories across different locations during production (Yeung, 2022), making production and trade difficult to monitor (OECD, 2025c). European production and demand focus on mature/legacy semiconductors (≥28 nm nodes), in particular analogue, discrete and micro-component devices primarily serving automotive and industrial sectors (Bonnet et al., 2025, pp. 41–42; Raj & Wuttke, 2026). Advanced logic and memory chips production for electronics assembly occur largely in Asia (Yeung, 2022).

Assessing capacity expansion gaps is complex as capacity (wafer starts/month) and demand (monetary value) use incomparable units. However, existing studies see no gap between current capacities and demand in global or Chinese mature semiconductor production (Benson et al., 2024; Ebrahimi, 2024; Goujon et al., 2024; Gunter et al., 2025; Irwin-Hunt, 2024; Rühlig, 2024; Schumacher, 2024; Triolo, 2024). With Chinese fab utilisation rates matching or exceeding global averages (Goujon et al., 2024), there is no evidence of market flooding or price dumping in global markets, and capacity increases are aligned with growing domestic demand driven by digitalisation and automotive electrification (Schumacher, 2024; Triolo, 2024).

Concerns centre on future Chinese investments potentially causing overcapacity (see Irwin-Hunt, 2024). Global mature-node capacity is expected to grow at 6.05 % of compound annual growth rate (CAGR) (2023–2030) according to SEMI World Fab Forecast data (Triolo, 2024).[11] Most automotive semiconductor demand forecasts suggest that overcapacity is unlikely (For Insights Consultancy, 2024; PwC, 2024; TechInsights, 2024; The Research Insights, 2025). However, recent forecasts are less optimistic (Tschiesner et al., 2025). One European semiconductor firm's Manufacturing Analytics Manager expects 'overcapacities in mature semiconductors across the board' based on more recent lower demand forecasts (Interview 1).

Even though European capacity is expected to grow at 4 % (CAGR, 2023–2030) versus 11 % in China (Triolo, 2024), the automotive semiconductor demand in Europe is forecast to grow at roughly the same rate as Asia-Pacific at 8–9 % CAGR 2025–2031 (Mordor Intelligence, 2025). This creates a mismatch where capacity outside of the EU, potentially from China, will have to meet the increasing European demand (see also Rühlig, 2024). This reflects, however, an internal European capacity shortfall, rather than Chinese overcapacity.

Overcapacities are unlikely to emerge despite the high relative capacity growth rate, because current Chinese production cannot meet domestic demand (Ebrahimi, 2024; Triolo, 2024). Capacity expansion will mainly help to fill this existing gap, in line with China's main policy objective of increasing semiconductor self-sufficiency (see below). Data, albeit limited, suggest China's mature semiconductor capacity met only 37 % of domestic demand in 2020 (Triolo, 2024), and that total semiconductor production met 23 % of demand in 2023 (Goujon et al., 2024). China produced 30 % of global mature semiconductors but 33.8 % of motor vehicles in 2024. Given that EVs contain 3 to 10 times more semiconductors than internal combustion engine (ICE) vehicles (see Lawrence and VerWey, 2019) and account for 64 % of China's production, this suggests under-proportionate capacity. Also, Rühlig (2024) finds no mismatch in China, while Taiwan and Japan, for example, show large capacity-demand overshoots.

Nevertheless, on a more detailed level at the type of mature semiconductors, capacity expansions in China are disproportionate in sensors and silicon-carbide power semiconductors (Interview 2). China's share in global output of sensors (as a sub-category of discrete semiconductors) grew from 10 % (2020) to 19 % (2025) and is expected to rise rapidly (ibid.). In silicon-carbide semiconductors (another discrete sub-category), China reached 42 % of global capacity by 2023, projected to hit 50 % by 2026 (ibid.).[12] Chinese firms are also "massively adding capacity" in mature-node microcontrollers, with future overcapacity likely (Interview 1).

[11] It needs to be noted that the SEMI World Fab Forecast, on which all of these different studies base their production capacity projections, collects its data by calling semiconductor firms and asking them about their current capacity and current/planned investments. There can thus be no certainty that the data is correct. There are some worries regarding the Chinese data in particular (Interview 1; Interview 2), which have intensified recently, as the important provider of semiconductor industry analysis TechInsights was blacklisted in China in October 2025 (Fujino, 2025).

[12] On the point about China's high share in silicon-carbide semiconductors, see also Office of the United States Trade Representative (2025).

Europe's technological landscape and global positioning

Figure 3 breaks down European capacity by semiconductor type and over time. Europe's capacity in 2023 is almost exclusively at mature nodes (≥28 nm), with exceptions being Intel in Ireland (microcomponent) and Globalfoundries in Germany (foundry[13]). Significant producers include NXP in the Netherlands (ASIC logic), STMicroelectronics in France (ASIC logic, discrete), Infineon in Germany (ASIC logic, analogue, discrete), STMicroelectronics in Italy (microcomponent, discrete), onsemi in Czech Republic (analogue), Nexperia in the UK (analogue, discrete), AMS-Osram in Austria (analogue), Texas Instruments in Germany (analogue), Analog Devices in Ireland (analogue), ELMOS in Germany (analogue), Infineon in Austria (discrete), Bosch in Germany (discrete), Nexperia in Hamburg (discrete), AMS-Osram in Germany (discrete), Vishay in Germany (discrete), X-Fab in Germany and France (foundry), as well as Newport Wafer Fab (today Vishay) in the UK (foundry).

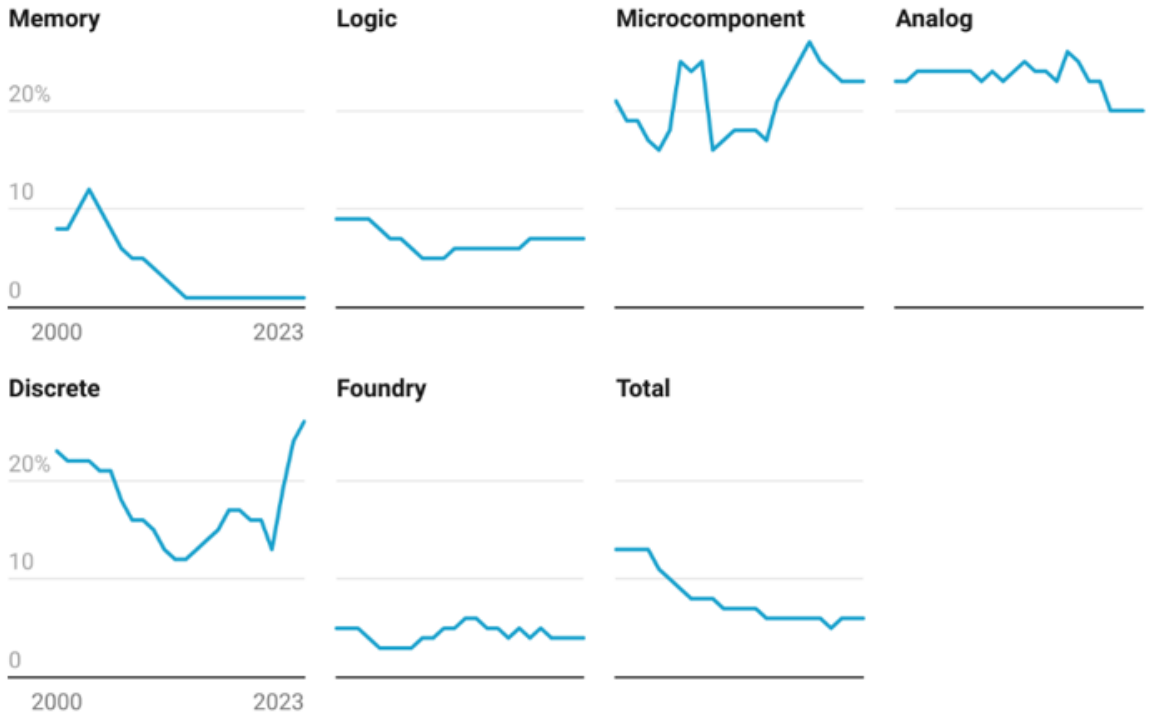
Many of these operations are long-established, and investment in European semiconductor fabs was very limited up to 2020 (EC, 2022). This has changed in recent years, as subsidies provided under the 'Important Projects of Common European Interest' (IPCEI) scheme and under the European Chips Act have improved the business case for investing in semiconductor production in Europe compared to other locations that have long offered such support (see Raj & Wuttke, 2026). Despite major investments since 2021, overall investment in other world regions, especially in China, outpaces Europe (see Triolo, 2024).

Given the European production structure, the largest risks for European production emerge from China's disproportionate capacity expansions in sensors and silicon-carbide power semiconductors. These can threaten European production significantly, as Europe produces substantial discrete and micro-component volumes (see Figure 3), with STMicroelectronics particularly invested in silicon-carbide through new Italian facilities (Interview 2). In analogue mixed-signal semiconductors, no threat from Chinese capacity is expected in the medium term (Interview 1).

Moreover, Europe holds strong global shares in semiconductor materials (19 %) and equipment (39 %) (ZVEI, 2025). While smaller than frontend in value-added, Europe's higher global share means substantial potential loss. Interviews confirmed no significant overcapacity risk in these categories currently (Interview 1; Interview 2), with European, US, and Japanese firms leading in key equipment and materials. ASML holds a de facto monopoly in advanced lithography equipment, although China is reportedly pursuing domestic capability development in this segment (Potkin 2025).

[13] Foundry is not a distinct semiconductor type *per se*. It includes semiconductors of several types, such as logic, microcomponents and analogue, but is defined by the production method, i.e. in a foundry. This means that the firm is a semiconductor contract manufacturer for other semiconductor firms.

Figure 3: Europe's share in global wafer capacity, by type, 2000–2023



Source: Created by the authors using wafer capacity data from a custom proprietary dataset assembled by the Global Production Networks Center at the National University of Singapore, ultimately derived from market research firms IHS Markit and Informa Tech and provided courtesy of Professor Henry Yeung, Co-Director of GPN@NUS (subsequently referred to as GPN@NUS).

Policy landscape and key drivers in China

China's semiconductor expansion is driven by state-directed industrial policy aimed at achieving technological self-sufficiency.[14] The 14th Five-Year Plan (2021–2025) prioritizes technological independence, with integrated circuits featuring prominently. Beginning with the 'Made in China 2025' policy, aimed at transforming the country into a technologically advanced global manufacturing powerhouse, semiconductors were designated critical to national security with targets of 49 % self-sufficiency by 2020 and 75 % by 2030. As one expert noted, "China's strategic objective for semiconductors is self-sufficiency above anything else. Overcapacity could be a by-product, but not a strategic objective" (Interview 2).

Government Guidance Funds (GGFs) serve as the primary support mechanism, with equity support the most important form of state support (OECD, 2019). The National Integrated Circuit Industry Investment Fund ('Big Fund') deployed CNY 138.7 bn (EUR 20.2 bn) in 2014, mobilising CNY 514.5 bn (EUR 83 bn) in private capital, and allocated 67 % on frontend manufacturing. Phase II launched in 2019 with CNY 204 bn (EUR 30 bn) in chip design and equipment. Phase III launched in 2024 with a CNY 334 bn (EUR 46 bn) budget. Beyond the national fund, numerous local semiconductor funds operate similarly at provincial and municipal levels. State-owned enterprises dominate among the leading Chinese semiconductor firms, with state ownership providing crucial equity. Additional support mechanisms since 2000 include tax incentives, direct R&D grants, below-market loans, and global talent attraction support schemes.

[14] Unless otherwise indicated, the claims about Chinese semiconductor policy made in this section are based on EC (2024a, pp. 548–602).

Policy-induced localisation pressures on the Chinese semiconductor market are intensifying. This is a major concern for European firms which often generate 20–30 % of their revenue through sales on the Chinese market (Interview 1; Interview 2). Informal local content requirements are emerging for Chinese products using semiconductors, conveyed informally during closed-door meetings as well as via industry associations (ibid.). The European semiconductor firm with operations in China we interviewed reported the existence of informal requirements mandating 30–50 % domestic semiconductor content in newly approved electric vehicles within a three- to five-year timeframe, as well as a 100 % domestic content requirement for cybersecurity applications encompassing semiconductor production and R&D (Interview 1). Whether foreign firms with Chinese factories qualify as 'domestic' is uncertain, prompting European firms to expanding local production or by increasing their domestic ownership through partnerships with local firms (Interview 3). There is little doubt among industry practitioners that eventually 'domestic' will mean indigenous Chinese firms only (Interview 1).

So far, and depending on the potentially limited quality of data, Chinese self-sufficiency in semiconductors is far below targets and imports remain very high. Triolo (2024) predicts that "foreign firms like NXP, Infineon, Renesas, and Texas Instruments will continue to hold dominant positions in the Chinese market" because of their significant technological lead compared to domestic Chinese firms in certain types of semiconductors, which is partly based on intellectual property rights (Interview 2). This is fundamentally different from the situation in EV and batteries where Chinese firms are the technology leaders (see Chapter 2.3.2.).

While China's state support for the semiconductor industry is substantial, it is not unique. The EU's IPCEI and Chips Act programmes, along with similar U.S. and Japanese initiatives, demonstrate that semiconductor subsidisation is a global phenomenon (other countries like Taiwan, South Korea, Singapore, and Malaysia also have a history of semiconductor subsidies, mainly in the form of tax incentives, see OECD, 2019). However, European subsidies are averaging 30 % of fab investment for capital expenditures (Raj & Wuttke, 2026), while China's full support levels remain opaque but are most likely higher, covering operational expenditures, while they have been provided for many years before the European programmes even began.

Outlook and implications for the EU

Currently, there are limited signs of Chinese capacity expansion beyond demand growth in semiconductors broadly or in mature-node frontend specifically, despite heavy state aid. Overcapacity as defined in 2.1., where subsidized Chinese capacity outgrows domestic demand, driving inventories and below-cost exports — is unlikely, as Chinese capacity must grow massively to close current domestic demand-production gaps, the main industrial policy objective, and global semiconductor demand is expected to grow rapidly with digitalization.

Nonetheless, two risks exist for European firms and production. First, there is a risk that increasing European demand for mature semiconductors will not be met by increased frontend production in Europe, but rather by Chinese production (although this is not a given, see discussion above). This presents a missed opportunity to grow the European semiconductor production footprint and ecosystem, rather than a problem of Chinese overcapacity.[15] To avoid this scenario, extensions and intensifications of the current subsidy programmes in Europe need to be discussed, at least to match the incentives provided elsewhere, particularly in China, to make the business case for

[15] This could potentially lead to a further increase of already high dependencies in the value chain, which are already problematic from a geopolitical (e.g. dependency on advanced logic chips from Taiwan, and high backend production of certain semiconductors in China and Taiwan) as well as a supply chain resilience perspective (no memory chip production and almost no backend production in Europe) – topics which are beyond the scope of this particular study, but of course tangentially relevant.

investing in Europe more attractive. Currently, each EU Chips Act subsidy undergoes detailed Commission analysis of non-European distortions, with subsidies approved accordingly. The gathered data could be used to develop a ballpark figure for acceptable subsidy levels, thereby avoiding the need to keep repeating this time-consuming exercise. State aid procedures in the EU are notoriously slow (see Schmitz et al., 2025), and as speed is of the essence to ensure that investments are not distorted away from Europe, pragmatic approaches should be considered.

The second risk concerns discrimination against European semiconductor firms on the Chinese market due to self-sufficiency drives, as evident by unofficial local content requirements. Currently, strong European firms like Infineon, STMicroelectronics, NXP, and Bosch generate significant revenues in China (Lee & Kleinhans, 2024), because China is the largest automotive and industrial production hub. If market share losses result from competition, there is no complaint. Nevertheless, discriminatory policy measures make a difference. Disentangling whether Chinese industrial customers shift from European to Chinese firms due to state requirements or their own initiative is difficult, as requirements are opaque. Nevertheless, monitoring this situation and discussing carefully crafted countermeasures if discrimination is credibly proven is essential.

Concerning potential EU local content rules, one interviewee representing an industry association was rather sceptical: 'We need to be very careful with this. Many European semiconductor firms are highly dependent on the Chinese market for revenue. So, only implement such measures as a retaliation in case they do it, and still be very careful' (Interview 2). A representative from a semiconductor firm, notably one with a high dependence in sales on the Chinese market, was more forceful: 'China is massively distorting with all its incentives and subsidies. Hence, we should already discuss a well-defined local content criterion…including strategic friends and safe countries of origin such as the US, Japan, and Korea. This should be differentiated by end market…already implemented for security/defence-related applications, more gradually for automotive. The United States Connected Vehicle Regulation could serve as a role model' (Interview 1). However, it should be mentioned that such an approach would not be in conformity with WTO law (see section 3.2).

The interviewees agreed on the need for an EU monitoring system for overcapacity risks — tracking capacity, demand, and market shares by semiconductor sub-type. This could occur under Pillar 3 of the EU Chips Act, which has monitoring components for supply security. The current Chips Act review is an opportunity for adding overcapacity monitoring, requiring appropriate indicators of development and industry coordination.

The interviewees pointed out that tariffs represent an inappropriate policy instrument for addressing semiconductor overcapacity concerns for several reasons. First, there exists significant ambiguity regarding the appropriate point of application — whether tariffs should target frontend products (wafers/dies), backend products (packaged/tested chips), or both stages of the supply chain. Moreover, semiconductors rarely enter the European market as semiconductors; rather, they are pre-integrated into final products or intermediate modules, complicating tariff implementation and enforcement. Second, the economics of mature-node semiconductors undermine the effectiveness of tariffs. Many mature semiconductor products are extremely low-value components, with unit prices as low as three to four cents. At such price points, even substantial ad valorem tariffs would have a negligible impact on total product costs and would be unlikely to alter market dynamics meaningfully.

2.3.2 Case study: Electric vehicles & batteries

Industry overview and capacity dynamics

The share of EVs increased drastically from 4 % of global vehicle sales in 2020 to 18 % in 2024 (IEA, 2025d) with sales in China accounting for 65 % of global EV sales.[16] Chinese EV production in 2024 represented 72 % of global output and exceeded domestic consumption by one million units. European and rest-of-world production and demand are roughly aligned, though production is slightly lower than sales. The gap between demand and production in Europe[17] and elsewhere corresponds to China's production-demand surplus. Capacity utilisation rates are more or less uniform across regions, with China and Europe roughly equal (see Table 4). The IEA projects EV sales of 40 million units by 2030, which exceeds current global EV production capacity by 37 % (IEA, 2023).

Table 4: Electric vehicles, demand/production/capacity in 1000 units, 2024

EVs	World	China	Europe	ROW
Demand	17 500	11 300 (65 %)	3 180 (18 %)	3 020 (17 %)
Production	17 300	12 400 (72 %)	2 500 (14 %)	2 400 (14 %)
Capacity	29 100	21 500 (74 %)	4 100 (14 %)	3 500 (12 %)
Capacity utilisation	59 %	58 %	61 %	69 %

Source: Demand data is 2024 sales data based on IEA Global EV Outlook 2025, Global EV Data Explorer. Production and capacity data is 2024 data based on IEA World Energy Outlook 2025, p. 75.

Battery packs constitute 30 % to 50 % of an EV's value, with battery cells representing roughly 70 % of the pack's value.[18] China's share in global EV battery cell demand is 60 %, aligned with its EV demand share (Table 5 5). However, its production share is much higher at 81 %. Europe's and the rest of the world's production shares are substantially lower than their demand shares. This aligns with China's USD 58.7 bn lithium-ion battery (LIB) trade surplus in 2024, while the EU had a corresponding trade deficit of USD 18 bn, with many LIB imports from China (UN Comtrade data, HS 850760).

Table 5: EV battery cells, demand/production/capacity in GWh, 2024

EV battery cells	World	China	Europe	ROW
Demand	996	566 (57 %)	170 (17 %)	260 (26 %)
Production	1 100	891 (81 %)	73 (7 %)	136 (12 %)
Capacity	3 364	2 827 (84 %)	173 (5 %)	364 (11 %)
Capacity utilisation	33 %	31.5 %	42 %	37 %

Source: Demand data is for all sorts of e-vehicles, including passenger cars, buses, trucks, and two/three wheelers. Demand data is 2024 sales data based on IEA Global EV Outlook 2025, Global EV Data Explorer. Production and capacity data is 2024 data based on IEA World Energy Outlook 2025, p. 75.

The overall gap between production and demand in Table 5 appears because while demand data only includes demand for battery cells for EVs, production plants can serve varying end markets,

[16] The definition of EVs used by the IEA and in this report includes battery electric vehicles, plug-in hybrid electric vehicles, fuel cell electric vehicles, and range-extended electric vehicles.
[17] For production/capacity, the data is only available for Europe, not for EU27. Europe is defined in the IEA as: "European Union regional grouping and Albania, Belarus, Bosnia and Herzegovina, Gibraltar, Iceland, Israel, Kosovo, Montenegro, North Macedonia, Norway, Republic of Moldova, Serbia, Switzerland, Türkiye, Ukraine and United Kingdom".
[18] Based on battery cell and pack price data from BloombergNEF between 2013 and 2025.

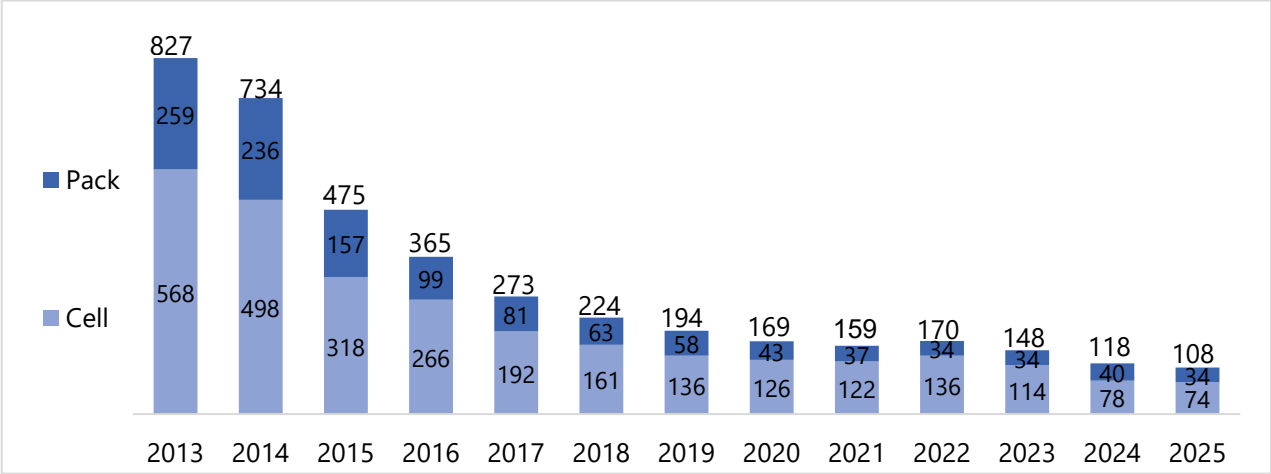
including energy storage and consumer electronics. Although EVs are by far the largest end market at more than 85 % (IEA, 2025d, p. 134) and this ratio is forecast to remain stable.

Table 4 shows very low average global battery cell factory capacity utilisation in 2024, ranging from 31.5 % in China to 42 % in Europe.[19] One major reason is slower EV demand growth than assumed. In 2022, battery cell demand was projected to reach 1700 GWh by 2025 (Fleischmann et al., 2023), contrasting with actual demand standing at less than 1000 GWh in 2024. Another reason is that battery factories cannot easily switch between different form factors (prismatic, pouch, cylindrical) or chemistries (NMC, NCA, LFP), creating demand-capacity mismatches when carmakers change preferred cell designs for new vehicle models (Interview 4; Interview 7).

Low capacity utilisation is not necessarily problematic for battery cell producers. According to interviews, battery firms build facilities only on the basis of off-take agreements with car producers per vehicle model (Interview 4). These agreements contain take-or-pay commitments where cell producers fine carmakers for taking fewer cells than contractually agreed. Such fines occurred frequently in 2024–2025 due to slower EV sales than anticipated when agreements were signed (Interview 5).

Capacity utilisation rates are expected to increase as EV demand rises (Interview 5). Nonetheless, the capacity-production/demand difference is remarkable. Current forecasts anticipate 3200 GWh EV battery cell demand by 2030, meaning current global installed capacity as of 2024 (see Table 5) already exceeds total anticipated 2030 demand, despite expected EV uptake growth.[20] The difference between capacity and demand is especially striking in China despite rapid recent EV uptake. Globally, battery cell and pack prices declined significantly due to technical innovations (see Figure 4).

Figure 4: Battery pack and cell prices, real 2025 USD/kWh



Source: Weighted average survey data from BloombergNEF.

EU's technological landscape and global positioning

The automotive industry is the EU's largest manufacturing sector, providing direct and indirect jobs for 13.8 million people (6.1 % of EU employment) and generating revenue equal to 7 % of EU GDP.[21]

[19] While there is no space here to discuss this in depth, there is also a large mismatch between production capacity and demand in the core battery materials, with global production capacity for cathode materials being 95% higher than global demand in 2024, and production capacity for anode materials being 100 % higher than demand (for more details, see Sirma and Wicke, 2025).
[20] The source for the 2030 demand forecast is IEA Global EV Outlook 2025, p. 136.
[21] Source: European Commission, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs.

In 2024, the EU produced 11.4 million and registered 10.6 million cars of all propulsion types. Of total EU production, 7.2 million vehicles were sold within the EU. As a result, the EU records a trade surplus, exporting 4.5 million cars while importing 3.4 million units (ACEA, 2025).

In EVs, the EU has a similar position with battery electric vehicles (BEV) accounting for 13 % of the EU's car production and plug-in hybrids (PHEV) for 6 % in 2024, and a trade surplus in electric cars (EC 2025). Europe produced more EVs than North America and all of Asia (excluding China) combined (Table 4) and roughly 75 % of European EV production occurred in Germany, making it the world's second-largest EV producer after China.[22] The EU faced a strong surge in imports and imported EVs mostly from China (EUR 9.9 bn or 38 % of EV imports), South Korea (EUR 3.1 bn), Japan (EUR 2.8 bn), and the US (EUR 2.7 bn) in 2024 (Eurostat data).

Following an investigation concluding Chinese EV production is unfairly subsidized, the EU introduced countervailing duties on Chinese EV imports in 2024 which increased significantly since 2020 (see also section 2.2.2). There are, however, no dumping signs, as Chinese brand EVs are offered in Europe at significantly higher prices than in China (Sebastian et al., 2024). While BEV imports from China to the EU declined until November 2025 by 18 % year-on-year, imports of PHEVs from China and BEVs from Korea and Japan increased significantly (Eurostat data). Nevertheless, the countervailing duties plus various benefits to local production for local sales (high transport costs for bulky components, just-in-time and just-in-sequence requirements, existing ecosystem strength) make domestic EV production likely to continue roughly tracking European sales. However, outcomes depend on European production competitiveness and EU policymaker reactions to changing circumstances.

The EV battery cell production and import situation is more concerning. Europe's current production only meets 43 % of its demand (Table 5), with the remainder mainly met by Chinese imports. The largest operating European EV battery cell plants as of November 2025 are LG Energy Solutions in Poland (86 GWh capacity), Samsung SDI in Hungary (40 GWh), SK On in Hungary (20 GWh), and CATL in Germany (14 GWh). South Korean firms entered earliest (LG and Samsung in 2017; SK On in 2020). CATL began German production in December 2022, currently constructing larger plants in Hungary (100 GWh) and Spain (50 GWh). CATL's first Hungarian unit (40 GWh) will start production in early 2026, with Spanish production starting in two years. EVE Energy is constructing a Hungarian plant (28 GWh) starting production in 2026. Other Chinese firms are building factories with production starting in two years or later: Sunwoda in Hungary, Gotion High-Tech in Slovakia, CALB in Portugal. European battery start-ups like Northvolt (now owned by US start-up Lyten), PowerCo, ACC, and Verkor announced big plants, but many were cancelled or downscaled, with actual production still very low.[23]

South Korean and Chinese battery cell firms continue to open European factories because supplying local EV assembly with local battery cells offers several benefits, mirroring long-known automotive component production geography tendencies (see Wuttke, 2023). Nevertheless, current European cell demand is met to a significant extent by Chinese imports. One major reason is that operational European factories all focus on nickel-based battery chemistries, for which demand has declined compared to lithium-iron-phosphate (LFP) batteries. LFP battery cells must currently be imported from China as Europe lacks production capacity (Interview 4). Once CATL Spain goes online, this will change. Various Chinese battery firm investments indicate that rather than supplying from China, a business case exists for producing cells in Europe for European EV

[22]Source: German EV production data based on EV plant locations provided by ACEA and author's own web research on plant production data.
[23]Source: Analysis by VDI-VDE IT, available at https://www.ipcei-batteries.eu/accompanying-research/market-updates.

assembly. This is why the European Automobile Manufacturers' Association is not seriously concerned (Interview 6). However, they emphasised that European carmakers will ultimately source from China if all factors considered (price, supply speed, etc.) make imported Chinese cells preferable to locally produced cells. Current European factory construction all based on off-take agreements with carmakers for European EV production (Interview 4), indicates, however, that this is not the case.

European EV producers like Volkswagen, BMW, and Renault are globally competitive in EV production (though no longer global leaders as in internal combustion engine (ICE) based vehicles), European battery firms are not. Their global market share is less than 1 %. EV battery cell production is globally dominated by Chinese firms (CATL, 37.9 %; BYD, 17.2 %; CALB, 4.4 %; Gotion, 3.2 %; EVE, 2.3 %; Sunwoda, 2.1 %), South Korean firms (LG ES, 10.8 %; SK On, 4.4 %; Samsung SDI, 3.3 %) and one Japanese firm (Panasonic, 3.9 %).[24] Supply of battery cell component, including cathode and anode active material, electrolytes and separators is also dominated by Chinese and South Korean firms, alongside a small number of Japanese firms. Chinese firms are global leaders in the production of EVs, battery cells, and battery cell components/materials. This position is facilitated by state subsidies (see below), but equally importantly by China's extensive LIB production and innovation ecosystem that had emerged over three decades. The Chinese EV battery firms are 10 to 20 years ahead on the learning curve compared to European start-ups PowerCo, ACC, and Verkor (Whitfield & Wuttke, 2025a).

Policy landscape and key drivers in China

China's dominance in global EV and EV battery cell production largely stems from its large domestic EV market, which is heavily driven by consumer purchase subsidies for 'new energy vehicles' (NEVs)[25] since 2009. This market and associated economies of scale created opportunities for Chinese EV and battery firms to learn, build capabilities, and eventually leapfrog into global technological leadership (H. Gong & Hansen, 2023; Whitfield & Wuttke, 2025a). Chinese NEV sector policies deliberately pursued technological leadership in ICE alternatives to circumvent the incumbency advantages of existing European, American, Japanese, and Korean players and to dominate the global automotive industry post-ICE (Whitfield & Wuttke, 2025b). EV R&D has benefitted from public funds since 2001. The Guiding Catalogue for Industrial Structure Adjustment, which is a key industrial policy document issued by China's National Development and Reform Commission (NDRC) (updated roughly every five years) that guides public investment direction at the central and regional/local levels, as well as policies on public finance, taxation, credit, land, and trade, has, since 2005, made consistent reference to NEVs. Made in China 2025 named NEVs as one of ten key sectors (EC, 2024a).

One of the most important policies was related to purchase subsidies. In 2009, the central government introduced subsidies for public-sector NEV purchases in selected cities, requiring certain cities and organisations to achieve specific NEV percentages in vehicle fleets. Purchase subsidies extended to private consumers in six cities in 2010, covering 40 to 60 % of EV costs. In 2013, consumer purchase subsidies expanded to 88 cities. Only after this expansion did EV sales grow considerably (Whitfield and Wuttke, 2025a). Purchase subsidies decreased gradually, fully ending in 2022. However, the purchase tax exemption for NEVs remains in effect today. The purchase tax (normally 10 %) was fully exempt for NEV purchases until December 2025, and will be reduced to 5 % in 2026–2027 (State Council of the People's Republic of China, 2023). Direct

[24] Global market share data from SNE Research.
[25] What the NEV definition covers has changed over time. Initially, it included BEVs, PHEVs, fuel-cell EVs and HEVs. HEVs were later excluded from the definition (see Whitfield & Wuttke, 2025a).

purchase incentives were supplemented by indirect support — quicker registration and access to scarce number plates in big cities for NEVs.

Having NEVs as a Guiding Catalogue priority sector since 2005 has also meant that EV and battery producers benefited from supply-side benefits such as cheap credit and land access, tax exemptions, and various other difficult-to-quantify benefits (EC, 2024a). An indirect supply-side subsidy was the dual-credit policy introduced in 2017. Chinese carmakers must meet fuel efficiency targets and NEV-share fleet targets. Overperforming carmakers accumulate credits sellable to underperformers, providing important income for Chinese EV producers (EC, 2024a, p. 699).

State ownership has occasionally played a supportive role. Generally, the most successful firms in the sector, such as BYD, CATL, and Geely are private companies. However, successful EV firms like Chery, Brilliance, GAC, SAIC, and JAC are state-owned, as is the battery firm CALB. For NIO and Brilliance, capital injections based on new or extended state ownership helped during financial troubles (EC, 2024a, p. 695, 698).

EV battery firms benefited from government purchase subsidies for NEVs, expanding EV demand and, in turn, battery demand beyond what would have occurred without incentives. As components of the NEV sector, batteries also benefited from various supply-side support measures. The most important policy, the 'Automotive Power Battery Industry Specification' (Battery Whitelist), however, was not discussed in the EC's assessment (EC, 2024a). This policy stipulated that between 2016 and 2019, only NEVs equipped with batteries manufactured by Chinese firms were eligible for purchase subsidies (Gong & Hansen, 2023, p. 11). This 'was classic "infant industry protection […] and it provided secure and protected market demand and thus scale economies required for Chinese battery firms to engage in learning-by-doing' (Whitfield & Wuttke, 2025a). Similar local content requirements (LCRs) were also in place for electric motors and inverters. LCRs were, however, always temporary to avoid local firm complacency.[26]

Outlook and implications for the EU

A complicated picture emerges. Chinese EV and battery firm capabilities are very real. These are not over-subsidised inefficient companies but highly competitive, cutting-edge technology firms that benefited from systematic state support over a decade while undertaking significant efforts to build knowledge and capabilities, innovate, and leverage China's extensive three-decade LIB production and innovation ecosystem. Attributing the competitiveness of Chinese EV and battery firms to subsidies alone would be wrong. Nor are there indications of Chinese firms flooding the European market with EVs and batteries at dumping prices. While Chinese EV firms made inroads into the European market, these EVs are sold at much higher prices than in China without capturing very a large market share. In EV batteries, EU imports from China are high, but production capacity is being built rapidly, driven by a business push for local EV battery production for local assembly. This regional production for regional sales is a dominant tendency in the global automotive industry (see Wuttke, 2023).

Nevertheless, significantly low utilisation rates in battery cell production currently exist. This is a global phenomenon not confined to China (Table 5), but since the industry is dominated by Chinese production and firms, it is inextricably linked to China. Lately, Chinese policymakers themselves identified EV battery overcapacity as a major problem, devising measures like capacity monitoring and to 'guide enterprises to scientifically plan production capacity' (Sina Tech Digital, 2025).

A major European concern with Chinese overcapacity is the risk of hollowing out the local automotive manufacturing chain as ICE component jobs disappear without replacement by battery

[26] On the incompatibility of LCRs with international law see section 3.2.

production jobs if battery cells were imported from China instead of making them in Europe. While LIB imports from China are currently substantial, this seems mainly related to current LFP cell production absence in Europe. This will be mitigated once the announced LFP production capacity in Europe is operational.[27] However, new facility production output should be closely monitored. If local factories are not ramped up as announced, policy countermeasures could be considered.

Another concern not directly related to overcapacity is that European battery firms are currently uncompetitive, having just begun their learning curve. Chinese battery firms offer higher quality and lower prices based on decades of accumulated knowledge and capabilities, and economies of scale from large market shares. This makes them preferable suppliers for carmakers, while European cell producers Northvolt, PowerCo, ACC, and Verkor have either failed (in the case of Northvolt), down-scaled ambitions, or are only kept afloat by customers willing to pay higher prices throughout initial scaling and learning phases. Significant overcapacity worsens their position as Chinese producers can increase production and further lower prices if European firms become serious competitors. Carmakers, conversely, are not worried about battery overcapacity and low prices, as it lowers crucial input costs (Interview 6). They might run out of patience to keep local cell producers afloat by accepting higher prices (Interview 5). However, overcapacity is not the major issue here. The issue is that European firms are latecomers in EV battery technology and vehicle connectivity systems, requiring significant industrial policy support if they are ever to become competitive with the current Chinese industry leaders. The policy challenge is not one of addressing overcapacity but of supporting firms in technological catch-up (for required policies, see e.g. Whitfield & Wuttke, 2025b).

2.4 Case studies: Potential overcapacities

In two further case studies, the hydrogen economy and robotics are assessed as prospective key technologies for sustainable and automated industrial production, where Chinese capacity expansion despite significant uncertainty over future demand could undermine the technological capabilities of EU companies.

2.4.1 Case study: Hydrogen economy

Overcapacity indicators in China's electrolyser industry

The hydrogen energy industry is a crucial pillar in the global green transition. Renewable hydrogen[28] — produced via water electrolysis using renewable electricity — offers a pathway to decarbonise hard-to-abate sectors such as heavy industry, long-haul transport, and energy storage. This case study focuses on the emerging renewable hydrogen sector and its equipment manufacturing base. Electrolysers and fuel cells form the technological backbone of this transition, with alkaline water

[27] Both demand forecasts as well as production capacity forecasts for EV battery cells are highly uncertain. The capacity investment announcements by established players like CATL, EVE, Samsung SDI, SK On, LG, Gotion High-Tech, and CALB, which are less uncertain than those by unproven start-ups, in Europe amount to roughly 380 GWh. All this capacity should be operational by the end of the decade, on top of the existing capacity of 173 GWh (see Table 5). If the announced (and not yet cancelled) capacity of the largest European startups PowerCo, ACC, and Verkor (170 GWh) as well as that of foreign less experienced firms AESC and ProLogium (163 GWh) is added with a probability value of 50 %, Europe will have a battery cell production capacity of 720 GWh by 2030, which exceeds the forecasted EV battery cell demand in Europe of 590 GWh (demand forecast based on IEA Global EV Outlook 2025, p. 136).

[28] The use of hydrogen terminology varies across reports. This study adopts a consistent wording: 'Low-emission hydrogen' refers to all production routes that significantly reduce CO_2 emissions, including hydrogen produced from fossil fuels with carbon-capture, utilisation and storage. 'Electrolytic hydrogen', widely used in the Chinese context, refers to hydrogen produced via water electrolysis using electricity from any source. 'Renewable hydrogen' is defined strictly as hydrogen produced through the electrolysis of water using 100 % renewable electricity, resulting in near net-zero emissions.

electrolysers (AWE) and proton exchange membrane electrolysers (PEM)[29] representing the most common and commercially mature production technologies, while proton exchange membrane fuel cells dominate transport applications, notably in fuel cell electric vehicles (PEM FCEVs).

Global demand of all types of hydrogen reached around 100 million tonnes in 2024, up 2 % from 2023. Global low-emission hydrogen production — which includes fossil-based hydrogen production coupled with carbon capture technologies as well as renewable hydrogen produced by water electrolysis — is only one million tonnes in 2024, still a meagre 1 % share of total output (Deasy-Millar et al., 2025, p. 7; IEA, 2025e). In the same year, 36 300 tonnes of low-emission hydrogen were used in Europe (EU27, The European Free Trade Association (EFTA) and the UK), corresponding to approximately 3.6 % of global low-emission hydrogen production (European Hydrogen Observatory, 2025b, p. 47).

Nevertheless, the low-emission hydrogen industry's market potential is considerable, with announced investments reaching EUR 607 billion through to 2030. Of those, projects in construction or past final investment decision make up EUR 98 billion (Hydrogen Council & McKinsey & Company, 2025, p. 8). China has made the largest investment commitments mainly for domestic demand, followed by the US and the EU. EU demand for renewable hydrogen implied by demand policies in 2030 is expected to account for two-thirds of global demand, to be met through a combination of domestic production and imports (Hydrogen Council & McKinsey & Company, 2025; European Commission, Joint Research Center, 2024). However, under current market and policy implementation conditions, this projected demand share appears overly optimistic. Total hydrogen demand in the EU declined slightly in the latest reported years (2023 and 2024) (European Hydrogen Observatory, 2025a), the global share of its low-emission hydrogen demand remained in the low single digits — as mentioned above —, and reported delays in renewable hydrogen policy enforcement on the national level, infrastructure deployment and electrolyser project realisation indicate that application continues to lag behind policy ambition (ACER, 2024; Burgess, 2025).

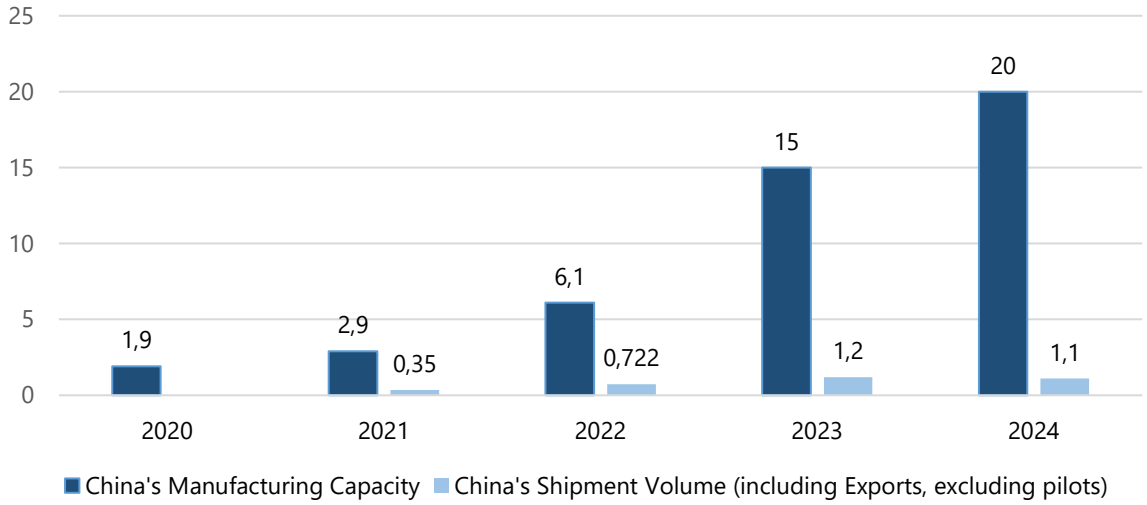
Despite growing investments, market uptake remains limited in all regions. A key factor is that, in 2023, renewable hydrogen production costs were approximately 1.5 to six times higher than those of fossil-based hydrogen (IEA, 2024b), resulting in depressed demand and creating the need for long-term off-take agreements (IEA, 2024b; Odenweller & Ueckerdt, 2025). While scaling is expected to reduce costs, the formation of an autonomous low-emission hydrogen market ultimately depends on stable deployment.

Regarding total hydrogen production, China remained the world's largest producer and consumer in 2024 with output reaching 36.5 million tonnes and annual growth of around 3 % (NEA & Guoneng Hydrogen Innovation, 2025, p. 8). In the same year, total hydrogen production capacity exceeded actual output by more than 14.5 million tonnes. This gap is particularly visible in hydrogen produced

[29] AWE technologies have the longest industrial track record and, like PEM technologies, operate at relatively low temperatures to split water into hydrogen and oxygen. Next to industrial maturity, AWE systems benefit from comparatively lower capital costs and longer lifetime due to the use of non-noble metal catalysts. In comparison, PEM technologies rely on platinum group metals, which are more capital intensive and generally show less durability due to faster material degradation. The main strengths of PEM technologies are their higher current density—allowing more hydrogen to be produced within a smaller installation—and their ability to adjust hydrogen output quickly when electricity supply changes. PEM's system design compactness and its flexibility in response to electricity fluctuations make them particularly suitable in the European context, where available space may be limited and electricity generation from wind and solar power varies over time. In contrast, AWE systems are better suited to cases, where land is readily available and installations operate with stable, continuous electricity loads. Other electrolysis technologies, including Anion Exchange Membrane (AEM), and Solid Oxide Electrolysis are currently emerging, whereas Proton Conducting Ceramic (PCC) electrolysis technologies are still in an early development stage (European Commission, Joint Research Center, 2024, p. 13f.).

via water electrolysis, where utilisation reached around 64.5 % based on an estimated annual production capacity of 500 000 tonnes and actual output of approximately 320 000 tonnes, accounting for about a third of global production. While output experienced a modest 3.6 % annual increase compared to 2023, capacity was up 9.7 % (NEA & Guoneng Hydrogen Innovation, 2025, p. 8f), in line with a global demand increase of 10 % in 2024, according to the IEA's Global Hydrogen Review 2025.

Figure 5: Chinese electrolyser manufacturing capacity and shipments in GW



Source: IEA Data (IEA, 2025b, 2025e, p. 9) and GGII Data (GGII High Tech Industry Institute, 2024, 2025a, 2025b; Pingan Securities Group, 2023).

In electrolyser manufacturing, the anticipated demand growth has led to massive capacity build-up as confirmed by comparing annual IEA Data on China's annual electrolyser manufacturing capacity with annual shipment data from Gaogong Industry Research Institute (GGII), a leading private industry research institute in China (Figure 5): In 2021, China had the capacity to produce 8 times more electrolysers than it delivered globally. In 2024, this ratio even reached 18 times the annual capacity of actual global shipments, securing China's dominant share of almost 60 % of global electrolyser manufacturing capacity (IEA, 2025e, p. 9).

In the case of China these imbalances may be temporary, as domestic demand is expected to increase in line with policy-driven hydrogen scale-up. China's Hydrogen Alliance's projection of 102 GW of electrolyser deployment in China by 2030, shows substantial growth expectations that could absorb today's oversupply (China Energy Media Research Institute, 2023). Current capacity-shipment gaps should therefore be interpreted as potential overcapacity risks in a rapidly scaling yet still policy-driven global market, where political strategies incentivise early supply expansion, with cost reductions expected to follow from learning and scale effects. Within this context, current severe electrolyser manufacturing overcapacity is a global phenomenon, including in Europe, not specific to China (European Commission, Joint Research Center, 2024, p. 34; Martin, 2024).

The pace of price erosion in China constitutes an additional indicator of overcapacity, which has been driven by high competition among electrolyser manufacturers. Between 2022 and 2025 prices for alkaline and PEM electrolysers in China decreased by respectively 53 % and 50 % (EnerScen, 2023; Sinolink Securities, 2025, p. 1). Falling equipment costs are pushing green-hydrogen projects in China closer to parity with fossil-based production.

On average, Chinese production and installation costs of electrolysers are 40 to 60 % cheaper than system costs outside China, including in Europe. However, overseas projects face additional costs related to shipping, tariffs, certification and standards compliance, and local installation, which

narrow the effective cost gap with European and US suppliers (IEA, 2025e, p. 106). In parallel, rapid price erosion in China's domestic and export markets has prompted leading electrolyser equipment manufacturers to adopt voluntary self-discipline measures. In 2025, they launched the Initiative for the Healthy Development of China's Electrolyser Industry, committing not to bid below production costs, including in export markets where buyers increasingly expect prices aligned with those prevailing in China (Hydrogen Energy Branch of the China Industrial Development Promotion Association, 2025).

Overcapacity indicators in China's fuel cell industry

China's industrial policies in the hydrogen sector over the past two decades have focused on transportation, positioning FCEVs as a complement to battery electric vehicles (BEVs). The global market for FCEVs remains small, with just 87 500 vehicles in operation worldwide by the end of 2024 (IEA, 2025c), geographically concentrated in Asia. Within this limited market, China has emerged as the key deployment country for commercial FCEV, particularly buses and light- to heavy-duty trucks. According to data from the China Association of Automobile Manufacturers, domestic FCEV sales reached 5 805 vehicles in 2023, with year-on-year market growth rates of 5.8 %, 112 %, and 72.4 % between 2021 and 2023 (CIECC Hydrogen Energy Center & EnerSen, 2024, p. 24f).

In contrast to Japan's leadership in technical fuel cell patent value (Li et al., 2023), and South Korea's dominance in passenger FCEV deployment (36 % of the global fleet in 2024) (IEA, 2025a, p. 10), China's industrial strength lies in manufacturing scale and cost reduction. The cost of vehicle fuel cell systems in China fell from 20 000 CNY/kW in 2018 to around 1 500 CNY/kW in 2024, with further reductions projected to reach 500 CNY/kW by 2030 (China Orange Group, 2025, p. 76). Nevertheless, the total cost of ownership for a fuel cell truck remains 15 % higher than that of a comparable battery electric truck (China Orange Group, 2025, p. 154ff).

Against this backdrop, Chinese fuel cell production capacity has grown much faster than its market (EU-China Energy Cooperation Platform, 2023, p. 35). In contrast to the renewable hydrogen production sector, Chinese government and industry sources do not publish systematic data on accumulated production capacity for fuel cell stacks or complete fuel cell systems. However, initial public offering (IPO) disclosures provide partial insight into downstream manufacturing conditions. In particular, the IPO prospectus of REFIRE and an industry report by Everbright Securities based on the IPO of Guohong Hydrogen, both among China's largest domestic fuel cell stack and system integrators, report CUR across key segments of the fuel cell value chain (Table 6).

These data show CUR have been frequently low, with only a few products and individual years exhibiting high utilisation rates. More upstream components such as membrane electrode assemblies (MEAs) and graphite bipolar plates exhibit higher and more stable utilisation. MEAs are the core and most technologically demanding component of fuel cells (China Orange Group, 2025, p. 84f), suggesting relatively high market-entry barriers. Like bipolar plates, MEAs also have a broader range of applications beyond vehicles (Everbright Securities, 2022, p. 6).

Taken together with the persistent net losses reported by both leading Chinese firms, this pattern indicates that manufacturing capacity for fuel cell systems and stacks has grown faster than market demand, pointing to emerging downstream overcapacity rather than upstream supply constraints. REFIRE acknowledges that demand remains volatile and policy-dependent, and explicitly states that future market demand is uncertain (Shanghai REFIRE Group Limited, 2024). Initial estimates indicate a global decline in the FCEV market in 2025 (Martin, 2025).

In contrast to China's electrolyser industry, excess supply in China's fuel cell sector presently benefits European producers that supply key components on which China's FCEV manufacturing value chain remains import-dependent (see the following Technological Landscape section).

Table 6: Firm-level CUR and net losses among leading Chinese fuel cell manufacturers

Company	Year	CUR: fuel cell systems	CUR: fuel cell stacks	CUR: graphite bipolar plates	CUR: membrane electrode assemblies	Net profit/loss (in EUR million)
REFIRE	2021	29.6 %	27.2 %	0 %	89.7 %	-85.8
	2022	26.2 %	26.5 %	0 %	49.4 %	-77.1
	2023	43.8 %	35.1 %	92.5 %	59.1 %	-75.4
Guohong Hydrogen	2019	–	21.1 %	21.1 %	–	+29.8
	2020	29.6 %	27.2 %	43.6 %	–	-246.5
	2021	61.1 %	78.8 %	66.7 %	–	-188.7

Sources: (Everbright Securities, 2022; Shanghai REFIRE Group Limited, 2024)

Policy learning in China's hydrogen sector

Political goals and measures to develop a renewable hydrogen industry

Embedded in China's energy system, the development of fuel cell and electrolyser industries reflect the role of the national energy system as a strategic channel for both green transformation and economic growth. Since the 1980s, China's energy production and consumption have been shaped by the overarching objectives of national energy security and economic competitiveness, later reinforced by emission-reduction pressures (C. Ma, 2023; World Bank, 2007). Political support for industries in the energy sector has been strategically calibrated along these three thresholds, visible in the reduced support for FCEVs in favour of BEVs, whose development promised faster and more substantial economic advantages for China. Furthermore, China's first National New and Renewable Energy Development Plan (1996–2010) (Shi, 1996), published in 1996, consolidated these three objectives and introduced policy instruments that remain central to China's industrial policy approach today: comprehensive planning, support for R&D and subsidies for pilot projects to reduce initial costs, the localisation, clustering and scaling of supply chains, as well as demand side policies (Banares-Sanchez et al., 2024; Hove, 2024; Shi, 2009).

This political framework raises two major concerns identified in interviews. First, China's pursuit of economic self-reliance conflicts with the mutual interdependencies inherent in a global market system (Interview 9). Second, this tension results in market distortions by systematically disadvantaging foreign companies in the Chinese market (Interview 8). Against the backdrop of European concerns about industrial overcapacity, China's central government has responded by strengthening its control over production capacity and market order as part of its industrial policy approach to the hydrogen sector. The following section discusses these issues in turn.

Self-reliance in the long-run, modest support for FDI in the short-run

Substantial political support for China's hydrogen industry first materialised in 2001, when FCEVs were included in a major government-funded R&D programme for NEVs (Chinese Academy of Sciences, 2005; China NEV, 2011). However, subsequent prioritisation of the more technologically mature BEVs side-lined hydrogen at the central political level (China Buses, 2021; Dugoua & Dumas, 2024).

Central policy attention returned in the mid-2010s amid concerns over technological dependence following advances in Japan's FCEV sector (Wei, 2025), showcasing China's strong government

commitment to technological self-reliance. In 2016, the Action Plan of Energy Technological Revolution and Innovation (2016–2030) identified the full hydrogen supply chain as a 'key task' (NDRC & NEA, 2016), an objective that was later specified at the local level as achieving 'self-reliance and industrialisation in key technologies across the full chain' (Shanghai DRC, Shanghai STC, Shanghai EIC, & Shanghai PNR, 2022).

Successively, national policies shifted the development focus to renewable hydrogen production and related equipment manufacturing (NDRC, 2023) in regions with low electricity prices (NDRC, NEA, & MOF, 2021). China's massive expansion of renewable electricity has given it a cost advantage across energy-intensive manufacturing (International Renewable Energy Agency, 2025), particularly in the operation of electrolysers for hydrogen production, an advantage that European firms currently struggle to match (Interview 8).

While Chinese policy goals of concrete breakthroughs in PEM electrolysers and solid oxide electrolysers (SOEC) were only included in the 14th Five-Year Plan of Energy Technology Innovation in 2021 (NEA & MOST, 2021), preferential treatment for foreign direct investment (FDI) in local production and joint ventures remains an interim tool to mitigate import dependence on key components (NDRC, 2022a) (see section 2.4.2). At the same time, China's procurement system applies explicit preferences for locally produced equipment and domestic intellectual property (General Office of the State Council, 2025), indicating that foreign firms and joint ventures face structural disadvantages in accessing publicly supported demand, a concern also raised by European companies active in joint ventures in China (Interview 8). European countermeasures are already in place: The EU Hydrogen Bank funding rules include a cap of 25 % in electrolyser stack sourcing from China (EC, Directorate General Climate Action, 2024). While intended to limit strategic dependencies, the effectiveness of this cap will depend on monitoring and enforcement, particularly given the potential for circumvention through indirect sourcing arrangements or third-country intermediaries.

National policy learning: Financial support and central control of capacity

In contrast to the central level's single and conservative renewable hydrogen production target of 100 000–200 000 tonnes per year by 2025 (NDRC, 2022), local governments have demonstrated earlier and more sustained long-term support. At the end of 2025, local governments had issued more than 600 hydrogen-related policy documents (China Hydrogen Alliance, 2025) and frequently adopted earlier timelines, longer planning horizons, and more ambitious targets (Fremery et al., 2025).

The cautious approach of central political support in the hydrogen sector suggests policy learning from earlier BEV price wars and solar PV overcapacity. According to a recent study by Zeng et al. (2025), central planning has shifted toward short-term, controllable instruments for incremental supply-side co-financing of R&D projects. Local governments increasingly bear the costs of initiation and the risks of implementation, while the central government strengthens control over capacity formation. For instance, national funding through the major FCEV Demonstration City Clusters (MOF, MIIT, MOST, NDRC, & NEA, 2020) requires local governments to pre-finance projects under capped, *ex post* performance-based national re-funding. This constrains their ability to continue supporting economically non-viable firms, which often rely on aggressive price undercutting to maintain market share, thereby increasing the likelihood of price-based competition and price wars. Alongside local governments, state-owned enterprises (SOEs) have been crucial in early market formation by acting as experimental actors and risk bearers in capital-intensive R&D segments (Fremery et al., 2025, p. 13–14; Gong et al., 2023; Gunter et al., 2025; Meng et al., 2022). While SOE-led projects still drive electrolyser demand, manufacturing is largely carried out by private firms, indicating that SOEs play a foundational role in market creation rather than

manufacturing dominance (China Power, 2023; GGII High Tech Industry Institute, 2025b). This role was further strengthened by China's 2024 Energy Law, which defines hydrogen as an energy source (37 Presidential Order 2024, Art. 2). The hydrogen industry is thereby integrated into the SOE-dominated energy sector, granting access to planning mechanisms, preferential electricity prices, and long-term purchasing agreements (Collins, 2024; EC, 2024a; NEA & Guoneng Hydrogen Innovation, 2025).

The upcoming 15th Five-Year Plan for Social and Economic Development is expected to continue this cautious industrial policy approach. First drafts introduce 'hydrogen energy' as a 'future industry' — earmarking a strategic industry with uncertain outcome and therefore more cautious political support (MIIT, MOE, MOST, & MOT, 2024) — and a source for future economic growth (Central Committee of the Communist Party of China, 2025). From a European perspective, this shift toward tighter central control and more cautious financial support has ambiguous effects. On the one hand, reduced tolerance for overcapacity and stricter budget discipline may limit subsidy-driven price wars, potentially easing competitive pressures on European hydrogen technology providers. On the other hand, the deeper integration of hydrogen into the SOE-dominated energy system may entrench structural barriers to market access for foreign firms.

Technological landscape and global positioning

China's globally dominant electrolyser manufacturing and deployment base remains heavily concentrated in AWE, representing over 90 % (IEA, 2025e, p. 104) of existing electrolyser production capacity and almost 70 % of operational water electrolysis domestically (IEA, 2025f). In comparison, in the EU PEM electrolysers account for approximately 65 % of installed electrolysis capacity (IEA, 2025f). In addition to their compact design, which facilitates deployment given Europe's limited availability of suitable land, PEM electrolysers' fast dynamic response is an advantage for renewable hydrogen production from volatile electricity sources such as wind and solar power. In contrast, besides higher capital costs for PEM technologies globally, constraints on domestic innovation in core components limit their deployment in China.

China's AWE supply chain is highly localised (around 95 %) and broadly aligned with international benchmarks. In contrast, despite reported localisation rates of around 80 % for PEM electrolysers, China continues to rely on imports of critical high-value components, including advanced membranes, precious-metal catalysts, and specialised corrosion-resistant materials, whose leading suppliers and intellectual property are headquartered in the US, Japan and Europe (Germany, UK, Belgium) (Hwabao Securities, 2023; Li et al., 2025, p. 12; U.S. Department of Energy, 2022, Appendix B).

Similarly, to produce PEM FCEV, China's 2025 Fuel-Cell Industry White Book (China Orange Group, 2025) finds that domestic production of PEM fuel cell stacks relies heavily on imported key stack-critical material, notably proton-exchange membranes supplied mainly by firms from the US and Japan, precious-metal catalysts dominated by suppliers from Japan and the EU, and gas-diffusion media/carbon paper sourced largely from Japan, Germany, and the US.

Outlook and EU implications

The global low-emission hydrogen market remains policy-driven and pre-commercial. Government targets and industrial policies aim to shape both supply and demand. However, manufacturers worldwide have scaled capacity in anticipation of demand that has yet to materialise. Thus, the capacity mismatch in China leading to declining prices for electrolysers is a potential risk for European electrolyser companies to supply their own EU market and projects outside the EU that are intended to export to Europe.

Against this backdrop, the EU has established strong support frameworks through its hydrogen strategy and Hydrogen Bank funding. This political support has contributed to Europe retaining its technological leadership in PEM and SOEC, as well as in advanced capabilities in anion-exchange membrane (AEM) electrolyser development, which combine the flexibility of PEM with lower material costs. However, European manufacturers producing domestically are currently not competitive on price in global markets.

While export markets for electrolysers remain small, according to IEA estimates, China could account for 95 % of global electrolyser exports by 2030 (IEA, 2024a, p. 247). At the same time, China remains dependent on imported core components for PEM electrolysers and PEM fuel cells, including from European firms and joint ventures such as John Cockerill, creating mutual interdependencies. By prioritising renewable hydrogen and using its potentially large market demand as leverage, the EU can foster global incentives for deploying PEM technologies that, even if manufactured and exported by China, include technology inputs from Europe. China's emphasis on national energy security is likely to reinforce domestic R&D efforts to reduce these dependencies.

A representative of a leading European hydrogen industry alliance argued for 'ring-fencing' European technologies, expanding 'Made in Europe' provisions, and ensuring implementation of set targets to secure demand (Interview 5). In parallel, a representative of the European Trade Union Institute cautioned that relations with China in emerging industries are highly complex and require context-sensitive assessment by political decision-makers that also take into account the social costs of a delayed green transformation (Interview 7). Drawing on these perspectives, a selective and time-limited EU approach could be warranted: protecting technology-leading segments (PEM, SOEC, AEM) to allow further development and commercialisation, while remaining open to imports of cost-competitive alkaline electrolysers to reduce hydrogen production costs and stimulate demand. Conceivable options include time-limited bilateral agreements with reciprocal off-take guarantees for electrolysers to support demand, focused on each side's respective strengths. Given Europe's geographic conditions and renewable integration needs, advanced electrolyser technologies, such as PEM and AEM, are likely to remain central to the development of its hydrogen industry and its export potential.

2.4.2 Case study: Robotics

Over the past decade, China has emerged as the world's largest producer and consumer of robotics, exhibiting historically unprecedented growth rates (International Federation of Robotics, 2020, p. 172). Since 2017, this expansion has been accompanied by concerns over excess capacity and 'price wars' (OECD, 2022a, p. 101; People's Daily, 2017; Yicai, 2025; Zhang, 2024). Yet, China has only recently emerged as a viable domestic and international producer in selected segments, and systematic assessments remain limited. While these debates cover industrial, service, and humanoid robotics,[30] this study focuses on industrial robots due to their strategic importance for European manufacturing and the presence of established European firms that still dominate, particularly in the high-end components segment. Given the high growth prospects of industrial

[30] Under the generally accepted definitions of the ISO 8373, the distinction between industrial robots and service robots is determined by their intended application: industrial robots are automatically controlled, reprogrammable, multipurpose manipulators with three or more fixed axes intended for industrial automation, whereas service robots are intended to perform useful tasks for humans or equipment outside industrial automation. Humanoid robots are generally subsumed under service robots (ISO, 2021).

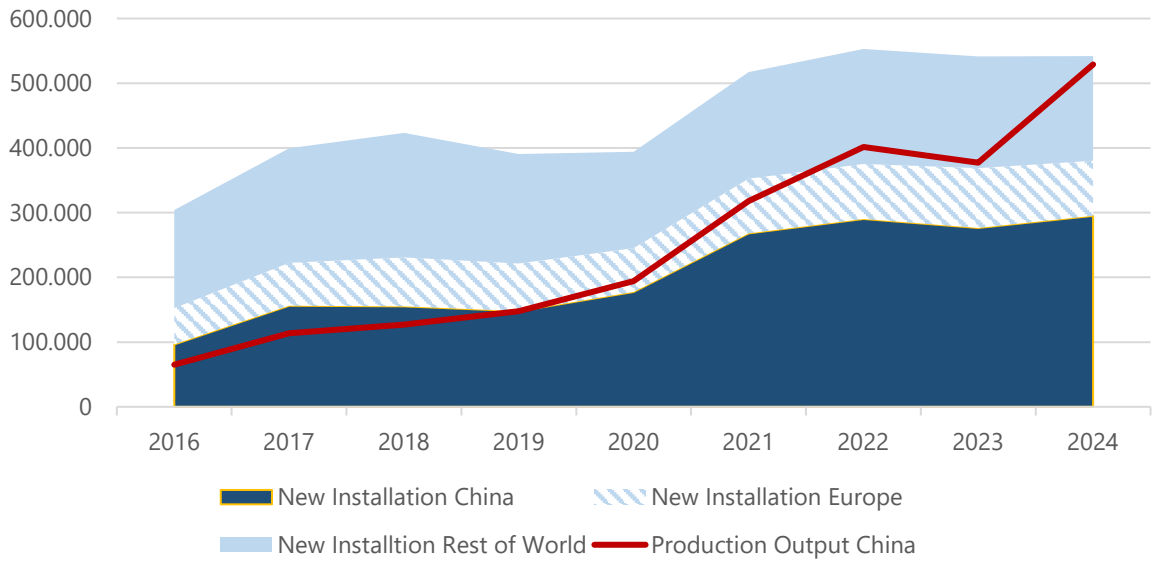
robots, the main question is whether European companies can defend their position in the coming years.[31]

Recent capacity and export expansion in China's industrial robot industry

Comparing China's industrial robot production output and global demand, proxied by annual new installations, reveals recent tendencies toward capacities exceeding demand (see Figure 6). After reaching about half of global demand in 2020, China's output surged sharply in 2024, almost matching total global installations. This production expansion appears to have continued into 2025. According to the China Machinery Industry Federation (CMIF), industrial robot output had already exceeded the full-year production level of the previous year in the third quarter of 2025 (Guo, 2025).

At the same time, a substantial share of China's rapidly expanding production has been absorbed by its domestic market. While China's share of global demand soared from almost 32 % in 2016 to over 54 % in 2024, Europe's declined from almost 19 % in 2016 to nearly 16 % over the same period. The surge in China's domestic demand proved highly beneficial for foreign robot manufacturers, allowing them to expand both exports to China and localised production to capture a large share of the growing market. However, over time, Chinese-owned manufacturers have strengthened their position in meeting domestic demand for industrial robot installations. Their share increased from 29 % of installations in 2019 to 57 % in 2024, marking the first year in which locally owned firms supplied the majority of industrial robots newly installed in China (International Federation of Robotics, 2020, p. 173, 2025a). This growth in market share is rooted in China's strength in low- and mid-tier industrial robots, while despite government support, it struggles to compete in high-end products (Blaugher & Gordon, 2025, p. 8).

Figure 6: Global industrial robot installations by region and China's industrial output (units)



Data source: International Federation of Robotics, 2022, p. 57, 2023, 2024, 2025b; National Bureau of Statistics of China, 2025; European figures for 2022 are derived from regional shares reported in press releases by the International Federation of Robotics (IFR). Data from China's National Bureau of Statistics (NBS) exclude observations for January and February; accordingly, annual figures for China's production output are calculated over the period from March to December.

[31] Service robotics remains an important field, but differs in market structure and technological trajectories, making it less directly comparable for the purposes of this analysis.

Based on annual trade statistics from China's General Administration of Customs, China's rapid expansion in industrial robot production has been accompanied by steadily rising export values, indicating that exports absorb part of the additional output. Table 7 reports annual import and export values for industrial robots (HS Code 8479.50) between China and global markets, with particular attention to Germany as Europe's leading manufacturer in the sector. China's exports increased almost fivefold between 2016 and 2024, reaching EUR 523 million.

Table 7: Chinese export and import values of industrial robots (in EUR million)

Year	Export overall	Import overall	Export to Germany	Import from Germany
2024	523	589	40	33
2023	415	1 147	29	97
2022	382	1 302	22	78
2021	289	1 301	25	91
2020	214	913	1	62
2019	214	880	14	115
2018	223	963	16	101
2017	18	1 174	8	202
2016	140	789	6	94

Data source: GAC, 2025.

At the same time, import value outpaced export value by a wide margin between 2016 and 2023. The imbalance was especially pronounced in 2017, when imports from Germany were 25 times larger than exports. This strong increase in imports reflects the continued reliance of China's industrial robot industry on foreign suppliers and the rising demand for higher-value, technologically advanced robots (Lin, 2018; OECD, 2022, p. 103). As a result, China remained a large net importer of industrial robots globally and, more markedly, in bilateral trade with Germany up to 2023. Germany remains China's third-most-important export partner, behind only the US and Vietnam (UN Comtrade data).

In 2024, amid a surge in local production, trade patterns shifted. China's exports approached parity with global imports and exceeded imports from Germany for the first time in the series. These export and import patterns remained stable in the first eleven months of 2025, with relative differences vis-à-vis both Germany and the wider world remaining largely unchanged. The closing of China's trade deficit shown in the trade data, however, is not indicative of the value capture by foreign-owned firms as producers in China and suppliers of inputs (see below).

Furthermore, in a study on the broader restructuring of global value chains, Musitz (2024a) compares customs-based trade statistics with Trade in Value Added (TiVA) data. The latter captures the value added from intermediate inputs in each country along the value chain used to produce final goods. Assessments of trade between Europe and China based on TiVA data reveal a much more balanced relationship, with Europe contributing substantially more value through intermediate inputs and China contributing correspondingly less than suggested by traditional customs-based trade statistics.

Indicators of emerging overcapacities

Average industrial robot unit prices rose by about 15 % between 2017 and 2020, before falling by roughly 35 % thereafter to CNY 178 800 (≈ EUR 25 254) in 2022 (Huafu Securities, 2023, p. 12). Price declines accelerated further in 2025. Selective compliance assembly robot arm (SCARA) prices nearly halved within one year to around CNY 10 000 (≈EUR 1 239) per unit, while standard six-axis 6 kg models, another core high-volume product category, sold for just over CNY 20 000 (≈EUR

2 472), intensifying margin pressure across the sector (Jiang, 2025). Amid falling prices in 2024, ESTUN Automation, the largest domestic industrial robot original equipment manufacturer (OEM)[32] in terms of shipment volume, reported net losses of almost CNY 836 million (≈ EUR 107 million). Its annual report explicitly attributes this huge profit collapse to advances into high-barrier product markets dominated by international brands, intensified price competition and rising depreciation associated with capacity expansion (ESTUN, 2025, p. 16). Notably, ESTUN's loss was confined to a single year, as the firm was profitable both before and after 2024, whereas two further comparable Chinese OEMs, SIASUN –as a spin-off of the Chinese Academy of Science the only state-affiliated company in the sample –and EFORT continued to make losses in 2025 (EFORT, 2024; Jiang, 2025; SIASUN, 2025).

In contrast to Chinese government statements and policy studies that emphasise overcapacity pressures on low-end manufacturers (Gunter et al., 2025; People's Daily, 2017), the available evidence only partially supports this claim. Drawing on the annual reports of three main Chinese low-end producers, only Topstar, which recently expanded its portfolio into high-end products, reported losses in 2024, but regained profitability by the third quarter of 2025 (Topstar, 2025a, 2025b). Kinco Automation and Zhongda Leader both show continued revenue declines but remain profitable (Kinco Automation (Shanghai), 2025; Zhongda Leader, 2025). Relative to the domestic OEMs focused on middle- to high-end manufacturing analysed above, these firms exhibit tightening profit margins, yet they continue to operate profitably. This confirms that Chinese firms face greater difficulties in reproducing and replacing high-end products dominated by foreign incumbents — a policy objective formulated as 'independent and controllable' technological capabilities (State Council of the PRC, 2021) — than in maintaining their competitiveness and dominance in low-end production.

This interpretation is consistent with the continued strong performance of foreign firms producing locally for the Chinese market. KUKA, which is regarded in China as a foreign-owned company due to its German-held intellectual property, reports in its 2024 annual report that sales revenue in China increased, though this growth was insufficient to offset a decline in global sales (KUKA, 2025, p. 22). In contrast, ABB announced a record year of overall revenue growth and described its revenues from China as stable, albeit suffering from a slight decline (ABB, 2025, p. 10 and 56). In its robot division, FANUC reports only limited weakening in the Chinese market, while demand developments in Europe and the Americas were the main drivers of overall performance amid a global demand slowdown (FANUC CORPORATION, 2024, p. 6).

Taken together, China's massive expansion of production output, which in 2024 almost equated to global demand, has intensified competitive pressure on domestic firms and is likely to drive export growth. At present, Chinese OEMs continue to struggle to compete with established high-end foreign companies in the sector.

Political support for a fast technological follower

Developing a domestic robotics industry has been one of China's core political priorities since the early 2010s (NDRC, 2017). The sector is one of ten key industries targeted by the Made in China 2025 policy in 2015. Political support for industrial automation reflects concerns over population ageing and rising labour costs, which puts pressure on China's price-sensitive global manufacturing competitiveness. At the same time, 'domestic substitution' to reduce significant reliance on foreign robotics technologies and components became a central objective of China's industrial policy (Lin,

[32] OEM (Original Equipment Manufacturer) refers to firms that design and manufacture products under their own brand name, distinguished from firms that primarily distribute or integrate equipment produced by others.

2018). In 2015, five key technological categories — robot bodies, reducers, servo motors, controllers, and sensors — were formally identified as bottlenecks constraining domestic capability development (NDRC, MIIT, & MOF, 2016; State Council, 2015).

By the end of the 2010s, Chinese companies had manufactured complete robot bodies and made advances in the four other key fields identified above (MIIT, NDRC, MOST, & Ministry of Public Security, 2021). Today, China has emerged as a viable competitor in the collaborative robot industry, primarily in domestic markets and increasingly in export markets. However, Chinese competitiveness remains largely price-driven, with continued reliance on foreign suppliers for high-precision and safety-critical components (Gaogong Industry Research Institute, 2023; Marketing Intelligence Resource, 2024). This dependence becomes more pronounced in high-load and high-precision industrial automation applications (Wu et al., 2025). China has gained significant momentum in the robotics industry, yet still follows European, Japanese, and U.S. technological leadership, notwithstanding the Chinese government's goal for domestic firms to become competitive on international markets in 2035 (MIIT et al., 2021).

Public funding has played a central role in reinforcing China's fast-follower trajectory by supporting innovation, capacity expansion, and market formation. On the supply side, firms in sectors prioritised by industrial policy, such as robotics, may qualify as High and New Technology Enterprises if they meet R&D and intellectual property requirements. They benefit from a reduced corporate income tax rate of 15 % (down from the standard 25 %), alongside R&D super-deductions that allow up to 200 % of eligible R&D expenditure to be deducted from taxable income (MOF and STA, 2023). Trade policy complements these measures through export VAT refunds (STA, 2026) and customs duty exemptions for high-precision equipment and key components addressing identified technological bottlenecks (MIIT, MOF, GAC, & STA, 2025).

Next to preferential bank loans — a major corporate finance channel used to enforce industrial policies in China (García-Herrero & Schindowski, 2024, p. 12) — capital supply is supported by the instrument of government-guided funds, an equity-based policy tool to pool state and private capital into strategic emerging industries (Y. Xu, 2024). In this context, a state backed venture capital fund is expected to mobilise CNY 1 trillion (≈ EUR 124 billion) from local governments and private investors over the next 20 years, with high tech sectors and robotics among the main beneficiaries (International Federation of Robotics, 2025c). Under special funds the government financially supports strategic supply-side capacity building (Qiushi, 2024). Within this national policy framework, the Nanshan District Government in Shenzhen offers firms financial support for the industrialisation of R&D, capped at CNY 10 million (≈ EUR 1.24 million) (People's Government of Nanshan District, Shenzhen, 2024). In addition, local governments such as Suzhou target domestic capacity-building and expansion of local components to fill supply chain gaps. Leading enterprises cooperating with upstream and downstream industrial-chain actors and research institutions that contribute to this goal may receive subsidies of up to 2 million CNY (≈EUR 247 000 EUR) (Several Measures of Suzhou City for Supporting the Innovative Development of the Embodied Intelligent Robotics Industry, Art. 2 and 4.)

Complementary demand-side measures reduce adoption costs and incentivise the formation of lead markets for domestic technology substitution. Between 2014 and 2018, local governments provided purchasing subsidies for industrial automation equipment, typically covering 10–20 % of the purchase price (China Business Industry Research Institute, 2019, p. 8-9). A study by Ma Runhong (2024) finds that demand-side measures alone drove a boost in innovation, increasing industrial robot patent applications by 13 % and the number of robot producers by almost 30 %, predominantly benefitting big enterprises (Ma, 2024, p. 23f. and 70f.). Demand-side policies remain in place today but have shifted toward subsidising projects that expand first-of-a-kind application

models and broaden their fields of use beyond manufacturing (e.g. General Office of the Beijing Municipal People's Government, 2023).

Outlook and EU implications

The Unified Vision of EU Robotics (eu ROBOTICS, 2024) emphasises that robotics plays a key role as an enabling technology for advanced manufacturing, thus underpinning its strategic relevance. In the case of industrial robotics, the EU faces the challenge of balancing increasing import potential from China on the one hand, and maintaining China as a key export destination on the other. China is the world's largest market for industrial robots in terms of annual installations and growth, making it a significant trade partner for European producers.

In 2024, the EU continued to export more industrial robots to China than it imported from China. While EU imports of Chinese industrial robots have stabilised in recent years, EU exports to China have shown declining growth dynamics (UN Comtrade data). However, EU import data from Eurostat confirms recent reports of surges in robot imports into the EU in 2025, notably in September 2025 (Klement, 2025).

Until 2023, China was a net importer of German industrial robots. However, China has recently entered a phase in which industrial robot production output approaches global demand, exports increase and prices decline. Taken together, these developments indicate excess capacity pressures, although these pressures have not yet stabilised over time to qualify as structural overcapacity. From the technological perspective, China's industrial robotics sector remains primarily competitive on price and continues to depend on high-precision components from Europe, the USA, and Japan. Nevertheless, China's government provides comprehensive funding aimed at overcoming these technological bottlenecks. High production output therefore implies substantial potential for further export expansion.

Germany remains one of the world's leading exporters of industrial robots and competes with China and Japan both in major third-country markets such as the US and Mexico and within the respective domestic markets, including the EU, China, and Japan, with China continuing to be the world's largest importer (UN Comtrade data). Drawing on the annual reports of leading high-end firms, the product differentiation between quality and price has so far enabled them to defend their market segments.

For Europe, this underlines the importance of strengthening innovation capacities. Chinese firms remain unable to compete effectively in high-end industrial robotics and, at times, lack the profits to cover R&D expenditures. Comprehensive public funding across R&D, industrialisation, and application is however likely to enable catch-up dynamics in the medium to long term, with direct implications for future EU–China trade relations in the robotics sector.

2.5 Preliminary conclusions on overcapacities in China

Building on insights from developmental state experiences in East Asia and the extensive China-focused literature, we propose a dynamic definition of industrial overcapacity. Industrial overcapacity is a condition in which productive capacity expands faster than economically viable demand over a prolonged period, leading to sustained overproduction pressure that is absorbed through inventory accumulation, price and profit margin declines, and trade spill-overs rather than through capacity exit. The prolonged duration of five to seven years of such conditions increases the likelihood that they are sustained by policy-induced incentives and institutional arrangements that shape investment behaviour and influence whether imbalances are resolved through market adjustment (e.g. market exit or downscaling of production capacity) or persist through inventory,

price and export adjustment mechanisms. Thus, industrial overcapacities exhibit greater persistence than cyclical capacity fluctuations.

The empirical application of this definition for 14 industrial sectors in China shows evidence of overcapacity. Ten sectors clearly meet the criteria for overcapacity under the proposed definition. While capacity expansion broadly tracked domestic demand in the early 2010s, the analysis shows a persistent divergence since the mid-2010s, with productive capacity growing faster than economically viable market absorption across most manufacturing sectors. This dynamic is evident in positive capacity expansion gaps (CEGs) at the aggregate industry level and most industrial sectors. Because of capacity expansion pressure on production, market adjustment has occurred primarily through the absorption channels identified in the definition. Inventory accumulation has emerged as the dominant buffer for elevated production, while price adjustment and margin compression play a minor role.

Six sectors see rising export shares. Thus, exports are not the primary adjustment channel for absorbing overcapacities. However, the six sectors with elevated export shares and, in particular, Automobiles and Electrical Machinery accounted disproportionately for the large increases in exports from China to the EU. The contribution of active state involvement to industrial overcapacity cannot be established conclusively by our empirical analysis, though there are indications that it has had an influence, notably through SOEs' adjustment behaviour. While the link between overcapacity and policy-induced incentives is widely recognised in the debate, including in China (see section 3.1.), robust empirical validation would require more comprehensive datasets capturing the full effects of policy interventions.

While widespread overcapacity in China's industrial sectors is evident, assessing the current status and potential emergence of new overcapacity requires a more detailed analysis. We examine four cases grouped into two categories: First, semiconductors and EVs/batteries illustrate fast-growing sectors with relatively developed production systems and potentially emerging overcapacity. Second, hydrogen and robotics represent high-growth sectors with substantial uncertainty regarding future demand, applications, and technologies. Across all cases, sectoral dynamics differ markedly, resulting in distinct assessments of overcapacity risks, varying implications for European firms, and differentiated policy implications.

In the semiconductor sector, European firms operate established production systems focused on mature (legacy) semiconductors for automotive and industrial uses, including exports to China. There are currently no clear signs of overcapacity. However, while global demand for mature semiconductors supports capacity expansion, China's rapid push toward greater self-sufficiency from a low base of around 40 % raises risks of emerging overcapacity if demand projections prove overly optimistic. However, two issues are more pressing for European producers: First, growing domestic demand should be met through expanded EU-based production to reduce future import dependence from China, warranting a reassessment and possible strengthening of existing subsidy schemes in the EU. Second, informal local content requirements in China may crowd out European firms despite their technological strengths.

In the EV and battery sector, European carmakers are competing with Chinese firms with strong technological and production capabilities. While China accounts for around 70 % of global EV production and exports to the EU have risen sharply — prompting EU countervailing tariffs — capacity utilisation in China is broadly comparable to other regions, including the EU. By contrast, battery cell production shows significantly higher overcapacity, largely driven by Chinese firms that dominate global supply and export to Europe. However, battery production is currently localised in the EU for European EV assembly, including investments by Chinese firms, which potentially shield European EV assembly from massive import competition from China. However, this intensifies

pressure on European battery producers, which are late entrants and remain uncompetitive in cost, scale, and experience. The core policy challenge for the EU is to support technological catch-up and industrial scaling of domestic battery producers.

The renewable hydrogen sector remains policy-driven and largely pre-commercial with government targets shaping both supply and demand. Electrolyser manufacturing overcapacity has already emerged globally, particularly in mature alkaline technologies dominated by Chinese producers, as market uptake lags behind capacity expansion. Europe, however, retains technological leadership in advanced electrolysis technologies, providing continued leverage through technological dependencies. As different electrolysis technologies serve distinct system needs, they can coexist. EU policy should therefore protect technology-leading segments whose scaling depends on policy-driven demand, while allowing imports of cost-competitive alkaline electrolysers to accelerate deployment and reduce costs. Ensuring demand certainty and safeguarding Europe's advanced electrolysis capabilities should remain central to EU hydrogen policy.

The robotics market is another example of a sector with substantial growth potential and high uncertainty. In the segment of industrial robots for the automation of industrial production processes, China has significantly expanded capacities and production and recently eliminated the import deficit also against the EU, which indicates a risk of overcapacity. However, growth has been in the production of low- and mid-tier industrial robots, while the high-tech segment and specific component production are still dominated by foreign companies, including European producers. Thus, China's industrial robotics sector remains primarily competitive on price, but continues to depend on high-precision components from Europe, Japan and the US, although Chinese policies aim to overcome these technological bottlenecks in the near future.

In sum, while overcapacity exists across many Chinese industrial sectors, the case studies' detailed analysis of products and sectors reveals that overcapacity might not be the sole key issue in the competition between the EU and China. In particular, in sectors with a technological gap between European producers and Chinese firms such as semiconductors, hydrogen, and industrial robots, the effects of overcapacity emerging in low- and mid-tier segments remain limited for European firms in the high-end segments of these sectors. The EV and battery sector represents a partial exception: while the restructuring and localisation of production networks may limit overcapacity risks for EV producers in the EU, they simultaneously increase technological and supply-chain dependence on Chinese battery producers. In other sectors, the greater challenge stems from China's push toward self-sufficiency and technological upgrading, which targets high-end segments and critical components where European firms remain strong, thereby threatening to increasingly restrict their access to the Chinese market. In this context, overcapacities may reinforce adverse effects on China's own upgrading strategies (see section 3.1). From an EU policy perspective, the priority should therefore be to safeguard technological leadership and support scale-up in strategically important industries.

3 Solutions to tackle overcapacities

3.1 China's internal policies

3.1.1 The Chinese debate on overcapacities

In economic negotiations with third countries, Chinese authorities reject allegations of overcapacity, while price wars among domestic producers have triggered internal policy debates and responses under the concept of 'involution'. In Chinese policy discourse, 'severe overcapacity' refers to traditional capital-intensive sectors such as steel, where the market exit of unviable firms is expected to help consolidate supply and demand imbalances. In stark contrast, 'involution' shifts

attention to persistent 'price wars' that draw firms into cut-throat competition, preventing profitability from functioning as an effective mechanism of market adjustment (Liu, 2025). In light of the EV case presented above, the concept of 'involution' helps capture phenomena in which declining prices are not necessarily driven by overcapacities.

Involution and overcapacity

For an English-language audience, the South China Morning Post (D. Wu, 2024) summarises the concept of 'involution' in terms broadly consistent with official policy discourse, defining it as follows:

> [...] excessive competition, where more resources are invested without increasing returns or reinvestment in research and development. For some industries, companies may attempt to gain a temporary competitive advantage by lowering prices at the cost of profit margin to undercut rivals. This behaviour may force all parties to invest greater effort, leading to a deterioration of the overall industry environment, increased pressure to survive, and stagnant or declining growth for all participants.

To analyse the core concept of 'involution' guiding China's central policymakers, this study draws on definitions published in Qiushi, the official flagship theoretical journal of the Chinese Communist Party (CCP), which is widely used by China analysts as a bellwether for understanding Party leadership ideology and policy priorities. In Qiushi's terms (Ju, 2025), the concept of 'involution' may include 'overcapacity' and 'overproduction', yet its central characteristics are excessive competition and delayed market exit by unsustainable firms. The article explicitly distinguishes between firm-level behaviour and local government action, without directly linking the two. Nevertheless, shared structural conditions point to sequential and mutually shaped dynamics. To better understand how China's policy decision-makers conceptualise the problem of 'involution', this study organises the CCP framework of 'involution' into a stylised process comprising three interrelated stages linking firm and government behaviour:

Stage 1: Capacity expansion and overproduction of homogenous products amid local government competition

In the first stage, capacity expansion neglects economically viable demand conditions. One major driver is 'blind' investment and subsidies from local governments. Although not explicitly mentioned in the Qiushi article, local governments' revenue bases are constrained within China's fiscal system, thereby incentivising competition among them to promote projects that can attract national funding (Lv, Liu, & Li, 2020). To cultivate local industries, local governments adopt 'irregular' forms of preferential tax, subsidy, and land arrangements and often support similar production facilities, neglecting technological and qualitative differentiation. Firms respond to these incentives by expanding output and over-investing in similar capacities to secure market access, thereby orienting their behaviour toward policy incentives and neglecting market demand signals. While these dynamics can generate excess capacity and, in some cases, oversupply, their crucial problematic effect is the concentration of production in homogeneous, low-differentiation and low-cost product segments.

Stage 2: Low-cost producers drive involutionary competition

In the second stage, excess supply of similar product lines leads to price as the predominant incentive to attract consumers. Firms engage in so-called 'vicious low-price competition' to maintain utilisation and secure their market presence. Market adjustment thus occurs primarily through declining margins, not output reduction, disadvantaging high-end and high-quality producers with higher production costs. From the perspective of Chinese policymakers, this margin

compression is particularly consequential. It squeezes firms' capacity for reinvestment and constrains R&D investment necessary for quality upgrading and product differentiation and thereby blocks alternative pathways of competition. As firms concentrate resources to keep pace with falling prices, this dynamic leads to what Chinese party discourse terms 'bad money driving out good'. Local governments contribute indirectly at this stage by tolerating distorted competitive practices.

Stage 3: Market-clearing mechanisms are delayed

In the final stage, local government intervention extends 'involutionary competition' by delaying the market exit of unprofitable firms and by delaying the reduction of backward and excess capacity. At the same time, bankruptcy and restructuring procedures are weakly implemented and time-consuming, discouraging timely market exit. Together, these factors lead to what *Qiushi* describes as firms being 'unable to live well, yet unable to die', allowing, in the long run, 'backward capacity to crowd out advanced capacity.'

Exports are no strategic remedy to China's domestic 'involution'

Read against the definition and analysis of overcapacity outlined in Chapter 2, the CCP acknowledges that government-driven capacity expansion contributes to persistent overcapacity. Yet, in the Chinese framing, the core problem is not overcapacity per se, but excess production by low-end manufacturers, which forces higher-end producers into price-based competition and leaves little room for qualitative differentiation. In turn, excessive competition cannibalises the investment needed for R&D and technological upgrading. In a Deutsche Bank research paper, Yi Xiong (2025, p. 4) adds that autonomous market-clearing mechanisms could be effective in the long run. Still, their slow adjustment would generate macroeconomic spill-over effects that could potentially lead to a downturn of China's national economy. This would undermine the 'output legitimacy' of China's one-party state, which is based on its capacity to implement swift measures to maintain economic and social stability (Yang & Zhao, 2015). Therefore, overcapacity is not framed as a basis for an export-led development strategy, as in the approaches of Japan and other East Asian countries discussed in section 2.1.3, but rather as a threat to China's global industrial competitiveness.

Moreover, China's industrial base has been increasingly restructured to meet domestic demand. This is evident in the decline of China's export share of its annual GDP, from 35.5 % in 2006 to 20 % in 2024 (World Bank, undated). The domestic market, as a pillar of China's economic development, is strongly supported by China's de-risking policy framework of dual circulation: taking 'domestic circulation as the primary focus', while international trade is relegated to a complementary role (State Council of the PRC, 2021). Consequently, government-supported industries today, such as electrolyser manufacturing, do not operate in an economic environment as strongly dependent on exports as China's photovoltaic sector some years ago (Gong, 2025, pp. 2 and 4).

Scaling of industries has been a strategic approach of state-led market creation to reduce production costs and thereby provide the basis for new products to compete with established ones (Gunter et al., 2025; Musitz, 2024b, p. 2). Even if industrial overcapacity has for some time supported China's export expansion and pursuit of global industrial leadership (Gunter et al., 2025), any analysis of China's future industrial policy approach must take into account the mounting domestic economic costs of price wars — at times resulting from overcapacities — including eroded profitability, and deflationary pressures (Watanabe, 2025). Overcapacities reliant on global absorption are increasingly conflicting with China's major goal of technological and economic 'self-reliance', again strengthened in first drafts of the upcoming 15th Five-Year-Plan — the blueprint for

policy priorities and institutional reforms over the five-year period 2026-2030 (Central Committee of the Communist Party of China, 2025).

Against the background of European concerns over price-competitive Chinese exports, overproduction and weak domestic demand may and often does drive firms to seek demand on export markets (OECD, 2022). However, fierce price competition taken abroad is not seen as a strategic remedy for domestic imbalances by Chinese policy makers. It poses instead obstacles to China's current politically salient development framework of 'high quality development', which emphasises technological advancement and product quality, discussed in detail below.

Strengthened market regulation to restructure government investments

China's core development paradigm of 'high-quality development' seeks to reorient the economy away from extensive, cost-driven growth toward higher value-added, innovation-led development. Rooted in political deliberations in 2017, the high-quality development framework was subsequently linked to the modernisation of China's industrial system and emphasises technological innovation, industrial upgrading, and upward movement along global value chains, alongside improvements in household income and social welfare (Xinhua, 2022). Within this macroeconomic framework, innovation-oriented investment, particularly in R&D, functions as the driver of this transition. 'Involutionary competition', as mentioned above, directly undermines the profit base of firms to invest in R&D to improve product quality and gain product differentiation.

Against this backdrop, China's President Xi Jinping (2025) has called for strengthened, centralised, and streamlined market regulation within the framework of 'building a unified national market.' This overarching concept is included in the recommendations for the upcoming 15th Five-Year plan, marking the new central government approach to correcting involution for the years to come (Central Committee of the Communist Party of China, 2025). Xi identifies six guiding principles that will help shape this new form of market regulation:

- strengthening laws and rule-based governance as the primary corrective mechanism;
- reforming government investment and public procurement practices, curbing local protectionism;
- tightening regulation of investment competition among local governments to prevent duplicative investments in similar product lines;
- aligning domestic product quality standards with export standards to facilitate the market exit of low-quality firms;
- unifying competition and bankruptcy laws at the national level; and
- recalibrating officials' performance incentives to discourage uncoordinated investment and local protectionism.

It is important to note that, against the backdrop of 'high-quality development,' this form of market regulation does not represent a break with an innovation- and investment-driven growth model but rather seeks to centralise and restructure investment coordination. Thus, it can be concluded that in the Communist Party's response to its own diagnosis, the government, as the architect of an investment-driven approach that enables overcapacities, must be the source of regulative corrections to avoid capacity expansion gaps and the resulting adverse effects.

3.1.2 Domestic policy measures to tackle overcapacities

Under the umbrella term 'rectifying involution', the Chinese government is currently implementing comprehensive measures that will be discussed in turn in the following section.

Scaling back export support

Export tax rebates are the most prominent direct measure through which the Chinese government supports industries' overseas sales. They are a fiscal instrument used to avoid double taxation on international markets, thereby lowering exporters' cost burdens. Introduced in 1985, this policy reflected China's strong dependence on export-led growth at the time (T. Xiao, 2024). In principle, exported goods are zero-rated for value-added tax (VAT), meaning that the standard 13 % VAT paid on inputs used in production can be refunded or exempted. At the same time, the government retains the authority to introduce differentiated rules for specific products, allowing it to recalibrate export incentives and influence export dynamics in line with broader policy objectives (Value Added Tax Law of the PRC [中华人民共和国增值税法] 2024, Art. 2(2).

To promote 'industrial transformation and upgrading' and address involutionary competition (Xinhua, 2026), export tax rebates on aluminium were cancelled, and rebate rates for photovoltaics and batteries were reduced from 13 % to 9 % in 2024 (MOF & SAT, 2024). In a subsequent adjustment, export tax rebates for photovoltaic products will be fully cancelled from April 2026, while rebate rates for battery products will be reduced to 6 % over the same period and eliminated entirely as of 1 January 2027 (MOF & SAT, 2026). Furthermore, the Chinese government has adopted a licensing system that introduces quality standards to restrict the export of certain steel goods, effective since 1 January 2026 (MOC & GACC, 2025). According to the China Photovoltaic Industry Association (2026), the cancellation of export tax rebates is intended to curb the 'externalisation of involution' by domestic manufacturers, which has led to falling prices and weakened profitability both domestically and globally.

From a trade perspective, the export of overcapacity into international markets contributed to Chinese producers rapidly expanding international market share, but at the cost of eroding profitability across the global industry. Taken together, these developments suggest that the Chinese government is increasingly willing to recalibrate its export-promotion policies and, in certain sectors, to tolerate or even support measures that stabilise international prices and improve global industry profitability. China's Ministry of Commerce (2026) framed the China–EU consultations on the electric-vehicle case as being conducted 'in a spirit of mutual respect,' and described the 'price-undertaking guidance' as a practical and targeted, WTO-consistent means of resolving trade issues between the two economies.

Consumption-stimulating measures in sectors with trade frictions

China complements the cancellation of export tax rebates with consumption-stimulating measures to redirect export goods to domestic markets amid trade tensions (K. Wang, 2025). In 2024, the State Council (2024) adopted the action plan on large-scale equipment renewal and consumer goods trade-ins, encouraging the replacement of outdated industrial equipment and consumer goods with more energy-efficient and environmentally compliant alternatives. This included product categories such as steel, building materials, NEVs, and electronics, which closely align with sectors characterised by price wars and rising export frictions.

Upgraded product quality benchmarks on energy efficiency, emissions, safety, and technical standards aim to steer demand towards high-quality producers. Manufacturers at the low end are thereby excluded from policy support that amounted nationally to CNY 170 billion (≈ EUR 22 billion) in 2024 (K. Wang, 2025). In the subsequent implementation rules for 2026, concrete monetary incentives are defined at the national level, signalling central policy support. For example, consumers who replace an old vehicle with a new-energy vehicle (NEV) can receive subsidies of up to CNY 20 000 (≈EUR 2 470) (NDRC & MOF, 2025). In the coming years, central and local

governments plan to expand trade-in programs and their funding scale to increase consumption across sectors.

Strengthening legal measures against below-cost sales

In 2025, the National People's Congress revised the Anti-Unfair Competition Law (Anti-Unfair Competition Law (2025 revision) [中华人民共和国反不正当竞争法], 2025, Art. 14) that explicitly outlaws price setting below costs by providers of e-commerce platforms. Administrative fines are up to CNY 500 000 (≈EUR 62 000) for ordinary violations, and up to CNY 2 million (EUR 247 200) for serious violations. Furthermore, a draft for revision of China's Price Law extends the prohibition on 'dumping at prices below cost [...] for the purpose of excluding competitors or monopolising the market' to enterprises using their pricing system to coerce others to dump prices below cost (NDRC & SAMR, 2025b).

The revisions to these laws primarily address price dumping on e-commerce platforms, while strengthening administrative measures to enhance government oversight across industries. The National Development and Reform Commission (NDRC) — China's macro-economic planning institution — together with the State Administration of Market Regulation (SAMR) issued an announcement that introduces an early warning system for enterprises that set prices below 'minimum competitiveness.' If an enterprise fails to raise its prices, the SAMR starts investigations and pursues legal measures (NDRC & SAMR, 2025a). In combination with the government using its investment leverage to steer behaviour, as discussed below, early warning systems derive their effectiveness as gatekeepers of market access.

The early warning system to curb price dumping is recent and its practical impact remains to be seen. Existing evidence of intensified antitrust actions following the establishment of the SAMR in 2018 suggests that Chinese competition authorities are capable of forceful enforcement when policy priorities are clear (Zhu, J. and Gu, J., 2024; Chen, Y. et al., 2026). At the same time, enforcement practices in China remain uneven across contexts (Zhang J., 2025), implying that the effectiveness of this measure will likely depend on political prioritisation and sectoral focus.

Controlling market access

In regulating overcapacity, China's government approaches traditional industries such as the steel sector differently from emerging industries characterised by 'involutionary competition.' In the case of the steel industry, the government has progressively introduced capacity cuts since 2016 through mandatory replacement ratios (MIIT, 2025), administrative output controls (MIIT, MNR, MEE, & MOC, 2025) and export licensing measures (MOC & GACC, 2025). However, circumvention at the local level, together with falling demand amid China's property market downturn, contributed to renewed increases in steel overcapacity following the initial phase of reductions (Shi et al., 2023; TransitionAsia, 2024). In contrast to the steel sector, regulatory governance through standards and lists is the predominant policy tool used to address supply and demand imbalances in 'Strategic Emerging Manufacturing Industries'. These contrasting governance strategies reflect differences in state capacity and ownership structure. Direct administrative controls are more feasible in SOE-dominated sectors, while globally integrated emerging industries are predominantly underpinned by private enterprises, in which state leverage operates primarily through standards and incentives (Shi et al., 2023; Xiao et al., 2025, p. 4; Xiong, 2025, p. 5).

Mandatory technological standards enable the Chinese government to set benchmarks constraining market access for certain products. According to industry modelling, a draft standard introducing stricter efficiency requirements for polycrystalline silicon production — (Resources and Environment Research Division, 2025) at the time of writing in review —, could curb up to one-third

of national production capacity (Industrial Bank Research Institute, 2025). These mandatory standards are complemented by 'guiding' catalogues, lists, or standards.

Central to this approach is the Guiding Catalogue for Industrial Structure Adjustment, introduced in 2005, which classifies industries and activities into three categories: 'encouraged,' 'restricted,' and 'eliminated' (NDRC Industry Department, 2023). Although formally a soft-law instrument, it has hard effects. It serves as the basis for government and enterprise investment decisions, as well as for the formulation and implementation of fiscal, taxation, credit, land-use, and import and export policies.

Within this policy framework, the government has established quasi-mandatory whitelists of compliant enterprises in recent years. As a requirement for inclusion, enterprises in the PV industry must invest at least 3 % of their annual sales in R&D, or a minimum of CNY 10 million (≈ 1.28 million EUR), demonstrate a capacity utilisation rate of at least 50 %, and meet technological benchmarks that are close to, or at par with, the international state of the art, which tends to disadvantage smaller firms. Similar conditions apply to the whitelist in the battery sector (MIIT, 2024a, 2024b). Through these mechanisms, the government uses its leverage over market access and market positioning to curb excess capacity, push low-end producers out of the market, and strengthen firms' innovation capabilities.

Other measures

The National Financial Regulatory Administration recently promulgated new rules to ease the consolidation of firms through mergers and acquisitions (M&A), supported by commercial banks. Credit access was significantly expanded as the minimum shareholding requirement for minority acquisitions was reduced to 20 %. Control-type acquisitions are eligible for loan financing of up to 70 % of the transaction value.

So far, bond lending mediated through local government financing vehicles has enabled local officials to incur unrestrained hidden debts to invest in industry capacity building, primarily through fixed assets such as industrial parks. This contributed to a massive government debt crisis amounting to about 290 % of GDP in 2024 (Kunath, 2025, p. 58). Thus, the central government enabled credit swaps to ease pressure, but at the same time increased control and constrained local governments from taking up bonds to invest in new projects (Jing, 2025). Between February and December 2025, accumulated fixed assets investment declined by 3.8 %, marking the first contraction since 2000, excluding Covid-19 related shutdowns in 2020 (National Bureau of Statistics of China, 2026). The consolidation of financial oversight at the central level weakens local governments' capability to support companies in low-profitability sectors, but may increase the likelihood of capacity exit, market clearing, and upward price adjustments.

3.1.3 Effects of China's 'anti-involution' measures

Measured against the numerous officially published policy targets aimed at curbing economic 'involution,' China's central government demonstrates political commitment to correct dumping-level prices and persistent supply–demand imbalances. Since these measures were only promulgated in 2024 and began to take effect toward the end of 2025, it is too early to assess their comprehensive effects. First signs in the Photovoltaic Industry indicate stabilisation. Upstream Polysilicon production declined by almost 30 % while prices increased by around 9 % in the fourth quarter of 2025 (J. Ma, 2025; Vincent, 2026; Wood Mackenzie, 2025). Following the announcement to cancel export rebates, upward pressure on solar module prices is expected.

Drawing on insights from China's supply-side reform to curb production overcapacity a decade ago, success largely relied on local governments' commitment to enforce capacity cuts.

Performance targets for officials, based on economic growth, employment, and social stability, shape career prospects and increase reluctance to implement socially and economically painful policies (Boulter, 2018). As central actors in the local implementation and financing of national industrial policies, local governments play a key role in balancing public investment with efforts to curb production. At the same time, China's tax system systematically pressures local officials to financially support business projects. Receiving about 45 to 50 % of the national tax while bearing almost 90 % of the country's tax burden (Kunath, 2025, p. 27), local governments are structurally underfinanced. To fill this gap, they politically and financially facilitate the growth of local businesses to broaden their tax base (Lv et al., 2020).

Against the backdrop of tighter financing conditions for local governments, the recommendations for China's 15th Five-Year Plan call for a stronger centralisation of fiscal authority and investment decisions. In this context, the National Development and Reform Commission (NDRC) announced last year the National Venture Capital Guidance Fund as a new, standard-setting model focused on long-term support for basic research and original innovation, seed- and early-stage ventures, and technological bottlenecks. Under this framework, the central government assumes a more direct role in managing equity investments and allocating projects in line with macroeconomic reasoning. As investment decisions become more centralised, policy emphasis shifts toward promoting early-stage research and innovation as building blocks for high-quality development (NDRC, 2026). With 'high-quality development' and 'substantially enhancing scientific and technological self-reliance and self-strengthening' placed at the top of the list of the 15th Five-Year Plan agenda, government investments will continue to play a critical role in China's future economic development.

3.1.4 Implications for international trade and EU negotiations

From the EU's perspective, subsidies and overcapacity in China are seen as pushing low-priced exports onto European markets, thereby distorting the competitive conditions faced by European firms in their domestic markets. While Chinese authorities officially reject allegations of overcapacity in negotiations with the EU, internal policy discourse acknowledges the problem, albeit framed as 'involution', particularly in reference to persistent price wars.

Importantly, involution — though not necessarily characterised by overcapacity — in the domestic market is regarded by Chinese policymakers as a serious challenge. Market exit of low-end firms is understood as the primary solution to excessive competition. Importantly, internal CCP discourse does not frame exports as a strategic solution to domestic supply-and-demand imbalances. Rather than resolving involutionary competition, exports are seen as a risky absorption channel that draws both foreign and Chinese producers into global price wars, thereby restricting the funds needed for financing industrial upgrading. From this perspective, exporting does not alleviate competitive pressures but risks reinforcing them across markets.

Against this backdrop, China's anti-involution policies aim to curb price wars and raise prices through regulation, the withdrawal of export advantages, higher technological requirements, company whitelisting for market access, incentives for M&A, and restrictions on local governments' investment capacity. At the same time, the central government centralises and reorients public funding to the initial stages of industrial development, omitting — at least in its explicit goals — industrial expansion. Controlling market access to reduce competitors and restricting local governments from taking on new debts to finance industrial projects are the most powerful levers among anti-involution measures. If successful, these efforts would also be beneficial from a European perspective, as higher prices would ease pressure on international markets. In this limited sense, China and the EU share an interest in stabilising prices, even if their underlying diagnoses differ.

This partial convergence is reflected in the comparatively positive media reception on the Chinese side of recent discussions with the EU on price undertaking mechanisms for electric vehicles (MOC, 2026). Such instruments do not address the structural causes of involution or overcapacity, but they offer a pragmatic means of mitigating price-based distortions. Both economies could benefit: European firms from reduced competitive pressure, and Chinese firms, insofar as minimum prices would allow some producers to regain profitability and facilitate the exit of those reliant solely on low-price competition as quality and technologies become more important.

At the same time, China remains firmly committed to a long-term strategy of technological self-reliance, a priority prominently emphasised in context of the upcoming 15th Five-Year Plan. Government investment in research and development, industrialisation, and the commercialisation of new technologies will therefore continue in sectors with technological bottlenecks. Measures that restrict exports redirect supply toward the domestic market. While these policies could reduce pressure on European markets, they also intensify competition within China. As a result, a tightening domestic market may increase competitive pressure not only for Chinese firms but also for European firms exporting to and operating in China.

Finally, China's industrial strategies increasingly aim to close the technological gap between domestic and international firms. Against this backdrop, European industrial strategies need to support the innovation capacity of domestic technological leaders, to remain indispensable in China's domestic market while also competing on quality in international export markets.

3.2 Unilateral measures by the EU

Having established the nature of overcapacity and the options for action from China, this chapter turns to measures the EU can take to address overcapacity issues unilaterally. Obviously, such unilateral measures must be compatible with the EU's international legal obligations, in particular, under WTO law. This is true for 'traditional' trade defence instruments (TDIs) as well as measures under more novel legal instruments aiming to ensure the EU's economic security.

While WTO law does not address 'overcapacity' explicitly, it permits members to protect their industries against concomitant unfair practices such as dumping and subsidies, or to enact safeguards when confronted with a sudden surge of foreign goods. Specific WTO agreements set out rules for the initiation, investigation and imposition of such TDIs. Building on these rules, the EU has adopted several regulations, which provide the legal basis for concrete trade defence measures. While EU trade defence activities have generally increased significantly over the past years,[33] prominent recent cases, such as the countervailing duties on battery electric vehicles (BEVs) from China or steel safeguard measures, often directly concern overcapacity issues. Similarly, the EC's 2025 proposal for a regulation addressing the negative trade-related effects of global overcapacity on the Union steel market (hereinafter 'steel measure') directly aims to tackle sector-specific import pressures resulting from 'unsustainable levels of global overcapacity' (EC, 2025e).

In the following section, we revisit the three traditional TDIs (anti-dumping, countervailing duties, and safeguard measures) and analyse their application to cases of industrial overcapacity, before turning to other instruments the EU could potentially use to tackle this issue.

[33] In total there were 199 measures in place at the end of 2024 (EC, 2025, p. 2).

3.2.1 Trade Defence Instruments

Anti-dumping measures

The WTO framework for anti-dumping measures is contained in Art. VI GATT 1994 and the WTO Anti-Dumping Agreement (ADA). Dumping is exporting a product at a price below its 'normal value' (Art. VI:1 GATT), i.e. below the price for consumption in the exporting country or a comparable export price or the cost of production and sale plus a reasonable profit (Arts 2.1 and 2.2 ADA). The method for determining whether dumping occurs has been a contentious issue, especially when involving countries whose status as market economies is disputed (see, for example, Eeckhout, 1997; Snyder, 2010, p. 266 ff.; Brauneck, 2018). In case of China, its 'Accession Protocol' (WT/L/432) allowed the use of alternative non-market economy methodologies for 15 years (cf. Section 15(a) Accession Protocol). The EU initially kept using these methods after this 15-year period, but eventually amended the methodology in 2017 (Regulation 2017/2321/EU).[34]

If dumping occurs and there is a material injury to an established domestic industry, a threat thereof, or a threat of materially retarding the establishment of such an industry caused by dumping, countries can apply anti-dumping duties offsetting the dumping margin, i.e. the price difference (Art. VI:2 GATT). To this end, the ADA contains rules on the calculation and comparison of the prices involved, the determination of injury (or threat thereof), and the causality between dumping and injury to the domestic industry.

At the EU level, the EU Anti-dumping Regulation (Regulation (EU) 2016/1036 as amended) provides the legal basis for the EC to investigate potential dumping and — if it finds that dumping occurs, injury is caused, and EU interest calls for intervention — impose anti-dumping duties. The EC may conduct investigations in response to complaints or *ex officio*. In addition to the criteria set out in the ADA, Regulation 2016/1036 provides that anti-dumping measures may not be applied where the investigation concludes that they are not in the EU's interest (Art. 21). Moreover, the regulation contains provisions against circumvention (e.g., by way of transhipment or a slight modification of a product to avoid anti-dumping duties, cf. Art. 13).

Regarding the calculation of the normal value, which is essential for the determination of dumping, particular issues arise where government interventions cause significant distortions. More specifically, significant distortions exist 'when reported prices or costs, including costs of raw materials and energy, are not the result of free market forces because they are affected by substantial government interventions' (Art. 2(6a)(b) Regulation (EU) 2016/1036 as amended). Whether this is the case, must be established in the anti-dumping investigation.[35] If so, according to the EU's amended methodology, the EC may disregard domestic costs and prices, and determine the normal value on the basis of other sources, such as the production and sale costs in an appropriate representative country with a similar level of economic development or undistorted international prices, costs, or benchmarks. Importantly, where there is more than one appropriate representative country, the EC must give preference to countries with an adequate level of social and environmental protection (Art. 2(6a)(a) Regulation (EU) 2016/1036 as amended).

Countervailing measures

At the WTO level, subsidies are 'subject to an intricate set of rules' (van den Bossche & Zdouc, 2022, p. 840). The Agreement on Subsidies and Countervailing Measures (ASCM) distinguishes

[34] Note that Section 15(b) Accession Protocol is still in force and allows deviations from standard methodologies in some cases in China; cf. Gao & Zhou (2022).
[35] As evidence for the individual assessments of significant distortions, the EC has published reports on the state of distortions in industries in Russia (EC, 2020) and China (EC, 2024a).

between (a short list of) unconditionally 'prohibited' subsidies[36] and so-called 'actionable' subsidies, which may only be challenged if they adversely affect other WTO members (cf. Art. 5 ASCM). A third category, 'non-actionable' subsidies, which created a safe harbour for specific types of subsidies (for example, certain R&D as well as regional subsidies), expired in 2000 and has not since been revived (cf. Art. 8 ASCM). The ASCM provides for two distinct remedies — challenging an alleged subsidy[37] in WTO dispute settlement or imposing countervailing measures to offset the effects of the subsidy.[38]

The rules enshrined in the ASCM, however, are increasingly viewed as suffering from severe shortcomings, such as the narrow understanding of what constitutes a 'public body' potentially granting a subsidy in WTO case law (Howse, 2020, pp. 384 ff; Wu, 2016, pp. 301 ff); the high evidentiary burden in proving the existence of a subsidy (Hillman & Manak, 2023, p. 12-13); the persistent lack of transparency, due to 'chronically low submission rates' of subsidy notifications (WTO, 2025); or lengthy proceedings and the ineffectiveness of the prospective remedies[39] of the ASCM (Grossman & Sykes, 2025; Hillman & Manak, 2023). The last issue is particularly relevant where a subsidy entails 'serious prejudice to the interests of another Member'. In such cases, even the removal of the adverse effects of the subsidy *pro futuro* will 'often [do] little to dismantle the capacity that has been built to produce the subsidised goods' (Grossman & Sykes, 2025, p. 570; Hillman & Manak, 2023, p. 15). Moreover, a recent panel report, which found that the EU had infringed WTO law when imposing countervailing duties on certain steel products from Indonesia, illustrates just how difficult it is to establish the nature of a state action as a subsidy, the necessary specificity of subsidies, and the nature of an entity as a 'public body' (WT/DS616/R). Against this background, WTO members and scholars alike have made various suggestions for improvement. For example, the 2020 trilateral initiative of the EU, Japan, and the US (Joint Statement, 2020) proposed expanding the category of prohibited subsidies;[40] reversing the burden of proof regarding the negative impacts of subsidies considered to have a particularly 'harmful effect';[41] adding the distorting capacity of subsidies to the instances of serious prejudice; or incentivizing notification of subsidies by 'rendering prohibited any non-notified subsidies that were counter-notified by another Member, unless the subsidizing Member provides the required information in writing within set timeframes' (Joint Statement, 2020, para 4). Leaving aside the issue of their political feasibility,[42] it should be noted that many of these modifications raise considerable legal concerns (Grossman & Sykes, 2025; Howse, 2020). In contrast, many suggestions made by prominent WTO scholars have a different thrust, ranging from re-creating a safe harbour for subsidies in the common international interest and/or other subsidies that have positive (or no negative) externalities, to creating incentives for complying with basic transparency obligations, closing the remedy gap, and increasing penalties for rules violations (Grossman & Sykes, 2025; Hillman & Manak, 2023).

[36] Notably export subsidies and subsidies contingent on domestic content requirements, cf. Art. 3.1 ASCM.
[37] Art. 1.1 ASCM defines 'subsidy' as 'a financial contribution by a government or any public body [...] where [...] a benefit is thereby conferred'. In addition, the applicability of the ASCM rules largely depends on the specificity of the subsidy (cf. Arts 1.2 and 2 ASCM). Note that according to Section 10.2 of China's Accession Protocol, subsidies are specific if state-owned enterprises are the predominant recipients or receive disproportionately large amounts of such subsidies.
[38] More precisely, according to Art. 19.4 ASCM, no countervailing duty can be levied 'in excess of the amount of the subsidy found to exist, calculated in terms of subsidisation per unit of the subsidised and exported product'.
[39] On the 'remedy gap', see Brewster (2011).
[40] Including 'subsidies to enterprises unable to obtain long-term financing or investment from independent commercial sources operating in sectors or industries in overcapacity', Joint Statement (2020) para 1.
[41] Including 'subsidies creating massive manufacturing capacity, without private commercial participation', Joint Statement (2020) para 2.
[42] NB. modifications of the ASCM would require consensus among the WTO members.

At the EU level, the EU Anti-Subsidy Regulation (Regulation (EU) 2016/1037 as amended) provides the legal basis for the EC to impose countervailing duties. The EC may initiate proceedings based on a complaint by or on behalf of the relevant EU industry or, in special circumstances, *ex officio* (Art. 10). Where the EC provisionally finds subsidisation and injury to exist, it may impose provisional duties, if these are in the EU interest (Art. 12(1)). Where the facts finally established show the existence of a countervailable subsidy and injury caused thereby, the EC imposes a definitive countervailing duty, not exceeding the amount of the countervailable subsidy, or a lower amount, if such a lesser duty is adequate to remove the injury to EU industry (Art. 15(1)). Importantly, both provisional measures and definitive countervailing duties, in principle, apply only to future imports.[43] That said, in specific circumstances definitive countervailing duties may be imposed with limited retroactive effect (Art. 16(4)-(5)). One requirement is that the imports have been registered. In this regard it is noteworthy that while previously such registration usually required a justified request from EU industry, the EC now automatically registers all imports of products under anti-subsidy investigations (Directorate-General for Trade and Economic Security, 2024). Finally, it deserves mention that the country of origin and/or export or individual exporters may offer voluntary undertakings, which — if accepted by the EC — exempt the relevant imports from any provisional or definitive countervailing duties (Art. 13 and Art. 15(2)).

Safeguard measures

The WTO framework for safeguard measures is set out in Art. XIX GATT and the Agreement on Safeguards (SA). Art. XIX:1(a) GATT allows states to take measures if, 'as a result of unforeseen developments [...] any product is being imported into the territory of that contracting party in such increased quantities and under such conditions as to cause or threaten serious injury to domestic producers [...] of like or directly competitive products'. In principle, safeguard measures must be applied irrespective of the source of the product and therefore cannot target selected origins (Art. 2.2 SA). That said, the SA provides some flexibility as regards the precise quota allocations (Art. 5.2).

Overall, the SA sets a high threshold for the application of safeguard measures. In particular, as regards causality, Art. 4.2(b) SA stipulates that '[w]hen factors other than increased imports are causing injury to the domestic industry at the same time, such injury shall not be attributed to increased imports'. If the conditions under the SA are met, states may choose the 'most suitable' measure, which must not go, however, beyond what is 'necessary to prevent or remedy serious injury and to facilitate adjustment' (Art. 5.1).[44] Moreover, Art. 9.1 SA limits the application of safeguard measures against products originating in developing country members. In this context it may be mentioned that China recently announced to forego Special and Differential Treatment benefits (though not its designation as developing country) (Le Poidevin, 2025). Transitional rules, which allowed for easier application of safeguard measures against China expired in 2013 (Section 16 Accession Protocol).

At the EU level, the regulation on common rules for imports (Regulation (EU) 2015/478) largely transposes the WTO rules, though it occasionally goes beyond the SA. For example, the Regulation's indicative list of factors that may indicate a consequent impact of increasing imports on EU producers goes beyond the SA, and explicitly lists trends in stocks, prices, return on capital employed, and cash flow (Art. 9(1)(c)). Where the requirements set out in the regulation are met, the EC may prescribe import authorisation requirements and quotas as safeguard measures (Art. 15).

[43] Art. 15(1) subpara 3 and Art. 16(1) EU Anti-Subsidy Regulation.
[44] NB. For provisional safeguard measures, states may only use tariffs (Art. 6).

A case in which the EU has famously applied safeguards concerns the steel sector: in 2018 the US introduced tariffs, which diverted Chinese and other steel exports to Europe, – with severe implications for the steel industry in Europe (Urlesberger, 2020). Based on evidence provided by the EU steel industry, the EC conducted an *ex officio* investigation resulting in the adoption of provisional and later definitive safeguard measures for 26 steel product categories for a three–year period. The safeguard measures allowed continued tariff-free entry for a quota set at 5 % above the average steel imports for the previous three years (Recital 144 EC Implementing Regulation (EU) 2019/159 'Definitive Safeguard Implementing Regulation'). For countries, which had been major exporters of steel to the EU in the previous three years, a specific quota was set individually, while other countries' steel exports to the EU fell into a global 'residual' quota (Recital 147 Definitive Safeguard Implementing Regulation). Steel imported beyond the quotas was subject to a 25 % tariff (Recital 180 Definitive Safeguard Implementing Regulation).

In its Definitive Safeguards Implementing Regulation, the EC explicitly identified 'an unprecedented steelmaking overcapacity [...] accentuated by distortive subsidies and government support measures, which led to price depression' (Recital 49). The finding that 'excess production capacity increased at a time when it was economically expected to decrease' was considered an 'unforeseen development' justifying the adoption of the safeguard measures (Recital 52).[45]

The steel safeguards were subsequently amended multiple times: In 2019, the EC placed automotive-grade steel products under the end-use procedure set out in Art. 254 of the Union Customs Code (Regulation (EU) 952/2013). As such, they could benefit from a special quota if found to be steel products needed by the European automotive industry (EC Implementing Regulation (EU) 2019/1590). This amendment was reversed in 2020, after it was found to have resulted in products being 'held at the Union's customs border awaiting customs clearance', 'seriously and negatively affect[ing], or even disrupt[ing] supply chains' (Recital 5 EC Implementing Regulation (EU) 2020/35). Another amendment in 2023, sought to bring the measures into compliance with the Panel Report of the WTO Dispute Settlement Body in a case brought by Turkey against the EU steel safeguards. The amendment elaborates on the justification for the measures in more detail, without, however, changing the measures *per se* (EC Implementing Regulation (EU) 2023/104). Moreover, following the reviews of the measures, various amendments to the quotas were made (EC Implementing Regulations (EU) 2020/894; 2022/664; 2022/978; 2023/1301; 2024/1782; 2025/612).

3.2.2 Application of Trade Defence Instruments to overcapacity

As already mentioned, the term 'overcapacity' is not defined in WTO or EU law and none of the existing TDIs were created to specifically deal with overcapacity issues. That said, capacity issues are legally not entirely irrelevant (cf. e.g., Art. 8(8) and 18 Regulation 2016/1037; Art. 3(9) and 18(5) Regulation 2016/1036). Thus, apart from the steel safeguards discussed in the previous section, overcapacity issues have also played a role in several other trade defence cases, for example, the decision to impose a countervailing duty on imports of new BEVs from China (as part of the threat of injury assessment, cf. EC Implementing Regulation (EU) 2024/2754) or the recent partial interim review of a definitive dumping duty on imports of trichloroisocyanuric acid originating in China (in the context of assessing the lasting nature of circumstances related to dumping, EC Implementing Regulation (EU) 2025/2216).

[45] Curiously, the EC assumed the existence of such 'excess capacity' based on a Global Trade Alert Report by Evenett & Fritz, which questioned whether excess capacity was indeed the root problem and warned that 'targeting steel excess overcapacity is a fool's errand' (Evenett & Fritz (2018).

That said, when assessing the application of existing TDIs to overcapacity issues, we find three aspects to stand out: (1) the trade-offs between targeted vs general measures; (2) their temporal limitations; and (3) their substantial limitations as purely defensive measures.

Trade-offs between targeted vs general measures

As outlined above, anti-dumping measures and countervailing duties (CVDs) can be applied in a highly targeted manner to specific sectors, countries, and even companies. In contrast, safeguard measures apply to all third countries, regardless of whether they contribute to overcapacity issues. These differences in terms of target accuracy, however, come with various trade-offs. Safeguard measures generally apply to products of all origins and do not imply any improper behaviour on part of affected states. Consequently, WTO members subject to safeguards can rebalance under certain conditions, which may make safeguards less attractive and increase the risk of a further downward spiralling (in the context of the steel industry, for example, see Evenett & Fritz, 2018; EC, 2025c, p. 2).

By contrast, anti-dumping measures and CVDs are more targeted. The former respond to the behaviour of specific firms, the latter to the behaviour of specific states. This means that these measures do not entail the risk of (directly) harming third countries that have not contributed to overcapacities. At the same time, their application requires time- and resource-intensive investigations since the legal requirements for imposing anti-dumping measures and CVDs are notoriously difficult to establish. This is particularly true for non-transparent markets. Moreover, anti-subsidy investigations appear politically sensitive, as they directly challenge the actions of foreign governments, rather than private companies (EPRS, 2025, p. 4). This may be another reason why CVDs have been comparatively rare. For example, in 2024, the EU conducted 29 AD investigations but only three anti-subsidy investigations. Moreover, as the EC points out in its latest annual report on TDI activities, China initiated three trade defence cases that may have been 'motivated by retaliation against the EU's [...] anti-subsidy investigation into BEVs' (EC, 2025a, p.2).

Another issue is the potential circumvention of TDIs. To avoid circumvention, measures are often extended to countries through which the products are shipped or where minor assembly operations take place. That said, in cases of overcapacity, anti-circumvention measures should be carefully calibrated, as they otherwise risk negatively affecting third countries that are not responsible for the overcapacity issue (Crochet & Zhou, 2023). Moreover, as highlighted by Leichthammer (2025) in an assessment of the application of TDIs to (Chinese) overcapacity, 'anti-circumvention investigations require detailed evidence on production processes, supply chains, and trade flows, which can be difficult to collect and verify'. The EU's Import Surveillance Task Force, which monitors trade diversion, collects important trade data, but, in its current form, cannot provide knowledge of the ownership of intermediary companies, component suppliers, or sub-tier manufacturing processes. Collecting such information (e.g., by obliging importers to disclose it) would require a legal basis.

In sum, while generally applicable measures may be politically less sensitive, less resource-intensive relative to the achieved effects, and less prone to circumvention, targeted measures will often entail a lower risk of negative spill-over effects and allow for a more focused allocation of resources to sectors actually affected by overcapacity.

Temporal limitations: *pro futuro* effect of TDIs

Importantly, TDIs were not designed to force third countries to adapt or change their domestic policies, but to remove the injurious effects of such policies on the domestic market of the importing countries. With a view to tackling structural overcapacity issues, major limitations of existing TDIs therefore result from their imposition at a rather late stage. While TDIs – in line with

their objective – can effectively reduce the volume of imports, they are not suited to address the root causes of such overcapacity. In this regard, it should be noted that systemic subsidisation and other market-distorting state practices often play a major role in the gradual build-up of structural overcapacity before it becomes a trade issue. The imposition of TDIs, by contrast, requires that an injury has already occurred or that the threat of such an injury is imminent. Importantly, such a threat must be based on facts, and 'not merely on allegation, conjecture or remote possibility' (cf., for example, Art. 8(8) EU Anti-Subsidy Regulation). Thus, trade remedies only become amenable at a stage, when industrial overcapacity has already developed. Moreover, trade remedies principally apply *pro futuro*.[46] This remedy gap, which is further exacerbated by lengthy proceedings and only marginally mitigated by the possibility of applying provisional measures,[47] means that trade remedies can address some symptoms, but not the structural causes of overcapacity. Taking the case of subsidies as an example, this is because even the removal of the adverse effects of the subsidy *pro futuro* will 'often [do] little to dismantle the capacity that has been built to produce the subsidised goods' (Hillman & Manak, 2023).

Substantial limitations: TDIs as purely defensive measures

Finally, as the name suggests, TDIs are purely defensive measures. However, as just mentioned, trade distortions are a symptom, not the cause, of overcapacities. Trade remedies are therefore a 'band-aid', rather than a 'solution' or long-term strategy when it comes to related issues such as competitiveness or technological leadership. In other words, trade remedies may be most effective when paired with active industrial and R&D policies. It has been pointed out that 'the EU's protective capacity is only as strong as the domestic institutions that complement it' (Ban & Liu, 2025, p. 5). Beyond 'external protection', European companies' competitiveness, especially in emerging technologies, also depends on 'internal protection' such as innovation policy, start-up support, loan guarantees for scaling up, etc. (Ban & Liu, 2025).

In this context, it is important to note that — even if only *pro futuro* — trade remedies can in principle be applied over longer periods of time. While safeguards cannot be extended beyond a maximum of eight years, anti-dumping measures and CVDs can be renewed as long as dumping/subsidy, injury and causality continue to exist (Art. 11 ADA; Art. 11 EU Anti-dumping Regulation; Art. 21.3 ASCM; Art. 17 EU Anti-Subsidy Regulation). But even though certain TDIs may be employed long-term, this does not mean they create the structural conditions for effectively tackling overcapacity issues. In fact, critics argue that TDIs are often used to shield existing European industries from competition rather than to incentivise them to become more competitive. For example, some European car brands, which profit from countervailing duties on BEVs from China, would push to keep producing combustion engines rather than BEVs because of higher margins in the short term (Cohen & Nelson, 2025). While the European car industry has maintained high profits, investments in industry transformation have decreased (Merk, González, Rodrigo, & Boris, 2025).

3.2.3 Other measures to address overcapacity issues

Beyond traditional trade defence instruments, there are options that WTO members have either relied on to tackle overcapacity issues or that are being discussed as potential remedies. At the WTO level, these include the modification of schedules (Art. XXVIII GATT) — which the EC pursues

[46] Both the EU Anti-Dumping and the Anti-Subsidy Regulations allow for strictly limited exceptions, where measures apply retroactively (Art. 16(4) EU Anti-Subsidy Regulation; Art. 10(4) EU Anti-Dumping Regulation). However, in both cases, retroactive measures may only cover products imported 90 days prior to the application of provisional measures and are therefore of limited impact.

[47] For the maximum duration of such provisional measures see Art. 12(5) EU Anti-Subsidy Regulation; Art. 7(6) EU Anti-Dumping Regulation; and Art. 7(1) EU Safeguards Regulation. On the automatic registration of imports of products under investigation see above; on room for potential improvement, cf. AEGIS (2025, pp. 18-19).

in connection with the proposed steel measure (EC, 2025e, see below) —, the national security exception (Art. XXI(b)(iii) GATT) and in the particular case of China, the use of special provisions in the Chinese Accession Protocol. Some also consider local content and/or technology transfer requirements for Chinese investments in the EU as potential means to further EU interests and strategic autonomy (Goreichy, Gunter, & Stec, 2025). However, it must be noted that local content requirements are largely incompatible with WTO law and will therefore not be discussed in the following section.[48] At the EU level, several novel instruments could be applied in certain situations of overcapacity. These include the proposed steel measure as well as the Foreign Subsidies Regulation, the FDI Screening Regulation, and the International Procurement Instrument.

Options at WTO level

Modification of schedules (Art. XXVIII GATT)

Art. XXVIII GATT allows WTO members to modify or withdraw tariff concessions. A recent example of the invocation of Art. XXVIII is the EC's proposal for the steel measure (see below), which relies on a renegotiation of the EU's bound tariffs in the steel sector to ensure the compatibility of the planned measure with its WTO obligations (EC, 2025e, recital 14).[49] In principle, the applicant WTO member must negotiate and agree on such modifications with the other 'contracting parties primarily concerned' (Art. XXVIII:1 GATT). That said, even if no agreement is reached, the applicant member can modify or withdraw the concessions at issue. In such a case, any party with initial negotiation rights, a principal supplying interest[50] or a substantial interest[51] can withdraw 'substantially equivalent concessions initially negotiated with the applicant contracting party' (Art. XXVIII:3(a) GATT).

Importantly, renegotiations of concessions were 'conceived as a tool to be used by Members in the ordinary run of multilateral trade, and not under the pressure of special or emergency domestic circumstances' (Fabbricotti, 2011, para. 3). Thus, compared to safeguard measures, the modification of schedules has two significant advantages: for one, it is not subject to a maximum duration and thus can be applied long-term; for another, it does not require the same exacting justification, as no injury to domestic production or threat thereof are needed.[52]

However, especially substantial modifications also come with considerable downsides. First, higher tariffs are likely to increase costs for EU companies that rely on the affected goods as inputs for their own production. Second, since the modification of schedules applies *erga omnes*, it does not provide a targeted measure to address specific overcapacity issues. Third, and relatedly, the strategy can contribute to an ever-fiercer race to the bottom, as 'compensation claims will be substantial and negotiations difficult' (García Bercero, 2025). Moreover, the prevailing view is that compensatory measures taken by other parties would in turn also apply *erga omnes* (Fabbricotti,

[48] Plain local content requirements are explicitly considered incompatible with the national treatment obligation under Art. III:4 GATT, see Art. 2.2 and the illustrative list in the Annex of the Agreement on Trade-Related Investment Measures (TRIMs) (cf. Sauvé, 2016). In contrast, local procurement requirements may under certain strict conditions be compatible with WTO law, cf. *Canada–Renewable Energy*, WT/DS412/AB/R, WT/DS426/AB/R. See also (Hestermeyer & Nielsen, 2014). Technology transfer requirements may raise issues under the General Agreement on Trade in Services (GATS) (depending on members specific commitments) and must be consistent the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS).

[49] It is noteworthy that – while renegotiations are, in principle, subject to strict time limits – the EU is among the WTO Members that have made reservations, allowing them to open such negotiations essentially at any time.

[50] The WTO Member with the highest rate of exports of the product affected by the concession; see in detail Fabbricotti (2011) paras 20 ff.

[51] Members with a 10 % market share of the products subject to the concessions are assumed to have substantial interest, see WTO (2024) para 21.

[52] Note how, for example, in the *EU–Safeguard Measures on certain steel products* Case, WT/DS595/R, the panel took up only criticism of the justification of measures, not the conceptualisation of the measures.

2011) and could therefore trigger further compensatory measures by other WTO members (*Canada/European Communities — Article XXVIII Rights*, DS12/R, Award by the Arbitrator, p. 6). While some of these effects can be mitigated by preferential bi- or plurilateral trade agreements, it still appears prudent to use modifications of schedules cautiously and only in strategic sectors where (1) the EU can meet demand without undermining broader policy goals, for example, decarbonisation, whether through domestic production or preferential trade, and (2) overcapacity truly poses a critical issue for European industries (Leichthammer, 2025).

National security exception (Art. XXI(b)(iii) GATT)

In exceptional circumstances, the EU could moreover seek to rely on the national security exception in Art. XXI(b)(iii) GATT, which has been increasingly invoked in recent years, not least by the US (Hufbauer 2024).[53]

However, it should be noted that Art. XXI GATT does not automatically immunise members' actions from scrutiny by WTO dispute settlement. First, the national security exception is not entirely 'self-judging' (WT/DS512 R para 7.102). Members are bound to apply the exception in good faith, which not only precludes a simple 're-labelling' of trade as security interests, but also requires the measures at issue to 'meet a minimum requirement of plausibility in relation to the proffered essential security interests' (ibid. paras 7.132 ff). Thus, when relying on Art. XXI GATT to justify measures regarding overcapacity issues, a WTO member would not only have to substantiate the existence of an emergency in international relations — a criterion panels have interpreted strictly[54] — but also sufficiently clearly and plausibly articulate the essential security interests arising from such an emergency. Second, it is contested whether Art. XXI GATT applies outside the GATT. While some other WTO agreements contain similar clauses (for example, Art. XIV bis GATS; Art. 73 TRIPS), neither the ASCM nor the ADA contain any reference to national security considerations. Whether states could rely on Art. XXI to justify measures inconsistent with those agreements remains an open question.[55] However, neither the wording of the provision ('*Nothing in this Agreement* shall be construed…', emphasis added), nor WTO case law on the general exceptions clause in Art. XX GATT readily suggests such an expansive reading (van den Bossche & Akpofure, 2020).[56] In sum, the national security exception in Art. XXI GATT appears to offer, at best, very limited room for justifying overcapacity-related restrictive unilateral measures.

China's commitments in the WTO Accession Protocol on SOEs

China's Accession Protocol (AP) and the Report of the Working Party on the Accession of China[57] *inter alia* set out specific obligations for China to ensure that its state trading enterprises are not

[53] Cf. Trachtenberg (2025) p. 1: Investigations under Section 301 of the Trade Act 1974 'may occur if the rights of the United States under any trade agreement are being denied, or an act, policy, or practice of a foreign government (1) violates, is inconsistent with, or denies benefits to the United States under a trade agreement; or (2) is 'unjustifiable' and 'burdens or restricts' U.S. commerce'. The WTO compatibility of Sections 301-310 leg. cit. was challenged—and ultimately upheld in a 1999 panel decision. However, the panel stated that WTO law clearly prohibits 'specific instances of unilateral conduct by WTO Members when they seek redress for WTO inconsistencies in any given dispute' (WT/DS 152 R, para 7.39).

[54] Cf. *US – Origin Marking Requirements* WT/DS597/R (appealed 'into the void') para. 7.289 ('truly grave character').

[55] NB. in *US – Certain Tax Credits Under the Inflation Reduction Act* WT/DS623 argue that the GATT exception applies in the context of the TRIMs and ASCM, cf. https://ustr.gov/sites/default/files/enforcement/DS/DS623/US.Sub1.(21Mar25).fin.pdf.

[56] The applicability of Art. XX GATT in the context of the ASCM and TRIMs was deliberately left open in the January 2026 report of the panel in *United States – Certain Tax Credits Under the Inflation Reduction Act*, WT/DS623/R, para. 7.208. The applicability of Art. XXI GATT as claimed by the US in an earlier written submission was not addressed in the panel report.

[57] Section 1.2 of China's Accession Protocol stipulates that the commitments referred to in paragraph 342 of the Working Party Report shall be an integral part of the WTO Agreement. Para. 342 of the Working Party Report lists among others the commitments of para. 46 that this study refers to. The commitments here outlined are therefore legally binding.

influenced by the state as to the 'quantity, value, or country of origin of goods purchased or sold' (Sect. 6.1 AP, WT/L/432), and that purchases and sales of SOEs and state-invested enterprises must be 'based solely on commercial considerations, e.g. price, quality, marketability and availability' (para. 46, Report of the Working Party on the Accession of China, 2001). It has been argued that in addition to existing TDIs these provisions would 'provide abundant flexibility for WTO Members to challenge the non-commercial activities of Chinese SOEs' (Gao & Zhou, 2022, p. 100). However, the exact scope of these obligations currently remains unclear. For one, the provisions leave ample room for interpretation, e.g. concerning the question of what exactly qualifies as 'commercial considerations'; for another there is no case law, since WTO members have not yet relied on these provisions to challenge activities of Chinese SOEs (Gao & Zhou, 2022). Thus, their effectiveness in complementing TDIs to address unfair trading practices for now remains unclear.

Specifically, with a view to overcapacity issues, Sect. 6.1 AP could indeed provide a legal basis for challenging certain practices of Chinese SOEs. This is particularly true for SOEs 'selling input to selected downstream industries for less than adequate remuneration' (Gao & Zhou, 2022, p. 124). Importantly, Sect. 6.1 AP would allow the EU and other WTO members to address structural overcapacity issues at an early stage, and thus before they materialize in increased exports posing a risk of injury to their markets. That said, meeting the necessary evidentiary requirements could pose a significant practical hurdle to the successful invocation of Sect. 6.1 AP.

Options at EU level

The proposed steel measure

In 2025, the EC proposed a new regulation addressing the negative trade-related effects of global overcapacity on the Union steel market (EC, 2025e) (subsequently: 'proposal' or 'steel measure'). While the proposal has not been adopted, it is an example of how the EU seeks to directly target overcapacity issues in critical sectors. Thus, we will discuss the steel measure before turning to existing EU instruments that may be relied on to deal with overcapacity issues more indirectly.

The steel measure aims to replace the steel safeguards which will expire in mid-2026 after eight years in force. As far as relevant here, the proposal — similarly to the steel safeguards — provides for tariff-free import quotas and the level of duty applicable beyond these quotas. That said, the steel measure would not only cut the volume of tariff-free quotas almost in half (minus 47 % compared to 2024 steel quotas) but also double the level of out-of-quota duty to 50 % (compared to 25 % under the steel safeguards) (Art.1 of the proposal) (EC, 2025f). Moreover, the proposal seeks to strengthen transparency and prevent circumvention by introducing a 'melt and pour' requirement (Art. 3 of the proposal).

As mentioned above, to ensure WTO compatibility, the internal (ordinary) legislative procedure is to be accompanied by a modification the EU's bound tariffs according to Article XXVIII GATT. While, in principle, this seems to be a feasible approach, some commentators have warned that in particular the implementation of the steel measure could raise issues in terms of most favoured nation treatment (Art. I:1 GATT) or the non-discriminatory administration of quotas (Art. XIII GATT) (Crochet and Zhou, 2025). The proposal tasks the EC with laying down the country allocation of the tariff quotas by way of implementing acts (Art. 4 of the proposal). Thus, it will be crucial that the EC exercises this power in conformity with WTO law.

Finally, it has been suggested that the steel measure should be complemented by a broader new instrument to deal with overcapacities (AEGIS, 2025, pp. 8–9). In our view, the steel measure can, in principle, serve as a model for similar measures, which could strategically provide protection in specific sectors in which overcapacity issues pose a serious threat to EU industries. By contrast, a

horizontal measure, which would target overcapacities across sectors seems more problematic. First, 'overcapacity' is a complex phenomenon, not a legal concept. Any legal definition of 'overcapacity' would therefore have to ensure that such an instrument applies in a non-arbitrary and non-discriminatory manner. More importantly, perhaps, a horizontal measure would require far-reaching modifications of a wide array of current EU GATT commitments. This would not only drastically raise the prices of imports but likely also lead to large-scale retaliatory actions by trading partners, including similar modifications under Art. XXVIII GATT, and therefore also affect EU exports in an unpredictable manner. Against this background, a more strategic, sector-specific approach that combines defensive measures with pro-active industrial and R&D policies appears better suited to protect EU competitiveness.

EU Foreign Subsidies Regulation

The EU Foreign Subsidies Regulation (Regulation (EU) 2022/2560, hereinafter: FSR) was adopted in 2022. Based on the FSR, the EC can review foreign subsidies and impose 'redressive measures' (Arts. 7(4) and 11(2)) if these subsidies lead to distortions in the internal market (Art. 4 f). The FSR was intended as a response to challenges posed by state-owned and state-sponsored companies from China (Mattiolo, 2023; see also Alfter et al., 2023). While it has not only been applied to Chinese subsidies (for example, Commission, C/2024/3970 concerning subsidies of the UAE in the telecommunication sector) or in sectors where overcapacity is expected (for example, Commission, C/2025/4547 concerning subsidies of the UAE in the chemicals sector), the FSR can be relevant in that context. The very first FSR *ex officio* investigation concerns Chinese manufacturers in the wind turbine sector (Renner, Bahous, & Plotka, 2024), and there will likely be more investigations in sectors where the EU is grappling with Chinese overcapacity (Noerr, 2024). In this regard, it should be noted that overcapacity is considered a factor indicating that a foreign subsidy is likely to cause distortions in the internal market (cf. Recital 19). Since the FSR targets individual undertakings, it is overall more targeted than most traditional trade defence measures. However, its application raises specific legal challenges. One such challenge concerns the finding of a 'distortion in the internal market', resulting from a foreign subsidy granted to a particular undertaking (Akritidis & Blancardi, 2023, p. 383). Another challenge concerns the identification of foreign subsidies, given the frequent lack of transparency and reporting in many third countries (Frank, 2023, p. 936).

In the Clean Industrial Deal, the EC announced its intention to make use of the FSR in 'strategic sectors', but did not specify what these strategic sectors are (EC, 2025d). Conflicts with third countries could particularly arise in sectors considered security-sensitive. One of the earliest investigations conducted by the EC under the FSR included ordering the company under investigation (a Polish company with a Chinese parent company that produced security-inspection scanners) to grant the EC access to employees' correspondence stored on a server in China. In 2024, the President of the General Court dismissed the company's application for interim measures to suspend the operation of the EC's decision (T-284/24 R, appeal pending). The case not only illustrates the far-reaching impacts that FSR investigations can have but has also drawn fierce criticism from the China Chamber of Commerce, further straining an already difficult relationship (CCCEU, 2024). Moreover, while principally 'defensive', the FSR has come under scrutiny, as its application often coincides with more 'offensive' measures, such as the increased use of state aid by Member States in affected sectors (Blockx & Mattiolo, 2023).

Finally, it should be mentioned that the compatibility of the FSR with WTO law is much debated. Potential issues concern the EC's broad mandate to act *ex officio* (which Art. 11.6 ASCM allows only in specific circumstances) (Akritidis & Blancardi, 2023, p. 353); the fact that redressive measures may go beyond the scope of Art. 32.1 ACSM (despite a provision stating the contrary, cf. Art. 44(9) FSR) without being justified under Art. XX GATT (Crochet & Gustafsson, 2020); and breaches of

relevant WTO standards in relation to subsidies concerning services (Crochet & Gustafsson, 2020; disagreeing: Nagy, 2023; implicitly disagreeing: Frank, 2023).

Investment screening

A more indirect means of redress is the screening of foreign direct investment (FDI) inflows into the EU. Twenty-five EU Member States already have mechanisms in place that allow them to assess potential threats posed by FDI inflows to their security or public order and to impose mitigation measures or, in extreme cases, prohibit a transaction. At the EU level, the current FDI Screening Regulation (Regulation (EU) 2019/452) primarily coordinates these efforts through information sharing. That said, in December 2025, a provisional political agreement was reached on a revised EU FDI Screening Regulation (Council of the EU, 2025). While the agreed text has not yet been published, the revised regulation will mandate that all EU Member States screen FDI in key sectors, including artificial intelligence (AI), semiconductors and quantum technologies (Council of the EU, 2025). Member States, however, remain free also to screen FDI in other sensitive areas. Moreover, possible risks may be investment or investor-related (Mayr, 2025). Investor-related risks can particularly arise where foreign investors are owned or controlled by a third country, including through significant funding (cf. EC, 2024b). Similar to the FSR, this instrument is not geared to react directly to overcapacities in third countries; taken together, however, the employment of these instruments can prevent investments in Member States by investors that contribute to overcapacities.

International Procurement Instrument (IPI)

The International Procurement Instrument (Regulation (EU) 2022/1031, hereinafter, IPI) came into force in 2022 and pursues a twofold purpose: (1) to improve EU access to procurement markets in third countries; and (2), where impairments exist abroad, empower the EC to impose restrictive measures regarding the access of economic operators, goods or services from third countries to EU public procurement procedures (Art. 1(1)). Importantly, the IPI only applies to third countries in the absence of existing international obligations (Art. 1(3)). This means the IPI applies to China, which has not (yet) acceded to the WTO Agreement on Government Procurement (GPA).

Under the IPI, the EC may conduct investigations into alleged third-country measures or practices on its own initiative or upon a substantiated complaint of a Union interested party or a Member State (Art. 5). If an issue cannot be resolved through consultations with the third country concerned, the EC may adopt an 'IPI measure' prescribing score adjustments or excluding tenders by economic operators from that country (Art. 6). Importantly, an IPI measure may be adopted only if the EC deems it in the interest of the Union.[58] While this means that the EC enjoys a wide degree of discretion, it must explicitly consider the availability of alternative sources to minimise negative impacts on contracting authorities as well as the proportionality of the measure (Art. 6(3)).

With a view to overcapacity issues, the IPI could provide another indirect means of redress. The regulation defines 'third-country measure or practice' broadly as 'any legislative, regulatory or administrative measure, procedure or practice, or combination thereof, adopted or maintained by public authorities or individual contracting authorities or contracting entities in a third country, at any level, that results in a serious and recurrent impairment of access of Union economic operators, goods or services to the public procurement or concession markets of that third country'. Arguably, this definition would cover, for example, economic advantages (regardless of their qualification as 'subsidy') granted to overcapacity-producing economic operators which provide goods for domestic and international markets. While the IPI does not apply to or aim to restrict purchases by

[58] The EC's determination on whether adopting an IPI measure is in the EU's interest 'shall be based on an appreciation of all the various interests taken as a whole, including the interests of the Union's economic operators' (Art. 6(2)).

private operators, it should be noted that public procurement accounts for approximately 15 % of the EU's GDP (EC, 2025b), underscoring the potential economic significance of IPI measures.

3.3 EU cooperation with third countries in tackling overcapacities

3.3.1 Cooperation with China

Dialogue with China

As shown in section 3.1., China is well aware of its potential overcapacity, albeit calling it by a different name (Davis, 2025; Meltzer & Pearson, 2024). The EU has framed overcapacities as relating both to industries and SOEs, and thus, implicitly, as a Chinese issue (Mattiolo, 2023). China, by contrast, has been 'repackaging its efforts as fighting involution rather than excess capacity', so that 'Beijing can argue it isn't caving in to Western pressure', which a Chinese Embassy spokesman described as 'naked economic coercion and bullying' (Davis, 2025).

Framing overcapacities as an exclusively Chinese phenomenon, however, would do injustice to the reality of a global competition for technology leadership that affects sectors such as battery production or the semiconductor industry. Other countries, including the US and India, have also pledged billions in subsidies for these sectors (Maurer, Ahmad, & Hagemejer, 2025).

If the EU wants to promote cooperation with China, the EU should acknowledge the Chinese framing of the problem as one of 'disorderly competition' or 'involution' rather than 'excess capacity' or 'overcapacity'. Also, the state of research on overcapacity in China indicates that it is not a phenomenon that Beijing can get rid of overnight, even if it wanted to, but an externality of the Chinese economic model.

Instead, there must be room for negotiations. It is of mutual interest to China and Europe to deepen channels of communication, as both are interested in access to each other's markets and are to a certain degree dependent on imports from each other. The most recent China-EU Summits have been characterised by long lists of EU demands and very few productive outcomes (García-Herrero, 2023; Garlick et al., 2025). Two issues can be identified as hurdles in past attempts to engage in dialogue:

On the one hand, the EU has tried to 'solve overcapacity' in rather general terms, rather than focusing on the central sectors where Chinese overcapacity actually harms the EU (Goreichy et al., 2025). Considering industrial policy measures that encourage cooperation with China in specific sectors, as detailed below, could give the EU leverage in future negotiations with China.

On the other hand, trade negotiations have increasingly been overshadowed by the (geo-)political divide (Garlick et al., 2025). One example is the failure to conclude the EU-China Comprehensive Agreement on Investment. In 2020, the EU and China had reached an agreement in principle. Soon after, the EU enacted sanctions against Chinese officials due to human rights abuses against China's Uyghur minority. When China responded with counter-sanctions against Members of the European Parliament (MEPs) and several European institutions, the European Parliament adopted a resolution stating that it would not consider the agreement until Chinese counter-sanctions were lifted, thus putting the agreement on hold (European Parliament, 2021). Moreover, with increasing tensions between Russia and Europe, the EU has become warier of China's close ties to Russia, while China is wary of Europe's loyalty towards the US. Negotiations have focused on Russia's full-scale invasion of Ukraine, and when the war in Ukraine eventually ends, Europe might find the momentum to prioritise its trade relationship with China. Given the current trade and geopolitical tensions with the US, attempts to intensify cooperation with China could present an opportunity for economic diversification (Goreichy et al., 2025).

When negotiations have focused on specific issues, EU-China negotiations have sometimes yielded results: Negotiations between the EC and China's Ministry of Commerce led to the EC recently issuing a Guidance Document on the submission of price undertaking offers for Chinese BEV exporters (Directorate-General for Trade and Economic Security, 2026). By agreeing to sell above a minimum price, Chinese BEV exporters may enter the European market without paying the countervailing duties imposed on BEVs from China. The news has been seen as an improvement on countervailing duties for industrial actors in both China and Europe, as well as for consumers (Khan, 2026; Lemaire, 2026). However, it should be borne in mind that this result does not change anything about industrial or broader economic policies underlying overcapacity. It is just a change in how the trade defence measure may be applied. The subsidy practices of China that form the basis of the EU's employment of countervailing duties are not changed by this switch from countervailing duties to voluntary price undertakings.

The recently concluded strategic partnership between China and Canada also suggests that negotiations with China eventually can lead to win-win outcomes. Focused primarily on collaborating in energy, clean technology, and climate competitiveness, including through granting a tariff-rate quota for Chinese EV imports in the hope of incentivising Chinese FDI in these sectors in Canada, the agreement establishes cooperation in public safety and security, as well as other fields such as tourism and culture (Government of Canada, 2026). Whatever the specific motivations at play, the agreement suggests that trade cooperation with China remains possible. However, China is not lenient in negotiations and critics warn that the EU's economic interests are considerably different from Canada's, and potentially more in conflict with Chinese interests (García-Herrero, 2026). In particular, as China cements its dominance in emerging technologies, its trade policies become more assertive (Sutton & Williams, 2025). This makes trade cooperation with China more difficult for the EU, though not impossible.

Cooperation in industrial development

Overcapacity, as explained in section 2, is regarded as a problem because it lowers prices, thereby also lowering profit margins for (international) competitors, eventually driving them out of business. However, there are sectors where lower prices are beneficial for European buyers, for example, in cases where Europe has not established a profitable industry in the first place. In addition, there are cases of positive externalities in some fields characterised by overcapacity, especially in green technologies such as wind turbines, solar panels, and EVs. Against this background, it has been suggested that a possible cooperative approach is to actively foster Chinese investment in Europe in these fields (Solheim, 2025). However, Chinese FDI in the EU may not only raise security concerns (see FDI screening above) but also be seen as a divisive instrument, pitting some Member States against others (Martinez, 2025). That said, some third countries such as India, Brazil, Turkey, or Zimbabwe, struggling to absorb excess products from China have found ways to use Chinese investments in their favour, for example, by imposing local content, joint venture or technology transfer requirements (Adachi, 2025; Godement, 2025). It should be noted, however, that such requirements raise several issues under WTO law. The fact that China has been lenient towards developing countries employing such measures in no way indicates that China would not sue the EU in the WTO over comparable measures (Soong & Gunter, 2024).

3.3.2 Multilateral cooperation

Taking Chinese overcapacity seriously may mean understanding that 'the 'shadow of scale' is not a uniquely Chinese challenge, but a global governance imperative' (Watanabe, 2025). However, countries worldwide affected by Chinese overcapacity have developed various uncoordinated responses (Soong & Gunter, 2024).

Multilateral trading system

In addition to its dispute-settlement system to ensure compliance, the WTO provides several fora for discussion on matters concerning international trade. In the context of overcapacity, important fora include the Committee on Anti-Dumping Practices (including the Working Group on Implementation and the Informal Group on Anti-Circumvention); the Committee on Subsidies and Countervailing Measures; and the Committee on Safeguards. However, the effectiveness of these fora depends on members transparently notifying their policies, draft laws, and, most crucially, subsidies — something that rarely happens (WTO News, 2018).

While the effectiveness of the WTO has suffered in recent years (in particular, since the Appellate Body crisis) and many recognise the need to reform the WTO system, also to better tackle overcapacity issues (Meltzer & Pearson, 2024; Zhou et al., 2025), politically, any major reform currently seems highly unlikely. Even many countries that share the EU's desire to maintain the WTO system view the current multilateral order with 'deep reservations [...] with respect to the biased structure of the current regime' (Raza & Tröster, 2025). China appears committed to maintaining certain aspects of the WTO system, as illustrated by its accession to the Multilateral Interim Appeal Arbitration Arrangement. At the same time, it has no incentive to change the lax WTO subsidy regime, which benefits it in various ways.[59] The US, by contrast, has advocated reform of the WTO's subsidies regime alongside the EU and Japan, but has single-handedly paralysed the WTO Appellate Body since 2019, undermining members' ability to enforce their rights under WTO law (EPRS, 2024).

Cooperation beyond the WTO

Especially in the US context, many ideas for overcoming overcapacity issues through cooperation have been discussed (Campbell & Doshi, 2025; Meltzer & Pearson, 2024; Rühlig, 2025). A crucial insight in this regard has been that China's main advantage is scale, which it leverages both through the size of its domestic market and its pro-active economic policy approach (Campbell & Doshi, 2025). If other countries want to leverage similar advantages, they must cooperate by liberalising trade and investment between themselves, going beyond the liberalisations agreed upon in the WTO (so-called *WTO-plus* model; Campbell & Doshi, 2025; Rühlig, 2025). However, in practice, the conclusion of free trade agreements (with and without investment chapters) by the EU has been sluggish,[60] and hopes of more transatlantic cooperation to 'beat' overcapacity have fizzled away under the US's preference for unilateral action (Rühlig, 2025).

In few other industries has the problem of overcapacity been so clearly observed and collectively addressed (within the OECD) as in the steel sector. This sector thus serves as a 'learning ground' for the opportunities of collective action beyond the WTO. The example of the Global Forum on Excess Steel Capacity demonstrates that cooperative approaches enable better data and policy sharing, as well as the option to coordinate policies. However, they also entail risks, namely, further

[59] The ASCM puts high evidentiary burdens on countries aiming to take countervailing measures. This makes procedures against countries that do not notify their subsidies (such as China: Frank, 2023, p. 936) lengthy and difficult. Furthermore, China benefits of the narrow understanding of 'public body' that has emerged in WTO jurisprudence. The remedies foreseen in the ASCM also crucially can only stop excess imports due to subsidies, but not undo the technological advance made possible by subsidies that benefits the economy providing subsidies even after the subsidies are no longer used.
[60] On the one hand, the EU has recently reached political agreements on new free trade agreements with Indonesia and India (EC, 2026, undated a). One the other hand, some negotiations proved difficult (see the negotiations with Australia that only recently reached a turning point: European Parliament, 2026b; McIlroy, 2026), and other times, the adoption and ratification process turned out to be very lengthy (see the halted conclusion process of the free trade agreement with Mercosur after more than two decades of negotiations: European Parliament, 2026).

alienating countries not at the table and overlooking policy solutions outside the frame of overcapacity.

Within the Global Forum on Excess Steel Capacity, countries were willing to share data on production and their policies to a much larger degree than within the technical committees of the WTO. This can be attributed to better framework conditions (for example, assurances that the data will not be used in WTO trade disputes), a clear benefit of this cooperative approach. However, states are still reluctant to report protectionist measures or subsidies (Evenett & Fritz, 2018). Beyond information sharing, the Global Forum formulated a list of policy recommendations (Global Forum on Steel Excess Capacity, 2017), the tenor of which is that governments must stop using subsidies and other market-distorting practices to enable overcapacity, and instead let the market take its 'natural' course. In later reports, the Global Forum claimed that its members adhere to these recommendations and lamented the lack of efforts of non-member states as the reason for the continued malaise of the steel industry (Global Forum on Steel Excess Capacity, 2021, para. 53f). However, other reasons for the struggling steel industry in the G20 member states of the Global Forum have either not been given serious consideration (Global Forum on Steel Excess Capacity, 2021, para 38 only mentioning, but not further analysing issues of lacking industry investments in innovation even in times of profitability) or have been ignored entirely (for example, high energy costs in many member countries). As a trend reversal has not yet been achieved (see Global Forum on Steel Excess Capacity, 2023), experts expect the Forum to agree on further protectionist measures in 2026, when the Forum will be chaired by the US (Foster, Bounds, & Gridneff, 2025).

In this regard, however, two important aspects should not be overlooked: For one, calls to 'team up against economic threats' — as democracies do in response to security threats, such as, for example, those posed in the wake of Russia's invasion of Ukraine (Evenett, quoted in Foster et al., 2025) — risk further undermining international law. The scope and application of legal tools such as countermeasures or TDIs are subject to strict conditions. While states may impose a wide range of sanctions in response to severe breaches of international law, such as a violation of the prohibition on the use of force, their options are much more limited in the case of a country subsidising, for example, its steel industry. In the latter case, if they go beyond the remedies provided for by WTO law (Art. 32.1 ASCM), they violate the multilateral order they seek to maintain.

For another, the harsher the TDIs that countries adopt in response to overcapacity, the harsher the economic effects on third countries may be due to trade diversion. In other words, countries must either be able to 'sustain a trade war with China' or see 'their interests [...] become collateral damage' (Zhou et al., 2025). The United Kingdom, caught between the high tariffs on steel imposed by the US and the EU' proposed new steel measure has proposed 'a Western steel alliance — potentially including Washington — that would align tariff policies and grant members preferential tariffs on steel trade' (Hug, Stone, & Lanktree, 2025). However, there are many other countries whose steel industries are similarly affected by global overcapacities and TDIs, which find themselves in a more difficult position vis-à-vis the EU, especially if they do not have any preferential trade agreements with the EU in place (cf. non-OECD countries with steel production mentioned in Global Forum on Steel Excess Capacity, 2021, e.g., Pakistan, Iran or Malaysia). Thus, if the EU is serious about seeking new like-minded partners to work on the multilateral trading system, it should not only keep the doors open for countries seeking to trade freely with the EU but also strive for cooperation in areas beyond trade policy. For example, the Global Forum's reports show that, in many countries worldwide, global steel production capacity is rising due to investments (and sometimes foreign subsidies) from China (Global Forum on Steel Excess Capacity, 2021). If Europe wants free trade, without these investments potentially distorting trade, there needs to be greater international cooperation on competition policy, including 'establishing common principles for cross-border mergers and acquisitions' (Watanabe, 2025).

Lastly, it should be noted that unilateral efforts to tackle overcapacity will often focus on industries that are politically sensitive and strategically important' (Zhou et al., 2025). A case in point is the steel industry. The 2025 European Steel and Metals Action Plan explicitly highlights the importance of steel production for military applications (EC, 2025c). In short, being able to produce steel domestically is a matter of national security. From this perspective, the 'goalposts' shift: from ensuring a level playing field in the steel market — for which the eradication of overcapacity, including the enabling subsidies, seems of utmost importance — to ensuring sufficient domestic production capacities for security purposes. With this shift of perspective, other policy options come into focus. One such option could be reviving the category of 'non-actionable' subsidies at the WTO level. Creating a carve-out from WTO disciplines for subsidies serving specific public policy goals, such as R&D, environmental protection, or national security, could be of interest to members far beyond Europe. Indeed, a major reason for China to tolerate the inefficiencies of its current industrial policies is the belief that 'technological security' is essential; thus, perhaps even China could be on board with such a proposal (Goreichy et al., 2025).

4 Main conclusions

4.1 Opportunities and limits of strategic responses

Overcapacity should be understood as a dynamic concept that is related to both structural economic factors and policy decisions. The emergence of overcapacity and its related effects depend on the specific context, including sector- and product-specific characteristics, demand dynamics, and institutional factors. What is more, overcapacities in one country only become a problem for other countries if the reduction of unused capacities is not pursued domestically but through exports to other countries.

To the extent that overcapacities have emerged in China, they are largely the result of competition between Chinese government entities at the regional and local levels to build manufacturing capacity. This race to build capacity is in line with targets set out in Chinese government strategies to promote self-sufficiency and leadership in advanced technologies, amid US and allied policies (including those of the EU) impeding China's access to advanced technologies from the West. So far as the rapid build-up of capacity has exceeded domestic demand, adjustment has primarily occurred through inventory buffers and lower profit margins as a consequence of price competition between companies (see section 2.2). Higher exports have appeared across all Chinese industrial sectors, although our analysis suggests that exports have not been the dominant adjustment channel, except for six specific sectors (medicines, rubber & plastics, general machinery, special machinery, automobiles, and electrical machinery). The conclusion that a deliberate objective of these policies was to build up overcapacity to flood international markets, however, is not supported by our findings.

Though our analysis of the recent policy debate within China confirms that Chinese authorities have recognised the issue and have started to implement policies to combat overcapacity under the concept of 'involution', the effectiveness of these policies remains to be seen. Thus, the EU should remain cautious and prepare itself against eventual overcapacity events in the near to medium-term future, for instance as the current absorption of surplus production in inventories might face limits.

Moreover, to the extent that anti-involution policies prove effective in curtailing overcapacity, further technological upgrading amongst Chinese companies should be expected to accelerate. Thus, EU companies will likely face stronger competition from Chinese firms in higher-tech segments of the industrial sectors in which they currently still have a technological lead, e.g. in industrial robotics. In other sectors, such as batteries for EVs and other applications, China already

gained technological leadership due to the successful policy-supported build-up of a sophisticated industrial ecosystem that combines a host of productive and institutional capacities and capabilities. Regardless of the issue of overcapacity, economic and trade policy relations with China will thus have to be assessed in the broader context of intensifying competitive pressures.

Furthermore, price competitiveness has been negatively affected by exchange rate movements. The EU experienced an increase in producer costs of some 35 %since 2022, while producer prices remained flat overall in China, also as a consequence of domestic overcapacities and related price competition. Given these increasing producer price differentials,-and a roughly unchanged nominal exchange rate between the EUR and the CNY since 2020, the EU suffered from a real appreciation of roughly 40 % between 2020 and 2025 vis-à-vis China. Though often overlooked, this is an important element in explaining the sharp increase-of the European trade deficit with China in 2024, which roughly doubled relative to the average of the period 2015-2019 (Matthes, 2025). Exchange rate management issues, in particular the deliberate undervaluation of a currency, thus need to be addressed by policy-makers as well.

What is more, although the EU and the US are still the largest and second-largest trading partners, respectively, China has made significant advances in diversifying its trade patterns towards other emerging economies and the Global South more generally. It is already the largest trading partner for 120 countries, and among the top three partners for 157 countries, with trade growth most pronounced with emerging markets, such as ASEAN, and with export growth drivers shifting to high-tech products, including electric vehicles, lithium batteries and solar cells. China today is thus in a much better situation to cope with protectionist tendencies in the US, the EU or other high-income countries. That said, with the US keen to reduce its bilateral trade deficit vis-à-vis China, it is likely that export pressure towards the EU will remain high in the years to come.

Against this background, the EU needs to carefully calibrate its trade policy approach towards China. Essentially, we can distinguish between three options: (1) 'building fortress Europe', i.e. abandoning the current commitment to multilateral trade rules and resorting to protectionist measures on a large scale; (2) 'maintaining the status quo', i.e. adhering to and advocating for a liberal and rules-based multilateral trading regime, and (3) 'inclusive regionalism', i.e. taking an intermediate position involving the tailored and strategic use of protective measures and support for strengthening key EU industries, where necessary, while maintaining adherence to WTO principles and a constructive approach to international rules-based cooperation. All options involve complex trade-offs, while the ultimate outcome is difficult to determine.

Option 1 is to push back against Chinese overcapacity and the growing bilateral trade deficit by resorting to large-scale protectionist measures (tariffs, etc.) to give EU industries 'breathing space', allowing them to regain competitiveness and improve the EU's bargaining position vis-à-vis an increasingly assertive China. While clearly in conflict with the EU's international legal obligations, the political sway of such a strategy would depend on the perceived scale of the 'Chinese threat' and other ongoing geopolitical challenges to the EU's industrial base, its economic security and strategic autonomy. While its effectiveness would depend on a number of assumptions and complementary policies, such large-scale protectionism would clearly entail significant short-term costs. First, as seen in the US-China trade war, China has become increasingly assertive, suggesting significant retaliatory action should be expected. Given EU dependencies and vulnerabilities — for example, critical raw materials (rare earths) and environmental technologies (batteries, PV modules) — bottlenecks in supply chains for EU companies would quickly emerge. Moreover, EU companies with major outlets in China would likely lose access to the large Chinese market, while facing stiff competition from Chinese companies in third-country markets.

Second, whether EU companies would use the 'breathing space' provided to upgrade their business models and — via investment into R&D and innovation — restore competitiveness in the medium to long term, or whether they would adopt a defensive strategy of cost-cutting and downscaling to maintain their short-term profitability, is uncertain. A future-oriented — albeit risky — strategy of industrial transformation would thus be predicated upon a strong and joint commitment between EU governments, the corporate sector and civil society (especially trade unions). In particular, *Option 1* would require large-scale public investment programmes from the EU and member state governments far above current levels, as outlined, for example, in the Draghi Report (Draghi, 2024). To these costs one must add the costs of: (i) a reduction of the economic welfare of the European population, in particular via higher consumer prices, reduced product variety and supply scarcities; (ii) negative political externalities for Europe, as not only its bilateral relations with China, but likely also with other countries negatively affected by EU protectionism would deteriorate, and make international cooperation on multiple fronts more difficult, if not impossible; (iii) a slowing-down of the green transformation in Europe, given its high dependency on importing green products and technologies from China.

Located towards the other end of the spectrum, spanning from protectionism to free and rules-based trade, *Option 2* would seek to preserve compliance with the existing rules-based trading regime without significantly changing the EU's current approach to and role in it. Market access and tariff levels would remain unaltered, TDI, FDI screening and related economic security instruments would be used restrictively. TDIs would be employed with the aim of 'levelling the playing field'. Industrial policy would be focused on horizontal support for R&D and innovation; the fiscal and monetary policy framework would remain largely unchanged. Social and labour market policies would be used to cushion import-related shocks, in particular through worker retraining. By pursuing this strategy, the EU would arguably uphold its reputation as a leading proponent of a liberal rules-based international order, which could facilitate cooperation with international partners, particularly in the Global South, on issues such as WTO reform or the conclusion of new bilateral trade and investment agreements. Relations with China would be primarily based on trade diplomacy and a cooperative approach. While economic inefficiencies due to market distortions would be largely avoided and consumer welfare from imports maintained, the model is vulnerable to supply chain shocks and leaves the EU exposed to the weaponisation of trade by other powers, including the ripple effects of foreign trade wars. Also, adjustment shocks from import surges, cost-cutting, and large-scale layoffs by EU companies would be more pronounced, potentially widening social and regional inequalities. As a result, political repercussions in the form of shifting voting patterns by affected local communities might increase, given recent evidence both from the US and Europe (Autor, Dorn, Hanson, & Majlesi, 2020; Colantone, Ottaviano, & Stanig, 2024; Scheiring, Serrano-Alarcón, Moise, McNamara, & Stuckler, 2024).

Finally, *Option 3* would boil down to an intermediate position. It would essentially consist of the further development and strengthening of current TDIs and economic security instruments including new sector-specific instruments similar to the proposed steel measure, although in a proportionate and circumspect manner. This would be complemented by targeted industrial policy support for strategically important high-tech sectors (including green and digital technologies), while support for EU companies would be conditional upon reaching defined industrial upgrading targets (Raza & Tröster, 2025). EU trade policy vis-à-vis China would be vigilant, not only regarding import surges but also with respect to market access and the treatment of EU companies in China. Trade monitoring and intelligence capacities at the EU level would be substantially expanded. While such a strategic approach would likely provoke more tenuous trade relations with China, it would arguably stop short of triggering an escalating downward spiral and allow for dialogue and a commitment to cooperation. Moreover, it would increase the EU's negotiating leverage. Short to

medium-term costs from welfare losses, supply bottlenecks and import surges should remain manageable. Reputational losses and associated political costs for the EU should be limited, especially if the EU opts for proactive trade diplomacy and cooperation with like-minded countries, other emerging economies and the Global South more generally. It should, however, be emphasised that even under this option, significantly higher public financial support and investment for industrial transformation and technological upgrading would have to be mobilised.

Given distinct political preferences across the EU both on external and economic policy issues, and in light of the EU's international legal obligations, a shift to Option 1 currently does not seem realistic nor desirable. That said, given the volatilities of the geopolitical landscape, including complex relations with China, worsening transatlantic relations, and pronounced EU economic security issues, maintaining the status quo, as described in Option 2, does not seem to be a feasible solution either. Rather, it would further exacerbate vulnerabilities and risk a deep crisis of the EU's productive base. Consequently, Option 3, which strategically relies on a mix of selective protective as well as pro-active industrial policy measures, appears to be the preferable model. However, it is arguably also the most demanding model in terms of managing trade-offs and conducting day-to-day trade policies.

We will thus, in the following, provide a set of policy recommendations to support moving towards a model of 'inclusive regionalism' that can not only cope with imports propelled by overcapacity from China and other trading partners, but contribute to strengthening EU competitiveness in the longer term.

4.2 Policy recommendations

Given distinct sector-specific issues, policies to address overcapacity should not follow a standard approach, but need to be tailor-made. Depending on the case, they may include TDIs, but also broader economic policy instruments, including industrial and technology policy. Similarly, it will be important to not only focus on short-term remedies, but to adopt a systemic perspective that addresses the structural constraints and problems of particular sectors. Overcapacity, as has become clear throughout the case studies in this report, is an eventual by-product of a broader process of the build-up of manufacturing capacity and technological upgrading in China, whose repercussions on European companies are profound and multi-facetted. Thus, the broader process of industrial restructuring, in which China, the EU and other economies are currently engaged, presents a set of long-term challenges for European industries, to which EU economic policies need to develop comprehensive responses.

Thus, in the following we provide a set of policy proposals that, while focussing on trade defence and other trade-related instruments, also embed these instruments in broader economic policies needed to address the overall competitive challenges, which the EU economy will be confronted with in the years to come.

a) Trade Defence Instruments

As shown above, EU TDIs operate within the limits provided by the WTO framework. Therefore, much of the literature has focused on potential changes to existing WTO rules to address grievances and enable more effective trade defence measures (Grossman & Sykes, 2025; Hillman & Manak, 2023). However, given the often diametrically opposed interests of WTO members, adopting many of the suggested changes, including those from the 2020 trilateral initiative of the EU, Japan, and the US, currently appears politically infeasible. That said, there may be some areas where the EU could advocate for finding common ground. Moreover, the EU retains some room for manoeuvre to unilaterally push for a more effective application of TDI rules. This might include, *inter alia*:

1) **Explore potential for common ground on subsidy issues under WTO law:** First, the EU could advocate reviving (and potentially extending) the category of 'non-actionable' subsidies in the ASCM, which would immunise subsidies pursuing public policy goals such as R&D or environmental protection from WTO scrutiny. Moreover, the EU could advocate for including a carve-out/exception in the ASCM, which would cover—security-related subsidies, for example, in the steel or semiconductor sectors (Grossman & Sykes, 2025).[61] Such exceptions could be in the interest of many WTO members, including China, but arguably also the EU, which increasingly uses subsidies as part of an active domestic industrial policy (Blockx & Mattiolo, 2023).

   Second, while there is currently no political will for stricter WTO rules on subsidies overall, finding consensus in specific sectors where concerns about negative externalities (for example, environmental) are widely shared may still be a possible step forward (Grossman & Sykes, 2025, p. 567).

2) **Fully use and continue testing the legal policy space provided by WTO law:** In some areas, EU TDIs are currently not utilising all possibilities permitted under WTO law. One such example is the 'lesser duty rule': Despite the ADA allowing the imposition of duties that offset the dumping margin entirely, the EU Basic Anti-Dumping Regulation requires the EC to set only lesser duties (i.e. lower than the dumping margin), unless they would be inadequate to remove the injury (Art. 7(2)). To increase the bite of this instrument in line with demands from industry representatives and the European Parliament, the EU could amend the Regulation in accordance with the more permissive ADA (European Parliament, 2017, Amendment 4; AEGIS, 2025, p. 12). The Anti-Dumping Regulation could be aligned with the Anti-Subsidy Regulation, which has been amended to use the lesser duty only when it can be clearly concluded that it is not in the Union interest to determine the measure on the basis of the subsidy amount (EC, 2023, p. 1). A similar change of the Anti-Dumping Regulation would still allow the EC to use lesser duties, where higher duties are not in the EU's interest, for example, when having negative effects on other segments of the value chain, making industry less competitive, slowing growth or negatively affecting consumers (Maurer et al., 2025). Another example specifically in the context of China, are China's commitments in the Accession Protocol to the WTO, which could be used to challenge practices of SOEs to provide below-market price inputs to downstream industries that can enable overcapacities (Gao & Zhou, 2022). The lack of WTO case law on this question, however, means that the outcome of such an approach is uncertain. Even so, it could provide an additional avenue to tackle unfair practices worth testing.

   In particular, regarding subsidies, some major grievances stem from WTO jurisprudence, rather than the agreements as such (for example, the overly narrow interpretation of 'public body' in the ASCM). To allow competent bodies to revisit and adapt their interpretations, the EU should continue to apply its preferred understanding of relevant concepts (e.g., applying a broad definition of 'public body', treating government purchases of services as 'financial contributions' for establishing subsidies, adopting a wide understanding of 'any form of income and price support' as used to establish subsidies that covers also measures indirectly distorting prices)[62]. Moreover, the EU may put forward legislation that strategically tests the legal limits of its policy space under WTO law (one recent example is the FSR).

[61] NB: This would be a change of the ASCM *de lege ferenda*, not an application of the security exception of Art. XXI(b)(iii) GATT as explored in section 3.2.3.

[62] See for these suggestions and jurisprudence on them so far: Gao & Zhou (2022)

Another example of the EU's more assertive approach explicitly seeking to target overcapacity issues is the proposed steel measure, which aims to replace the steel safeguards upon its expiry in mid-2026. To ensure its WTO compatibility, the steel measure would be accompanied by a modification of the EU's bound tariffs in the steel sector in line with Article XXVIII GATT. In our view, the steel measure can serve as a model for similar instruments which could strategically provide sector-specific protection where overcapacity issues pose a serious threat to EU industries. Such a focused approach may be particularly promising if it combines defensive measures with pro-active industrial and R&D policies. By contrast, a horizontal measure, which would target overcapacities across sectors would not only raise complex legal issues but likely also lead to large-scale retaliatory actions by trading partners and should therefore not be pursued.

3) **Expand monitoring and trade intelligence capacities:** Investigating trade distortions in order to apply trade defence instruments is notoriously difficult, as evidentiary thresholds are high. This requires expertise and staff (AEGIS, 2025, p. 6). Even after measures have been taken, all trade defence measures entail a risk of circumvention. Effective enforcement depends on companies knowing — and disclosing — their own value chains. In the context of Chinese rare earth export controls, for instance, all companies are required to disclose information about their entire value chains, if these include any input from China, 'akin to building a massive 'panopticon' monitoring the entire global rare earth value chain' (Jin, 2025).

   In contrast, the EU's Import Surveillance Task Force focuses on changes in trade data. On a monthly basis, it publishes detailed lists of products with significant changes in import quantities and prices as well as heat maps combining surges in imports on a sectoral basis and the exporting countries/regions with the latest available trade data (EC, undated b). Although initial criticism regarding methodological transparency has reportedly been addressed (cf. Evenett, 2025), the outputs remain of limited analytical value without contextualisation in terms of affected EU industries, global value chain dynamics, and other structural drivers of trade changes. The task force encourages the industry to provide input on a voluntary basis, but it is unclear how such insights are used systematically. More detailed value chain and sector analysis require more data, including the ownership of intermediary companies, component suppliers, or sub-tier manufacturing processes. Obliging importers to disclose such information would be important, but also requires a legal basis. Moreover, strengthened monitoring at both the macroeconomic and industry or product levels is required to allow for the systematic detection of overcapacity risks. Macroeconomic analysis must overcome significant data and comparability challenges, as industrial sector classifications and statistics are highly country-specific, as illustrated by the case of China. At the industry level, effective monitoring depends on detailed knowledge of value-chain dynamics and sector-specific expertise. Dedicated monitoring units, therefore, need adequate analytical capacity and technical competence to assess overcapacity risks in a timely and reliable manner.

b) Other trade-related instruments

When it comes to resolving overcapacity issues, the effectiveness of TDIs is limited. Usually, they are ex-post remedies, cushioning the effects of import surges, but not addressing the root causes and thus not providing a sustainable economic response to problems related to economic competitiveness and other systemic challenges. Paradoxically, extensive use of TDIs might even produce adverse effects, as in the case of rent-seeking. Thus, there are good reasons to complement a tendentially more intensive use of TDIs with other economic measures. These might include, *inter alia:*

4) **Combine trade defence with industrial upgrading conditionalities:** the academic and empirical literature on rent-seeking suggests that while TDIs might shield industry actors from unfair competition, tariff protection might supply an adverse incentive to companies. Instead of investing in transforming their business model and upgrading their technology to become more competitive, they stick to their prevailing business model and cash in on the windfall rents offered by tariff protection. In other words, purely defensive measures such as TDIs do not, per se, raise competitiveness but rather solidify a problematic status quo (Raza & Tröster, 2025). Thus, in strategic sectors, TDIs need to be complemented by active industrial policy measures to foster competitiveness, including conditionalities for benefitting companies to actively upgrade their business models (Mazzucato & Rodrik, 2023).

5) **Use rebalancing measures to neutralise adverse exchange rate effects:** international trade dynamics, including import surges, can be significantly exacerbated by exchange rate movements, driven by deliberate currency management policies. In cases, where such practice leads to longer term effects on the value of currency vis-à-vis to other currencies, this may have significant effects on price differentials between domestic and imported products, which by and large represent an unwarranted competitive distortion to the benefit of producers in the country with the undervalued currency. A case in point are the Chinese government's currency policies. The real depreciation of the CNY vis-à-vis the EUR since 2022 has arguably had a strong effect on the doubling of the EU's bilateral trade deficit with China. In the absence of adjustment via market-driven currency movements, compensatory EU policies should be considered. The first-best solution to address the problem would be to negotiate an agreement with China (and potentially other countries) on jointly stabilising the EUR-CNY exchange rate. Alternatively, the European Central Bank (ECB) could intervene in the currency market, which would, however, require a change to the prevalent ECB approach of letting the EUR float against other currencies. If neither of the two options were to be achieved, the EU could consider invoking the impairment of its benefits pursuant to Art. XXIII GATT and seek to secure a satisfactory adjustment or launch a non-violation complaint due to the disadvantages from large CNY depreciations. While large and persistent exchange rate movements may have a substantial trade distorting effect, potentially outweighing the benefits conveyed by tariff reductions and other trade benefits, whether such a complaint could be successful is an open question. In any case, a closer monitoring of exchange rate policies thus appears to be warranted.

Finally, policy debates on overcapacity should not lose sight of the above-mentioned broader structural issues relating to increasing the competitiveness of the European economy in the face of intensifying competition in high-tech sectors emanating from both China and other countries, including the US. In China, recent policy efforts to curb overcapacity ('involution') are closely linked to ambitions to further move up the technological value chain. As a result, innovation and R&D support should remain a central objective for EU economic policy. Effective responses, therefore, require a comprehensive approach that addresses overcapacity risks while simultaneously engaging with the wider set of economic and industrial policies shaping technological competition.

Bibliography

Legal sources

Accession Of The People's Republic Of China. (2001). WT/L/432 (23 November 2001). https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/WT/L/432.pdf

Agreement on Government Procurement (GPA). (1994). 1915 UNTS 103.

Agreement on Safeguards. (1994). 1869 UNTS 154.

Agreement on Subsidies and Countervailing Measures. (1994). 1867 UNTS 190.

Agreement on the Implementation of Article VI of GATT 1994 (Anti-Dumping Agreement). (1994). 1868 UNTS 201.

Anti-Unfair Competition Law of the People's Republic of China (2025 Revision) [中华人民共和国反不正当竞争法], 50 Presidential Order (2025).

https://www.spp.gov.cn/spp/fl/202506/t20250627_699862.shtml

Canada/European Communities—Article XXVIII Rights, DS12/R - 37S/80, Award by the Arbitrator (26 October 1990). https://www.wto.org/english/tratop_e/dispu_e/90wheat.pdf

Commission Implementing Regulation (EU) 2019/159 of 31 January 2019 imposing definitive safeguard measures against imports of certain steel products (Definitive Safeguards Implementing Regulation), OJ L 31/27.

Commission Implementing Regulation (EU) 2019/1590 of 26 September 2019 amending Implementing Regulation (EU) 2019/159 imposing definitive safeguard measures against imports of certain steel products, OJ L 248/28.

Commission Implementing Regulation (EU) 2020/35 of 15 January 2020 amending Implementing Regulation (EU) 2019/159 imposing definitive safeguard measures against imports of certain steel products, OJ L 12/13.

Commission Implementing Regulation (EU) 2020/894 of 29 June 2020 amending Implementing Regulation (EU) 2019/159 imposing definitive safeguard measures against imports of certain steel products, OJ L 206/27.

Commission Implementing Regulation (EU) 2022/664 of 21 April 2022 amending Implementing Regulation (EU) 2019/159 imposing a definitive safeguard measure against imports of certain steel products, OJ L 121/12.

Commission Implementing Regulation (EU) 2022/978 of 23 June 2022 amending Implementing Regulation (EU) 2019/159 imposing a definitive safeguard measure on imports of certain steel products, OJ L 167/58.

Commission Implementing Regulation (EU) 2023/104 of 12 January 2023 amending Implementing Regulation (EU) 2019/159 imposing a definitive safeguard measure on imports of certain steel products following a report adopted by the World Trade Organization's Dispute Settlement Body, OJ L 12/7.

Commission Implementing Regulation (EU) 2023/1301 of 26 June 2023 amending Commission Implementing Regulation (EU) 2019/159 imposing a definitive safeguard measure on imports of certain steel products, OJ L 161/44.

Commission Implementing Regulation (EU) 2023/2840 of 14 December 2023 amending Implementing Regulation (EU) 2019/159 imposing a definitive safeguard measure on imports of certain steel products, OJ L 2023/2840.

Commission Implementing Regulation (EU) 2024/1782 of 24 June 2024 amending Implementing Regulation (EU) 2019/159, including the prolongation of the safeguard measure on imports of certain steel products, OJ L 2024/1782.

Commission Implementing Regulation (EU) 2024/2754 of 29 October 2024 imposing a definitive countervailing duty on imports of new battery electric vehicles designed for the transport of persons originating in the People's Republic of China, OJ L 2024/2754.

Commission Implementing Regulation (EU) 2025/612 of 24 March 2025 amending Commission Implementing Regulation (EU) 2019/159 imposing a definitive safeguard measure on imports of certain steel products, OJ L 2025/612.

Commission Implementing Regulation (EU) 2025/2216 of 4 November 2025 amending Implementing Regulation (EU) 2023/2757 imposing a definitive anti-dumping duty on imports of imports of trichloroisocyanuric acid originating in the People's Republic of China following a partial interim review pursuant to Article 11(3) of Regulation (EU) 2016/1036 of the European Parliament and of the Council, OJ L 2025/2216.

Commission, C/2024/3970. Summary notice concerning the initiation of an in-depth investigation in case FS.100011 –EMIRATES TELECOMMUNICATIONS GROUP / PPF TELECOM GROUP pursuant to Articles 10(3)(d) of Regulation (EU) 2022/2560, 21 June 2024, OJ C/2024/3970.

Commission, C/2025/4547. Summary notice concerning the initiation of an in-depth investigation in case FS.100156 – ABU DHABI NATIONAL OIL COMPANY (ADNOC) P.J.S.C. / COVESTRO pursuant to Article 10(3)(d) of Regulation (EU) 2022/2560, 12 August 2025, OJ C/2025/4547.

Energy Law of the People's Republic of China [中华人民共和国能源法], 37 Presidential Order (2024). https://www.nea.gov.cn/2024-11/09/c_1310787187.htm

EU–Measures Related to Price Comparison Methodologies, WT/DS516/9. Request for the Establishment of a Panel by China (10 March 2017). https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/WT/DS/516-9.pdf&Open=True

General Agreement on Tariffs and Trade (GATT). (1994). 55 UNTS 194.

General Agreement on Trade in Services (GATS). (1994). 1869 UNTS 183.

General Agreement on Trade-Related Aspects of Intellectual Property (TRIPS). (1997). 1869 UNTS 299.

Order of the General Court of the European Union of 12 August 2024, Case T-284/24 R, *Nuctech Warsaw Company Limited and Nuctech Netherlands v Commission*, ECLI:EU:T:2024:564.

Regulation (EU) 2015/478 of the European Parliament and of the Council of 11 March 2015 on common rules for imports, OJ L 83/16.

Regulation (EU) 2016/1036 of the European Parliament and of the Council of 8 June 2016 on protection against dumped imports from countries not members of the European Union, OJ L 176/21.

Regulation (EU) 2016/1037 of the European Parliament and of the Council of 8 June 2016 on protection against subsidised imports from countries not members of the European Union OJ 2016 L 176/55 as last amended by Commission Delegated Regulation (EU) 2020/1173 of 4 June 2020 OJ L 259/1.

Regulation (EU) 2017/2321 of the European Parliament and of the Council of 12 December 2017 amending Regulation (EU) 2016/1036 on protection against dumped imports from countries not members of the European Union and Regulation (EU) 2016/1037 on protection against subsidized imports from countries not members of the European Union, OJ L 338/1.

Regulation (EU) 2018/825 of the European Parliament and of the Council of 30 May 2018 amending Regulation (EU) 2016/1036 on protection against dumped imports from countries not members of the European Union and Regulation (EU) 2016/1037 on protection against subsidized imports from countries not members of the European Union, OJ L 143/1.

Regulation (EU) 2019/452 of the European Parliament and of the Council of 19 March 2019 establishing a framework for the screening of foreign direct investments into the Union, OJ L 79 I/1.

Regulation (EU) 2022/1031 of the European Parliament and the Council of 23 June 2022 on the access of third-country economic operators, goods and services to the Union's public procurement and concession markets and procedures supporting negotiations on access of Union economic operators,

goods and services to the public procurement and concession markets of third countries (International Procurement Instrument – IPI), OJ 2022 L 173/1.

Regulation (EU) 2022/2560 on the European Parliament and of the Council of 14 December 2022 on foreign subsidies distorting the internal market, OJ L 330/1.

Regulation (EU) 952/2013 of the European Parliament and of the Council of 9 October 2013 laying down the Union Customs Union, OJ L 269/1.

Report of the Working Party on the Accession of China. (2001). WT/ACC/CHN/49 (1 October 2001). https://www.worldtradelaw.net/document.php?id=misc/ChinaWorkingPartyReport.pdf

Value Added Tax Law of the PRC [中华人民共和国增值税法], 41 Presidential Order (2024). https://shanghai.chinatax.gov.cn/zcfw/zcfgk/zzs/202412/t474694.html

WTO. (1999). *United States–Sections 301-310 of the Trade Act of 1974*, WT/DS152/R, Report of the Panel (22 December 1999). https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspxf?ilename=Q:/WT/DS/152R.pdf&Open=True

WTO. (2019). *Russia–Measures Concerning Traffic in Transit*, WT/DS/512/R, Report of the Panel (5 April 2019). https://www.wto.org/english/tratop_e/dispu_e/512r_e.pdf

WTO. (2022). *EU–Safeguard Measures on certain steel products* Case, WT/DS595/R. Report of the Panel (29 April 2022). https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/WT/DS/595R.pdf&Open=True

WTO. (2022). *US – Origin Marking Requirements,* WT/DS597/R. Report of the Panel (21 December 2022). https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/WT/DS/597R.pdf&Open=True

WTO. (2025). *European Union–Countervailing and Anti-dumping Duties on Stainless Steel Cold-rolled Flat Products from Indonesia*, WT/DS616/R. Report of the Panel (2 October 2025). https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=Q:/WT/DS/616R.pdf&Open=True

WTO. (2026). *US – Certain Tax Credits Under the Inflation Reduction Act*, WT/DS623. Report of the Panel (30 January 2026). https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/WT/DS/623R.pdf&Open=True

Scholarly and other sources

ABB. (2025). *Integrated Report 2024*. https://library.e.abb.com/public/ab32fda35c6f4a3cb62f8c5e2e4476c6/Integrated%20Report%202024_ENGLISH.pdf?x-sign=GyZtAQkTWRNUIEs6y9UCoOcsPu0EIBiie70qIRRf7IwdmuDEkAM890fpbuOQc16W

ACEA. (2025). *Economic and Market Report Full year 2024*. Automobile Manufacturers' Association. https://www.acea.auto/files/Economic_and_Market_Report-Full_year_2024-rev.pdf

ACER. (2024). *European Hydrogen Markets. 2024 Market Monitoring Report*. https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER_2024_MMR_Hydrogen_Markets.pdf

Adachi, A. (2025). China's Overcapacity and the Global South: An Opportunity for the EU. *Internationale Politik Quarterly*, (2). https://ip-quarterly.com/en/chinas-overcapacity-and-global-south-opportunity-eu

AEGIS. (2025). *Fourth AEGIS Europe Report on the EU's Trade Defence Instruments*. AEGIS Europe

Akritidis, V., & Blancardi, J. B. (2023). Analysis of the Foreign Subsidies Regulation from an International Trade Law Perspective on Trade in Goods. *Global Trade and Customs Journal*, *18*(10), 380–383. https://doi.org/10.54648/GTCJ2023048

Alfter, M., Balachandar, V., Carlotta, B., & Amar Breckenridge. (2023). The EU Foreign Subsidies Regulation: What does it really mean? *frontier economics*. https://www.frontier-economics.com/uk/en/news-and-insights/articles/article-i20093-the-eu-foreign-subsidies-regulation-what-does-it-really-mean/

Amsden, A. H. (1992). *Asia's Next Giant: South Korea and Late Industrialization* (1st ed.). Oxford University Press: New York. https://doi.org/10.1093/0195076036.001.0001

Autor, D., Dorn, D., Hanson, G., & Majlesi, K. (2020). Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure. *American Economic Review*, *110*(10), 3139–3183. https://doi.org/10.1257/aer.20170011

Ban, C., & Liu, I. T. (2025). *The Emperor has No Batteries: Europe's Uneven Bid for Battery Strategic Autonomy* (No. ax56z_v1). Center for Open Science. https://doi.org/10.31235/osf.io/ax56z_v1

Banares-Sanchez, I., Burgess, R., Laszlo, D., Simpson, P., Reenen, J. V., & Wang, Y. (2024). *Chinese Innovation, Green Industrial Policy and the Rise of Solar Energy (Working Paper)*. Università Bocconi.

Baumol, W. J. (1982). Contestable Markets: An Uprising in the Theory of Industry Structure. *The American Economic Review*, *72*(1), 1–15.

Benson, E., Mouradian, C., & Alvarez-Aragones, P. (2024). *Evaluating Chip Overcapacity and the Transatlantic Trade Tool Kit*. Center for Strategic & International Studies. https://www.csis.org/analysis/evaluating-chip-overcapacity-and-transatlantic-trade-tool-kit

Berndt, E. R., & Hesse, D. M. (1986). Measuring and assessing capacity utilization in the manufacturing sectors of nine OECD countries. *European Economic Review*, *30*(5), 961–989. https://doi.org/10.1016/S0014-2921(86)80001-0

Berndt, E. R., & Morrison, C. J. (1981). Capacity utilization measures: Underlying economic theory and an alternative approach. *The American Economic Review*, *71*(2), 48–52.

Berry, S., Levinsohn, J., & Pakes, A. (1999). Voluntary Export Restraints on Automobiles: Evaluating a Trade Policy. *American Economic Review*, *89*(3), 400–431. https://doi.org/10.1257/aer.89.3.400

Blaugher, D., & Gordon, B. (2025). *Made in China 2025: Evaluating China's Performance*. US-China Economic and Security Review Commission. https://www.uscc.gov/research/made-china-2025-evaluating-chinas-performance

Blockx, J., & Mattiolo, P. (2023). The Foreign Subsidies Regulation: Calling Foul While Upping the Ante? *European Foreign Affairs Review*, *28*(Special Issue), 53–74. https://doi.org/10.54648/EERR2023014

Bonnet, P., Ciani, A., Molnar, J., & Nardo, M. (2025). *EU's strengths and weaknesses in the global semiconductor sector* (JRC No. 141323). Luxembourg: Publications Office of the European Union. https://data.europa.eu/doi/10.2760/6302476

Boullenois, C., Katz, A., & Rosen, D. H. (2024). *Overcapacity at the Gate*. Rhodium Group. https://rhg.com/wp-content/uploads/2024/03/Overcapacity-at-the-Gate.pdf

Boulter, J. (2018). China's Supply-side Structural Reform. *Reserve Bank of Australia Bulletin.* https://www.rba.gov.au/publications/bulletin/2018/dec/chinas-supply-side-structural-reform.html

Brauneck, J. (2018). New EU Anti-dumping law: Market distortion instead of non-market economy ["Neues EU-Antidumpingrecht: Marktverzerrung statt Nicht-Marktwirtschaft"]. *EWS*, *1*, 18–25.

Brenner, R. (2001). The world economy at the turn of the millennium toward boom or crisis? *Review of International Political Economy*, *8*(1), 6–44. https://doi.org/10.1080/09692290010013828

Brewster, R. (2011). The Remedy Gap: Institutional Design, Retaliation, and Trade Law Enforcement. *George Washington Law Review*, *80*(1), 102–158.

Burgess, J. (2025). Green hydrogen developers reckon with reality check in Europe. *S&P Global Energy.* https://www.spglobal.com/energy/en/news-research/latest-news/energy-transition/110525-green-hydrogen-developers-reckon-with-reality-check-in-europe

Campbell, K. M., & Doshi, R. (2025). Underestimating China: Why America Needs a New Strategy of Allied Scale to Offset Beijing's Enduring Advantages. *Foreign Affairs*, 10.05.2025. https://www.foreignaffairs.com/china/underestimating-china

Cassels, J. M. (1937). Excess Capacity and Monopolistic Competition. *The Quarterly Journal of Economics*, *51*(3), 426. https://doi.org/10.2307/1884835

CCCEU. (2024). CCCEU expresses serious concern over the European side's unannounced raids on Chinese companies Operating in Europe. *CCCEU*, 24.04.2024. http://en.ccceu.eu/2024-04/24/c_4219.htm

Central Committee of the Communist Party of China. (2025). Recommendations on the 15th Five-Year Plan for National Economic and Social Development [中共中央关于制定国民经济和社会发展第十五个五年规划的建议], 28.10.2025. https://www.gov.cn/zhengce/202510/content_7046050.htm

Chao, A. C., & Phillips, F. Y. (2023). The Chinese zombie firm: Dilemma and resolution. *Economic Research-Ekonomska Istraživanja*, *36*(3), 2153718. https://doi.org/10.1080/1331677X.2022.2153718

Chen, J., Chen, X., Hou, Q., & Hu, M. (2021). Haste doesn't bring success: Top-down amplification of economic growth targets and enterprise overcapacity. *Journal of Corporate Finance*, *70*, 102059. https://doi.org/10.1016/j.jcorpfin.2021.102059

Chen, W., Gong, W., Jiang, Z., & Jin, P. (2022). Regional Economic Competition, Fiscal Subsidies, and Overcapacity. *Journal of Global Information Management*, *30*(6), 1–19. https://doi.org/10.4018/JGIM.313410

Chimits, F. (2023). *What Do We Know About Chinese Industrial Subsidies?* (Policy Brief No. 42). CEPII.

China Buses. (2021). Twenty years of hydrogen energy: Yu Zhuoping's review and outlook on China's hydrogen development [盘点中国氢能20年发展历程], 15.12.2021. https://m.chinabuses.com/news/14/article_100433.html?v=1765872004

China Business Industry Research Institute. (2019). *China Industrial Robot Industry: Market Outlook Research Report [2019年中国工业机器人行业市场前景研究报告]*. https://co-image.qichacha.com/upload/chacha/att/20190523/1558598837982601.pdf

China Energy Media Research Institute. (2023). Research Report: Outlook and Challenges for Commercialising China's Hydrogen Energy Industry [研究报告: 我国氢能产业商业化探索前景及挑战]. 中国能源新闻网, 31.05.2023. https://www.cpnn.com.cn/news/baogao2023/202305/t20230531_1605570.html

China Hydrogen Alliance. (2025). Big Data Platform for Hydrogen Industry [氢能产业大数据平台]. https://www.chinah2data.com/#/client/database/PolicyBaseList

China NEV. (2011). Retrospective on China's New Energy Vehicle Policies [中国新能源汽车政策回顾], 23.12.2011. http://www.chinanev.net/news/newscontent/id/609

China Orange Group. (2025). China's Fuel Cell Vehicle Industry Development White Book 2025 [中国燃料电池汽车产业发展白皮书 （2025 年）]. https://file.jgvogel.cn/125/upload/resources/file/600225.pdf

China Power. (2023). China Hydrogen Enterprise Ranking [中国氢能企业排行榜2023], 19.09.2023. http://mm.chinapower.com.cn/qingneng/dongtai/20230919/217715.html

Chinese Academy of Sciences. (2005). Science Time's Weekly Focus [科学时报每周聚焦] [CAS], 12.09.2005. https://www.cas.cn/xw/kjsm/gndt/200906/t20090608_645684.shtml

CIECC Hydrogen Energy Center, & EnerSen. (2024). *2024 Global Hydrogen Industry Outlook Report [2024全球氢能产业展望报告]*. https://www.sdyanbao.com/detail/772378?cid=82

Cohen, P., & Nelson, E. (2025). Europe May Roll Back Combustion Engine Ban. *The New York Times,* 16.12.2025. https://www.nytimes.com/2025/12/16/business/economy/europe-auto-emissions-ban.html#:~:text=Premium%20carmakers%20like%20Porsche%2C%20Mercedes,($833%20million)%20a%20year

Colantone, I., Ottaviano, G., & Stanig, P. (2024). A Popular Backlash against Globalization? *Critical Sociology*, *51*(4–5), 721–742. https://doi.org/10.1177/08969205241261331

Collins, L. (2024). Major boost for hydrogen in China as new law classes it as energy resource, rather than hazardous chemical. *Hydrogeninsight,* 11.11.2024. https://www.hydrogeninsight.com/policy/major-boost-for-hydrogen-in-china-as-new-law-classes-it-as-energy-resource-rather-than-hazardous-chemical/2-1-1737843

Council of the EU. (2025). *Foreign direct investment: Council and Parliament reached political agreement to improve FDI screening*. Council of the EU, 12.11.2025. https://www.consilium.europa.eu/en/press/press-releases/2025/12/11/foreign-direct-investment-council-and-parliament-reached-political-agreement-to-improve-fdi-screening/

Crochet, V., & Gustafsson, M. (2020). *Lawful Remedy or Illegal Response? Resolving the Issue of Foreign Subsidisation under WTO Law* (Working Paper No. 226). Leuven: KU Leuven.

Crochet, V., & Zhou, W. (2023). Preventing the Anti-Circumvention Instrument from Undermining Development through Investment. *International and Comparative Law Quarterly*, *72*, 601–634.

Dai, X., & Zhao, Z. (2021). Can exporting resolve overcapacity? Evidence from Chinese steel companies. *Economic Modelling*, *102*, 105578. https://doi.org/10.1016/j.econmod.2021.105578

Davis, B. (2025). The New Buzzword That's Scaring China. *The New York Times*, 23.09.2025. https://www.nytimes.com/2025/09/23/opinion/china-economy-deflation-neijuan.html

Deasy-Millar, A., Fraile, D., Muron, M., Pawelec, G., Santos, S., & Staudenmayer, O. (2025). *Clean Hydrogen Monitor 2025*. Hydrogen Europe. http://hydrogeneurope.eu/?media_download=17908&dl=1759217493

Directorate-General for Trade and Economic Security. (2024). *Commission to register imports of all products under trade defence investigations in bid to fight unfair competition.* European Commission, 24.09.2024. https://policy.trade.ec.europa.eu/news/commission-register-imports-all-products-under-trade-defence-investigations-bid-fight-unfair-2024-09-24_en

Directorate-General for Trade and Economic Security. (2026). *Commission issues Guidance Document on submission of price undertaking offers for battery electric vehicles from China.* European Commission, 12.01.2026. https://policy.trade.ec.europa.eu/news/commission-issues-guidance-document-submission-price-undertaking-offers-battery-electric-vehicles-2026-01-12_en

Dixit, A. (1980). The Role of Investment in Entry-Deterrence. *The Economic Journal*, *90*(357), 95. https://doi.org/10.2307/2231658

Dong, K., & Sun, W. (2022). Would the market mechanism cause the formation of overcapacity?: Evidence from Chinese listed firms of manufacturing industry. *International Review of Economics & Finance*, *79*, 97–113. https://doi.org/10.1016/j.iref.2022.01.002

Draghi, M. (2024). *The future of European competitiveness*. European Commission. https://commission.europa.eu/document/download/ec1409c1-d4b4-4882-8bdd-3519f86bbb92_en?filename=The%20future%20of%20European%20competitiveness_%20In-depth%20analysis%20and%20recommendations_0.pdf

Dugoua, E., & Dumas, M. (2024). Coordination dynamics between fuel cell and battery technologies in the transition to clean cars. *Proceedings of the National Academy of Sciences*, *121*(27). https://doi.org/10.1073/pnas.2318605121

Ebrahimi, A. (2024). *China's Mature Node Overcapacity: Unfounded Fears*. French Institute of International Relations. https://www.ifri.org/en/memos/chinas-mature-node-overcapacity-unfounded-fears

EC. (2020). *Commission Staff Working Document on significant distortions in the economy of the Russian Federation for the purposes of trade defence investigations* (SWD (2020) 242). European Commission. https://ec.europa.eu/transparency/documents-register/api/files/SWD(2020)242_0/de00000001040304?rendition=false

EC. (2022). *European Chips Act: Staff Working document* (SWD(2022) 147 final). https://digital-strategy.ec.europa.eu/en/library/european-chips-act-staff-working-document

EC. (2023). *Communication to the European Parliament and the Council on the Revised Lesser Duty Rule in Anti-Dumping and Anti-Subsidy Investigations in the EU* (COM (2023) 294 final). Brussels: European Commission.

EC. (2024a). *Commission Staff Working Document on Significant Distortions in the Economy of the People's Republic of China for the Purposes of Trade Defence Investigations* (SWD (2024) 91 final). Brussels: European Commission. https://ec.europa.eu/transparency/documents-register/detail?ref=SWD(2024)91&lang=en

EC. (2024b). *Proposal for a Regulation of the European Parliament and of the Council on the screening of foreign direct investments in the Union and repealing Regulation (EU) 2019/452 of the European Parliament and of the Council* (COM (2024) 23 final). Brussels: European Commission. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52024PC0023

EC. (2025a). *43rd Annual Report from the Commission to the European Parliament and the Council on the EU's Anti-Dumping, Anti-Subsidy and Safeguard activities and the Use of Trade Defence Instruments by Third Countries targeting the EU in 2024* (COM (2025) 428 final). Brussels: European Commission. https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52025DC0428

EC. (2025b). *Commission Staff Document: Executive Summary of the Evaluation of the Public Procurement Directives* (SWD (2025) 333 final). Brussels: European Commission. https://ec.europa.eu/transparency/documents-register/detail?ref=SWD(2025)333&lang=en

EC. (2025c). *Communication to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions: A European Steel and Metals Action Plan* (COM (2025) 125 final). Brussels: European Commission.

EC. (2025d). *Communication to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions 'The Clean Industrial Deal: A joint roadmap for competitiveness and decarbonisation'* (COM (2025) 85 final). Brussels: European Commission. https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52025DC0085

EC. (2025e). *Proposal for a Regulation of the European Parliament and of the Council addressing the negative trade-related effects of global overcapacity on the Union steel market* (COM (2025) 726 final).

EC. (2025f). *Commission proposes plan to protect EU steel industry from unfair impacts of global overcapacity.* European Commission, 07.10.2025. https://ec.europa.eu/commission/presscorner/detail/en/ip_25_2293

EC. (2026). *EU and India conclude landmark Free Trade Agreement.* European Commission, 27.01.2026. https://ec.europa.eu/commission/presscorner/detail/en/ip_26_184

EC. (undated a). *Factsheet: EU-Indonesia Comprehensive Economic Partnership Agreement—Main benefits.* https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/countries-and-regions/indonesia/eu-indonesia-agreements/factsheets/factsheet-eu-indonesia-comprehensive-economic-partnership-agreement-main-benefits-0_en

EC. (undated b). *Monitoring trade diversion – Frequently asked questions.* https://policy.trade.ec.europa.eu/enforcement-and-protection/trade-defence/monitoring-trade-diversion/monitoring-trade-diversion-frequently-asked-questions_en

Eeckhout, P. (1997). European Anti-dumping Law and China. *EloP*, *7*. https://doi.org/10.2139/ssrn.302667

EFORT. (2024). *Annual report 2023 [2023 年年度报告]*. https://www.100est.com/res/financial-report/r2023/SH688165_202404291631624546.pdf

EnerScen. (2023). Electrolyser demand reached 920 MW in the first half of 2023 [2023上半年电解槽需求达920MW], 21.07.2023. http://danchenchen11s.web.97jindianzi.com/doc_27635373.html

EPRS. (2024). *International trade dispute settlement: World Trade Organisation Appellate Body crisis and the multi-party interim appeal arbitration arrangement*. European Parliamentary Research Service. https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/762342/EPRS_BRI(2024)762342_EN.pdf

EPRS. (2025). *Understanding the EU trade defence toolbox* (Briefing No. PE 775.919). EPRS. https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI(2025)775919

ESTUN. (2025). *Annual report 2024 [2024 年年度报告]*. https://notice.10jqka.com.cn/api/pdf/8e9f6a7ce2f0065d.pdf

EU Chamber of Commerce in China. (2016). *Overcapacity in China—An Impediment to the Party's Reform Agenda*.

eu ROBOTICS. (2024). *A Unified Vision for European Robotics*. https://eu-robotics.net/wp-content/uploads/December-2024-V1-A-Unified-Vision-for-European-Robotics.pdf

EU-China Energy Cooperation Platform. (2023). *Opportunities in LNG, CCUS and Green Hydrogen sector in China for EU Business: Enabling policies and financing needs.* http://www.ececp.eu/wp-content/uploads/2023/12/LNG-Green-H2-CCUS-EN.pdf

European Comission, Directorate General Climate Action. (2024). *Innovation Fund IF24 Auction Terms and Conditions*. https://climate.ec.europa.eu/document/download/b996825e-cd36-44c1-895d-a780062f626d_en?filename=2024%2009%2025%20Final%20TC_2nd%20Round%20RFNBO%20H2_IF_TCforPUBLICATION_Clean.pdf

European Commission, Joint Research Center. (2024). *Water electrolysis and hydrogen in the European Union: Status Report on Technology Development, Trends, Value Chains & Markets*. https://dx.doi.org/10.2760/0461008

European Hydrogen Observatory. (2025a). Hydrogen Demand. https://observatory.clean-hydrogen.europa.eu/hydrogen-landscape/end-use/hydrogen-demand

European Hydrogen Observatory. (2025b). *The European hydrogen market landscape.* https://observatory.clean-hydrogen.europa.eu/sites/default/files/2025-11/The%20European%20hydrogen%20market%20landscape_November%202025.pdf

European Parliament. (2017). *Report on the proposal for a regulation of the European Parliament and of the Council amending Regulation (EU) 2016/1036 on protection against dumped imports from countries not members of the European Union and Regulation (EU) 2016/1037 on protection against subsidised imports from countries not members of the European Union* (A8-0236/2017). European Parliament. https://www.europarl.europa.eu/doceo/document/A-8-2017-0236_EN.pdf

European Parliament. (2021). MEPs refuse any agreement with China whilst sanctions are in place. *European Parliament News*, 20.05.2021. https://www.europarl.europa.eu/news/en/press-room/20210517IPR04123/meps-refuse-any-agreement-with-china-whilst-sanctions-are-in-place

European Parliament. (2026a). EU-Mercosur: MEPs demand a legal opinion on its conformity with the EU treaties. European Parliament News, 21.01.2026. https://www.europarl.europa.eu/news/en/press-room/20260116IPR32450/eu-mercosur-meps-demand-a-legal-opinion-on-its-conformity-with-the-eu-treaties

European Parliament. (2026b). EU-Australia Free Trade Agreement. *Legislative Train Schedule—European Parliament*, 22.01.2026. https://www.europarl.europa.eu/legislative-train/theme-a-global-europe-leveraging-our-power-and-partnerships/file-eu-australia-fta

Evenet, S. J. (2025). *No Smoking Gun from the European Commission's First Trade Diversion Monitoring Report* (Zeitgeist Series Briefing No. 66). Global Trade Alert, 05.06.2025. https://globaltradealert.org/reports/European-Commission-Trade-Diversion-Monitoring-Report

Evenett, S. J., & Fritz, J. (2018). *Going Spare: Steel, Excess Capacity, and Protectionism* (22nd Global Trade Alert Report). London: CEPR Press.

Everbright Securities. (2022). *Guohong Hydrogen Energy Technology: A leading company in fuel cell stacks and systems [证券研究报告 国鸿氢能科技：燃料电池电堆及系统龙头公司]* [Securities Research Report]. https://pdf.dfcfw.com/pdf/H3_AP202212271581399709_1.pdf

Fabbricotti, A. (2011). Article XXVIII: Modification of schedules. In *WTO—Trade in Goods* (pp. 692–715). Netherlands: Martinus Nijhoff Publishers.

Fang, H., Li, M., & Lu, G. (2025). Decoding China's Industrial Policies. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.5078043

FANUC CORPORATION. (2024). *Consolidated Annual Financial Results For the Year Ended March 31, 2024*. https://www.fanuc.co.jp/en/ir/announce/pdf/2024/financialresult202403_e.pdf

Fleischmann, J., Hanicke, M., Horetsky, E., Ibrahim, D., Jautelat, S., Linder, M., … van de Rijt, A. (2023). *Battery 2030: Resilient, sustainable, and circular*. McKinsey & Co. https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/battery-2030-resilient-sustainable-and-circular

For Insights Consultancy. (2024). *Automotive Semiconductor Market Research Report—Global Forecasts to 2024 to 2030*. For Insights Consultancy. https://www.forinsightsconsultancy.com/reports/automotive-semiconductor-market/

Foster, P., Bounds, A., & Gridneff, I. (2025). Western allies seek to forge joint defence against cheap Chinese steel. *Financial Times,* 21.11.2025. https://www.ft.com/content/768abafe-320c-4183-845e-48ee5c395e5f

Frank, M. (2023). The EU's New Foreign Subsidy Regulation on Collision Course with the WTO. *Common Market Law Review*, *60*, 925–958.

Fremery, V., Yin, Y., Hu, N., & Flickenschild, J. (2025). *China's Hydrogen Sector 2025: Balancing Growth and Challenges.* Deutsche Gesellschaft Für Internationale Zusammenarbeit. https://energiepartnerschaften.org/fileadmin/global/publications_docs/8.22_The_German_energy_transition_and_impulses_for_China_s_carbon_peaking_and_carbon_neutrality_action_plans_-_targets__status_and_prospects.pdf

Fujino, I. (2025). China cuts chip analyst TechInsights off from data after Huawei report. *Nikkei Asia,* 10.10.2025. https://asia.nikkei.com/business/technology/china-cuts-chip-analyst-techinsights-off-from-data-after-huawei-report2

GAC. (2025). Custom Statistics: HS 847950. *海关统计数据查询平台.* http://stats.customs.gov.cn/indexEn

Gao, H., & Zhou, W. (2022). *Between Market Economy and State Capitalism: China's State-Owned Enterprises and the World Trading System* (1st ed.). Cambridge University Press. https://doi.org/10.1017/9781108908795

Gaogong Industry Research Institute. (2023). *The Blue Book of Robots Industry 2023 [2023 协作机器人产业发展蓝皮书]*. https://www.scribd.com/document/813924754

García Bercero, I. (2025). The EU should moderate its steel protection plan. *Bruegel*, 09.10.2025. https://www.bruegel.org/first-glance/eu-should-moderate-its-steel-protection-plan

García-Herrero, A. (2023). EU-China summit remains a 'dialogue of the deaf' for EU trade concerns. *Bruegel,* 22.12.2023. https://www.bruegel.org/newsletter/eu-china-summit-remains-dialogue-deaf-eu-trade-concerns

García-Herrero, A. (2026). Why Europe must reject Canada's tack towards China. *Euractiv*, 17.02.2026. https://www.euractiv.com/opinion/why-europe-must-reject-canadas-tack-towards-china/

García-Herrero, A., & Schindowski, R. (2024). *Unpacking China's Industrial Policy and its Implications for Europe*. Bruegel Working Paper 11/2024. Bruegel. https://www.bruegel.org/system/files/2024-05/WP%2011%202024_4.pdf

Garlick, J., Ferenczy, Z., Soula, E., Lehrer, J., Szatters, K., & Leonte, A. (2025). *The EU-China Inter-Parliamentary Meeting: Perspectives from CHOICE.* China Observers in Central and Eastern Europe (CHOICE), 20.10.2025. https://chinaobservers.eu/the-eu-china-inter-parliamentary-meeting-perspectives-from-choice/

Ge, P., Zhu, R., Chen, Y., & Huang, X. (2024). Selective industrial policy and overcapacity: Evidence from a quasi-natural experiment in China. *Economic Systems*, *48*(3), 101191. https://doi.org/10.1016/j.ecosys.2024.101191

General Office of the Beijing Municipal People's Government. (2023). *Beijing City Innovation Action Plan for the Robotic Industry《北京市机器人产业创新发展行动方案（2023—2025年）》* (No. 京政办发(2023) 17号). https://www.beijing.gov.cn/zhengce/zhengcefagui/202306/t20230628_3148572.html

General Office of the State Council. (2025). *Notice on Implementing Domestic Product Standards and Related Policies in Government Procurement [关于在政府采购中实施本国产品标准及相关政策的通知]* (No. 国办发〔2025〕34号). https://m.mof.gov.cn/czxw/202509/t20250930_3973725.htm

GGII High Tech Industry Institute. (2024). GGII Ranking: Top 10 of Chinese Electrolyser Shipments in 2023 [GGII榜单：2023年中国电解槽出货量TOP10]. *Sohu,* 29.01.2024. https://www.sohu.com/a/www.sohu.com/a/754242075_120717004

GGII High Tech Industry Institute. (2025a). Behind Peric Hydrogen Energy's Newly Signed Orders Exceeding RMB 1.4 Billion in 2024 [派瑞氢能2024新签订单超14亿背后]. *Sohu*, 22.01.2025. https://www.sohu.com/a/www.sohu.com/a/851036491_120717004

GGII High Tech Industry Institute. (2025b). GGII Ranking: China's Top Ten Electrolyser Shipments in 2024 [GGII榜单：2024年中国电解槽出货量TOP10]. *Sohu*, 02.04.2025. https://www.sohu.com/a/www.sohu.com/a/878622376_120717004

Global Forum on Steel Excess Capacity. (2017). *Global Forum on Steel Excess Capacity Report*. Global Forum on Steel Excess Capacity. https://www.steelforum.org/content/dam/steel-forum/en/publications/global-forum-on-steel-excess-capacity-report_301117.pdf

Global Forum on Steel Excess Capacity. (2021). *2021 GFSEC Ministerial Report*. Global Forum on Steel Excess Capacity. https://www.steelforum.org/content/dam/steel-forum/en/publications/gfsec-ministerial-report-2021.pdf

Global Forum on Steel Excess Capacity. (2023). *The Global Forum on Steel Excess Capacity (GFSEC) 2023 Results Report*. Global Forum on Steel Excess Capacity. https://www.steelforum.org/content/dam/steel-forum/en/publications/GFSEC-results-report-2023.pdf

Godement, F. (2025). Can Europe Do a 'Reverse Deng' With China? *Institut Montaigne,* 21.03.2025. https://www.institutmontaigne.org/en/expressions/can-europe-do-reverse-deng-china

Gong, H., & Hansen, T. (2023). The rise of China's new energy vehicle lithium-ion battery industry: The coevolution of battery technological innovation systems and policies. *Environmental Innovation and Societal Transitions*, *46*, 100689. https://doi.org/10.1016/j.eist.2022.100689

Gong, X. (2025). *Hydrogen in China: Why China's success in solar PV might not translate to electrolyzers* (OIES Paper No. CE 16). Oxford Institute for Energy Studies. https://www.oxfordenergy.org/wpcms/wp-content/uploads/2025/02/CE16-Hydrogen-in-China.pdf

Gong, X., Quitzow, R., & Boute, A. (2023). *China's Emerging Hydrogen Economy: Policies, Institutions, Actors.* RIFS Study. https://doi.org/10.48481/rifs.2023.001

Goodwin, R. M. (1951). The Nonlinear Accelerator and the Persistence of Business Cycles. *Econometrica*, *19*(1), 1. https://doi.org/10.2307/1907905

Goreichy, E., Gunter, J., & Stec, G. (2025). *China's overcapacity and the EU – All you need to know.* MERICS, 16.05.2025. https://merics.org/en/merics-briefs/chinas-overcapacity-and-eu-german-china-policy-under-merz-eu-china-trade

Goujon, R., Kleinhans, J.-P., & Gormley, L. (2024). *Thin Ice: US Pathways to Regulating China-Sourced Legacy Chips*. Rhodium Group. https://rhg.com/research/thin-ice-us-pathways-to-regulating-china-sourced-legacy-chips/

Government of Canada. (2026). Backgrounder—Preliminary Agreement in Principle to Address Economic and Trade Issues between Canada and the People's Republic of China. *Canada.ca,* 17.01.2026. https://www.international.gc.ca/news-nouvelles/2026/2026-01-16-china-chine.aspx?lang=eng

Grossman, G. M., & Sykes, A. O. (2025). Industrial Policy and Subsidies: Assessment of Current Rules and Possible Reforms. *Journal of World Trade*, *59*(4), 531–578. https://law.stanford.edu/wp-content/uploads/2025/07/TRAD2025030.pdf

Gunter, J., Brown, A., Chimits, F., Hmaidi, A., Vasselier, A., & Zenglein, M. J. (2025). *Beyond overcapacity: Chinese-style modernization and the clash of economic models*. MERICS. https://merics.org/sites/default/files/2025-04/MERICS%20Report%20Overcapacities_2025%20final.pdf

Guo, Q. (2025). In the first three quarters, nationwide revenue in the robotics industry increased by 29.5% year-on-year [前三季度全国机器人行业营收同比增长29.5%]. *Xinhua,* 04.11.2025. https://www.news.cn/fortune/20251104/f96a22d8346140f9849b6d43bd4caa73/c.html

Hestermeyer, H. P., & Nielsen, L. (2014). The Legality of Local Content Measures under WTO Law. *Journal of World Trade*, *48*(3), 553–591. https://doi.org/10.54648/TRAD2014019

Hicks, J. R. (1950). *A Contribution to the Theory of the Trade Cycle*. Oxford: Clarendon Press.

Hillman, J. A., & Manak, I. (2023). *Rethinking International Rules on Subsidies* (Council Special Report No. 96). New York: Council on Foreign Relations. https://cdn.cfr.org/sites/default/files/report_pdf/Rethinking%20International%20Rules%20on%20Subsidies_0.pdf?_gl=1*qsjbal*_gcl_au*MTA1ODM2NDkzLjE3NjgyMTc2MTE.*_ga*MTE4MDgxNzkwMi4xNzY4MjE3NjEx*_ga_24W5E70YKH*czE3NjgyMTc2MTEkbzEkZzAkdDE3NjgyMTc2MTIkajU5JGwwJGgw

Holz, C. A. (2020). *Understanding PRC Investment Statistics* (CESifo Working Paper No. 8110).

Hove, A. (2024). *Clean energy innovation in China: Fact and fiction, and implications for the future* (Oxford Institute for Energy Studies Paper No. CE 14). https://www.oxfordenergy.org/wpcms/wp-content/uploads/2024/07/CE14-Clean-energy-innovation-in-China-Final.pdf

Howse, R. (2020). Making the WTO (Not So) Great Again: The Case Against Responding to the Trump Trade Agenda Through Reform of WTO Rules on Subsidies and State Enterprises. *Journal of International Economic Law*, *23*(2), 371–389. https://doi.org/10.1093/jiel/jgaa017

Huafu Securities. (2023). *Deep-Dive Report on Industrial Robots in the Machinery Industry [机械行业工业机器人深度报告]* [Securities Research Report]. https://pdf.dfcfw.com/pdf/H3_AP202312221614811813_1.pdf?1703236913000.pdf

Hufbauer, G. (2024). If Chinese industrial capacity is a problem the US has better measures to deal with it. *East Asia Forum*, 04.08.2024. https://eastasiaforum.org/2024/08/04/if-chinese-industrial-capacity-is-a-problem-the-us-has-better-measures-to-deal-with-it/

Hug, C., Stone, J., & Lanktree, G. (2025). Britain seeks EU steel pact to counter China—And swerve tariffs. *Politico,* 27.10.2025. https://www.politico.eu/article/britain-eu-steel-pact-china-tariffs/

Hwabao Securities. (2023). *Green hydrogen scale-up is imminent; electrolysers are expected to be the first to benefit [绿氢放量在即，电解槽有望率先受益]* [Special Research Report on the Hydrogen Energy Industry]. https://pdf.dfcfw.com/pdf/H3_AP202304281586021109_1.pdf

Hydrogen Council, & McKinsey & Company. (2025). *Global Hydrogen Compass 2025: Industry progress and lessons learned from the first wave of mature clean hydrogen projects*. https://compass.hydrogencouncil.com/wp-content/uploads/2025/09/Hydrogen-Council-Global-Hydrogen-Compass-2025.pdf

Hydrogen Energy Branch of the China Industrial Development Promotion Association. (2025). *The Hydrogen Energy Industry Promotion Association published the 'Initiative for the Healthy Development of China's Electrolyser Industry' [我会发布《中国电解槽行业健康发展倡议书》]*, 17.11.2025. https://cn-heipa.com/newsinfo/10818592.html

IEA. (2023). *Global EV Outlook 2023*. IEA. https://www.iea.org/reports/global-ev-outlook-2023/prospects-for-electric-vehicle-deployment

IEA. (2024a). *Energy Technology Perspectives 2024*. https://www.iea.org/reports/energy-technology-perspectives-2024

IEA. (2024b). *Global Hydrogen Review 2024*. https://www.iea.org/reports/global-hydrogen-review-2024

IEA. (2025a). *Advanced Fuel Cell Technology Collaboration Programme: Annual Report 2024*. https://ieafuelcell.com/wp-content/uploads/2025/06/IEA-AFC-TCP-Annual-Report-2024.pdf

IEA. (2025b). *Announced electrolyser manufacturing capacity by region and manufacturing capacity needed in the Net Zero Scenario, 2021-2030*. https://www.iea.org/data-and-statistics/charts/announced-electrolyser-manufacturing-capacity-by-region-and-manufacturing-capacity-needed-in-the-net-zero-scenario-2021-2030

IEA. (2025c). *Distribution of FCEVs at the end of 2024.* IEA. https://ieafuelcell.com/output/fcev/

IEA. (2025d). *Global EV Outlook 2025*. IEA. https://www.iea.org/reports/global-ev-outlook-2025

IEA. (2025e). *Global Hydrogen Review 2025*. https://www.iea.org/reports/global-hydrogen-review-2025

IEA. (2025f). *Hydrogen Production and Infrastructure Projects Database [IEA Dataset].* IEA. https://www.iea.org/energy-system/low-emission-fuels/electrolysers

International Federation of Robotics. (2020). *World robotics 2020: Industrial robots*. Frankfurt am Main: VDMA Services GmbH.

International Federation of Robotics. (2022). *World robotics 2022: Industrial robots*. Frankfurt am Main: VDMA Services GmbH.

International Federation of Robotics. (2023). *World Robotics 2023 Report: Asia ahead of Europe and the Americas.* https://ifr.org/ifr-press-releases/news/world-robotics-2023-report-asia-ahead-of-europe-and-the-americas

International Federation of Robotics. (2024). Record of 4 Million Robots in Factories Worldwide. *IFR Press Room*, 24.09.2024. https://ifr.org/ifr-press-releases/news/record-of-4-million-robots-working-in-factories-worldwide

International Federation of Robotics. (2025a). *Global Robot Demand in Factories Doubles Over 10 Years.* IFR. https://ifr.org/ifr-press-releases/news/global-robot-demand-in-factories-doubles-over-10-years

International Federation of Robotics. (2025b). *World Robotics: Industrial Robots 2025—Executive Summary*. Frankfurt am Main: VDMA Services GmbH. https://ifr.org/img/worldrobotics/Executive_Summary_WR_2025_Industrial_Robots.pdf

International Federation of Robotics. (2025c). China to Invest 1 Trillion Yuan in Robotics and High-Tech Industries. *IFR,* 25.03.2025. https://ifr.org/ifr-press-releases/news/china-to-invest-1-trillion-yuan-in-robotics-and-high-tech-industries

International Renewable Energy Agency. (2025). *Renewable Power Generation Costs in 2024*. International Renewable Energy Agency IRENA. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2025/Jul/IRENA_TEC_RPGC_in_2024_2025.pdf

Irwin-Hunt, A. (2024). China semiconductor capex adds to overcapacity risks. *FDI Intelligence,* 24.01.2024. https://www.fdiintelligence.com/content/dde451f9-f7df-5129-89fa-b60ae8b189f7

ISO. (2021). *Robotics—Vocabulary*. https://www.iso.org/obp/ui/en/#iso:std:iso:8373:ed-3:v1:en

Itoh, M., & Nagaoka, S. (1997). VERs, VIEs, and Global Competition. In E. M. Graham & J. D. Richardson (Eds.), *Global competition policy*. Washington, DC: Institute for International Economics.

Jiang, X. (2025). China's robot makers boost output but struggle to lift profits. *Jiemian Global,* 13.11.2025. https://en.jiemian.com/article/13632808.html

Jin, B. (2018). China's Industrial Transformation and Upgrading in Globalization's New Era. In S. Binhong (Ed.), *Political Economy of Globalization and China's Options* (pp. 97–108). BRILL. https://doi.org/10.1163/9789004383944_008

Jin, U. (2025). 'No missiles or semiconductors without China'... US 'choked by rare earths. *bloomingbit,* 13.10.2025. https://bloomingbit.io/en/feed/news/98650

Jing, C. (2025). China's 10 trillion yuan debt clean-up falls short. *The Business Times,* 05.12.2025. https://www.businesstimes.com.sg/international/global/chinas-10-trillion-yuan-debt-clean-falls-short

Johnson, C. (1982). *MITI and the Japanese Miracle: The Growth of Industrial Policy, 1925–1975*. Stanford, CA: Stanford University Press.

Joint Statement of the Trilateral Meeting of the Trade Ministers of Japan, the United States and the European Union. (2020). Office of the United States Trade Representative, 14.01.2020. https://ustr.gov/about-us/policy-offices/press-office/press-releases/2020/january/joint-statement-trilateral-meeting-trade-ministers-japan-united-states-and-european-union

Ju, L. (2025). Profound understanding of 'involutionary' competition and its comprehensive rectification [深刻认识和综合整治"内卷式"竞争]. *Qiushi*, *13*. https://www.qstheory.cn/20250629/6d682af56d64487f817084e890fbcdfd/c.html

Kalecki, M. (1937). A Theory of the Business Cycle. *The Review of Economic Studies*, *4*(2), 77. https://doi.org/10.2307/2967606

Khan, S. A. (2026). BYD and Xpeng pop as EU hints at a minimum-price alternative to China EV tariffs. *Techstock2,* 14.01.2026. https://ts2.tech/en/byd-and-xpeng-pop-as-eu-hints-at-a-minimum-price-alternative-to-china-ev-tariffs/

Kinco Automation (Shanghai). (2025). *Annual Report 2024 [年年度报告]*. https://notice.10jqka.com.cn/api/pdf/7c91ac010a9ad894.pdf

Klein, L. R. (1960). Some Theoretical Issues in the Measurement of Capacity. *Econometrica*, *28*(2), 272. https://doi.org/10.2307/1907721

Klement, J. (2025). Europe's second 'China shock' is blessing in disguise. *Reuters,* 09.12.2025. https://www.reuters.com/markets/global-market-data/industrials/europes-second-china-shock-is-blessing-disguise-2025-12-09/

KUKA. (2025). Annual Report 2024. https://www.kuka.com/-/media/kuka-corporate/documents/ir/reports-and-presentations/en/annual-report/annual-report-2024.pdf?rev=a60419c9671a4cdc8c5a380faad83009

Kunath, G. (2025). Debt-fuelled growth in China and local government indebtedness. *IW Report*, *35/2025*. https://www.iwkoeln.de/fileadmin/user_upload/Studien/Report/PDF/2025/IW-Report_2025-China-Local-Government-Debt.pdf

Lawrence, A., & VerWey, J. (2019). *The Automotive Semiconductor Market – Key Determinants of U.S. Firm Competitiveness*. U.S. International Trade Commission. https://www.govinfo.gov/content/pkg/GOVPUB-ITC1-PURL-gpo190243/pdf/GOVPUB-ITC1-PURL-gpo190243.pdf

Le Poidevin. (2025). China to forego Special and Differential Treatment in future WTO negotiations. *Reuters*, 24.09.2025. https://www.reuters.com/world/china/china-forego-special-differential-treatment-future-wto-negotiations-2025-09-23/

Lee, J., & Kleinhans, J.-P. (2024). *Automotive Chips: A European Chokepoint?* In T. Rühlig (Ed.), Reverse Dependency: Making Europe's Digital Tehcnological Strengths indispensible to China (pp. 112–125). online: Digital Power China. https://dgap.org/system/files/article_pdfs/DPC%20-%20GESAMT_Final.pdf

Leichthammer, A. (2025). *Not a Flood but a Fault Line: The case for refined trade defence measures against Chinese overcapacity*. Berlin: Hertie School, Jacques Delors Centre. https://www.delorscentre.eu/en/publications/detail/publication/not-a-flood-but-a-fault-line

Lemaire, C. (2026). EU replaces China EV tariffs with a 'price floor'. Who benefits? *Brusselssignal*, 14.01.2026. https://brusselssignal.eu/2026/01/eu-replaces-china-ev-tariffs-with-a-price-floor-who-benefits/

Li, J., Liang, H., Zheng, Z., Teng, L., Zhang, Z., Gao, Z., ... Jiang, L. (2025). A critical review of China's hydrogen supply chain and equipment. *International Journal of Hydrogen Energy*, *180*, 151737. https://doi.org/10.1016/j.ijhydene.2025.151737

Li, Y., Zhao, J., & Yan, J. (2023). Technological Innovation and the Development of the Fuel Cell Electric Vehicle Industry Based on Patent Value Analysis. *Economic Research Institute for ASEAN and East Asia Discussion Paper*, *477*. https://www.eria.org/uploads/Technological-Innovation-and-the-Development-of-the-FCEV-Industry-Based-on-Patent-Value-Analysis-.pdf

Lin, I.-T. S. (2018). The Robotics Industry in China. *China Briefing,* 14.05.2018. https://www.china-briefing.com/news/chinas-robot-industry/

Liu, Z. (2025). Structural mechanisms of 'involutionary' competition and its comprehensive rectification "内卷式"竞争的生成机理与综合整治. *People's Daily,* 01.09.2025. https://paper.people.com.cn/rmlt/pc/content/202509/01/content_30104675.html

Lu, F. (2024). The Characteristics of the New Wave of Overcapacity and Its International Impact. *National School of Development,* 08.04.2024. https://en.nsd.pku.edu.cn/news/535391.htm

Lv, B., Liu, Y., & Li, Y. (2020). Fiscal incentives, competition, and investment in China. *China Economic Review*, *59*, 101371. https://doi.org/10.1016/j.chieco.2019.101371

Ma, C. (2023). The historical evolution of China's energy-security concept and policies since the founding of the People's Republic of China [新中国成立以来我国能源安全观及能源安全政策的历史演进]. 湖北社会科学, *2*, 76–82. https://doi.org/10.13660/j.cnki.42-1112/c.016046

Ma, J. (2025). China's solar industry enters a deeper adjustment, as losses narrow and policy discipline tightens | 界面新闻. *Jiemian News,* 19.12.2025. https://m.jiemian.com/article/13785203.html

Ma, R. (2024). *How Do Robot Subsidies Affect Aggregate Productivity and Firm Dispersion? Theory and Evidence from China (Working Paper).* LSE Department of Economics. https://www.lse.ac.uk/economics/Assets/Documents/job-market-candidates-2024-2025/robot-ma.pdf

Marketing Intelligence Resource. (2024). *The Global White Book of Robots 2024 [2024年全球协作机器人产业发展白皮书]*. https://pdf.dfcfw.com/pdf/H3_AP202412231641403661_1.pdf

Martin, P. (2024). 'Severe overcapacity': The global supply of electrolysers far outstrips demand from green hydrogen projects: BNEF. *Hydrogeninsight,* 27.03.2024. https://www.hydrogeninsight.com/electrolysers/severe-overcapacity-the-global-supply-of-electrolysers-far-outstrips-demand-from-green-hydrogen-projects-bnef/2-1-1618327

Martin, P. (2025). Sales of hydrogen fuel-cell vehicles continued to fall around the world in first half of 2025. *Hydrogeninsight*, 08.08.2025. https://www.hydrogeninsight.com/transport/sales-of-hydrogen-fuel-cell-vehicles-continued-to-fall-around-the-world-in-first-half-of-2025/2-1-1855331

Martinez, M. (2025). EU will protect its market if needed, German finance minister says in China. *Reuters,* 18.11.2025. https://www.reuters.com/world/china/eu-will-protect-its-market-if-needed-german-finance-minister-says-china-2025-11-18/

Matthes, J. (2024). China's Trade Surplus – Implications for the World and for Europe. *Intereconomics*, *59*(2), 104–111. https://doi.org/10.2478/ie-2024-0022

Matthes, J. (2025). *Yuan Undervaluation against the Euro: Unfair Cost Advantages for China?! (IW-Report No. 36).* Cologne: Institut der deutschen Wirtschaft.

Mattiolo, P. (2023). *Quelling the Flames or Fighting Fire with Fire? The EU's Two Approaches to Foreign Subsidies*. CELIS, 26.10.2023. https://www.celis.institute/celis-blog/quelling-the-flames-or-fighting-fire-with-fire-the-eus-two-approaches-to-foreign-subsidies/

Maurer, A., Ahmad, Z., & Hagemejer, J. (2025). *Shortening and simplifying EU anti-dumping and anti-subsidy investigations with a view to practices favourable for SMEs* Brussels: European Parliament's Committee on International Trade. https://www.europarl.europa.eu/thinktank/en/document/EXPO_STU(2025)754470

Mayr, S. (2025). Wirtschaftliche Sicherheit in Zeiten geopolitischen Wandels: Investitionskontrollen in Europa. *Europäische Zeitschrift Für Wirtschaftsrecht*, *36*(14), 676–685.

Mazzucato, M., & Rodrik, D. (2023). *Industrial Policy with Conditionalities: A Taxonomy and Sample Cases*. London: Institut for Innovation and Public Purpose. https://drodrik.scholar.harvard.edu/sites/scholar.harvard.edu/files/dani-rodrik/files/conditionality_mazzucato_rodrik_0927202.pdf

McIlroy, T. (2026). Australia and EU on verge of striking free trade deal long stalled by beef, parmesan and prosecco. *The Guardian*, 14.02.2026. https://www.theguardian.com/australia-news/2026/feb/14/australia-eu-free-trade-deal-negotiations-brussels-parmesan-and-prosecco

Meltzer, J. P., & Pearson, M. M. (2024). *How the US should address Chinese overcapacity and its impact on international trade*. Brookings. https://www.brookings.edu/articles/how-the-us-should-address-chinese-overcapacity-and-its-impact-on-international-trade/

Meng, Z., He, Q., Hu, H., & Shen, Y. (2022). Development Status and Reflections on China's Hydrogen Energy Industry [我国氢能产业发展现状与思考]. *Modern Chemical Industry*, *42*(1), 1–12. https://www.xdhg.com.cn/CN/abstract/abstract10974.shtml

Merk, J., González, A., Rodrigo, F., & Boris, S. (2025). *Manufactured crisis: Europe's car industry hoards funds and delays climate action*. SOMO. https://www.somo.nl/manufactured-crisis/

MIIT. (2024a). *Announcement on the Release of the 'Lithium-ion Battery Industry Standard Conditions (2024 Edition)' and the 'Interim Measures for the Administration of Lithium-ion Battery Industry Standard Announcements' (2024 Edition) [《锂离子电池行业规范条件（2024年本）》和《锂离子电池行业规范公告管理暂行办法（2024年本）》的公告]* (No. 工业和信息化部公告2024年第14号). https://policy.mofcom.gov.cn/claw/clawContent.shtml?id=101135

MIIT. (2024b). *'PV manufacturing industry standard conditions (2024 version)' and management measures《光伏制造行业规范条件（2024年本）》及管理办法*(No. 工业和信息化部公告〔2024〕33号). http://gxj.zaozhuang.gov.cn/gxwj/202411/t20241125_1971068.html

MIIT. (2025). *Implementation Measures for Capacity Replacement in the Steel Industry (Draft for Public Consultation) [钢铁行业产能置换实施办法（征求意见稿）]* (p. 2025). https://policy.mofcom.gov.cn/claw/opinionInfo.shtml?id=6391

MIIT, MNR, MEE, & MOC. (2025). *Working Plan for Stabilising Growth in the Steel Industry for the years 2025 and 2026 [钢铁行业稳增长工作方案（2025－2026年）]* (No. 工信部联原〔2025〕171号). http://m.law-lib.com/law/law_view.asp?id=782807&page=1

MIIT, MOE, MOST, & MOT. (2024). *Implementation Guidelines on Advancing the Innovative Development of Future Industries [推动未来产业创新发展的实施意见]* (No. 工信部联科〔2024〕12号). https://www.gov.cn/zhengce/zhengceku/202401/content_6929021.htm

MIIT, MOF, GAC, & STA. (2025). *Catalogue related to the Import Taxation Policy for Important Technology Eqipment [调整重大技术装备进口税收政策有关目录]* (No. 工信部联重装〔2025〕26号). https://wap.miit.gov.cn/zwgk/zcwj/wjfb/tz/art/2025/art_418dfba9c70449d7956b321665a4c31d.html

MIIT, NDRC, MOST, & Ministry of Public Security. (2021). *14th Five-Year Plan for Robotic Industry Development ["十四五" 机器人产业发展规划]* (No. 工信部联规〔2021〕206号). https://www.beijing.gov.cn/zhengce/zhengcefagui/qtwj/202204/W020220408555510821175.pdf

Ministry of Foreign Affairs, People's Republic of China. (2025). *Foreign Ministry Spokesperson Guo Jiakun's Regular Press Conference*, 25.07.2025. https://www.fmprc.gov.cn/mfa_eng/xw/fyrbt/202507/t20250725_11677269.html?utm

Minsky, H. P. (1982). The Financial Instability Hypothesis: Capitalist Processes and the Behavior of the Economy. In C. P. Kindleberger & J.-P. Laffargue (Eds.), *Financial Crises: Theory, History, and Policy* (pp. 13–39). Cambridge: Cambridge University Press.

MOC. (2026). Ministry of Commerce Update on Consultations in the China–EU Electric Vehicle Trade Case [商务部关于中欧电动汽车案磋商进展的通报], 12.01.2026. https://www.mofcom.gov.cn/syxwfb/art/2026/art_75ab0aa242094122abb5c518ccc23cc1.html

MOC, & GACC. (2025). *Announcement on the Implementation of Export Licensing Administration for Certain Steel Products [公布对部分钢铁产品实施出口许可证管理]* (No. 商务部 海关总署公告2025年第79号). https://www.mofcom.gov.cn/zwgk/zcfb/art/2025/art_045b556ed8da4978af8eb5c162b65b61.html

MOF, MIIT, MOST, NDRC, & NEA. (2020). *Notice on Launching the Demonstration Application of Fuel Cell Vehicles [开展燃料电池汽车示范应用的通知]* (No. 财建〔2020〕394号). https://www.gov.cn/zhengce/zhengceku/2020-09/21/content_5545221.htm

MOF, & SAT. (2024). *Notification on the Adjustment of the Export Tax Rebate [关于调整出口退税政策的公告]* (No. 财政部 税务总局公告2024年第15号). https://szs.mof.gov.cn/zhengcefabu/202411/t20241115_3947628.htm

MOF, & SAT. (2026). *Notification on the Adjustment of the Export Tax Rebate[调整光伏等产品出口退税政策的公告]* (No. 财政部 税务总局公告2026年第2号). https://fgk.chinatax.gov.cn/zcfgk/c102416/c5246745/content.html

Mordor Intelligence. (2025). *Automotive Semicondcutor Market Size & Share Analysis—Growth Trends and Forecast*. https://www.mordorintelligence.com/industry-reports/automotive-semiconductor-market

Musitz, L. (2024a). *China-a challenging partner for the German economy [China—Ein schwieriger Partner für die deutsche Wirtschaft]*. Düsseldorf: Hans Böckler Stiftung. https://www.boeckler.de/de/faust-detail.htm%3Fsync_id=HBS-008982

Musitz, L. (2024b). *China's green technological leadership [Chinas grüne Technologieführerschaft]* (Working Paper-Reihe der AK Wien No. 259). Kammer für Arbeiter und Angestellte für Wien. https://emedien.arbeiterkammer.at/viewer/fulltext/AC17389463/4/

Nagy, C. I. (2023). The EU's New Regime on Foreign Subsidies: Has the Time Come for a Paradigm-Shift? *Journal of World Trade*, *57*(6), 889–908.

National Bureau of Statistics of China. (2025). *Output of Industrial Robots*. https://data.stats.gov.cn/english/easyquery.htm?cn=A01

National Bureau of Statistics of China. (2026). *Investment Actually Completed in Fixed Assets, Accumulated Growth Rate (%).* https://data.stats.gov.cn/english/easyquery.htm?cn=A01

NBS China. (2026). *National Economy Pushed Forward with Innovation-led and High-quality Development and Expected Targets Achieved Successfully in 2025*. https://www.stats.gov.cn/english/PressRelease/202601/t20260119_1962328.html

NDRC. (2017). *Review of the Robotics Industry Development during the 12th Five-Year Plan ["十二五" 期间机器人产业发展回顾]*, 21.12.2017. https://www.ndrc.gov.cn/xwdt/gdzt/xyqqd/201712/t20171221_1197824.html

NDRC. (2022). *Catalogue of Industries Encouraging Foreign Investment (2022 Edition) (Draft for Public Consultation) [鼓励外商投资产业目录(2022 年版) (征求意见稿)]*. https://www.ndrc.gov.cn/yjzxDownload/20220510fj1.pdf

NDRC. (2023). *Mid- and Longterm Hydrogen Industry Development Plan [氢能产业发展中长期规划（2021-2035 年）]*. https://www.ndrc.gov.cn/xxgk/zcfb/ghwb/202203/P020220323314396580505.pdf

NDRC. (2026). *Expert Interpretation, Article No. 5: The National Venture Capital Guidance Fund Should Serve as a Model for the Standardized Development of Government Guidance Funds [专家解读文章之五 | 国家创业投资引导基金要成为政府引导基金规范发展的示范标杆]*, 05.01.2026. https://www.ndrc.gov.cn/fggz/202601/t20260105_1403044.html

NDRC Industry Department. (2023). *A responsible official of the National Development and Reform Commission answers reporters' questions on the Guiding Catalogue for Industry Structure Adjustment (2024 Edition) [国家发展改革委有关负责同志就《产业结构调整指导目录（2024年本）》答记者问]*, 29.12.2023. https://www.ndrc.gov.cn/xxgk/jd/jd/202312/t20231229_1363020.html

NDRC, MIIT, & MOF. (2016). *Robotics Industry Development Plan (2016-2020) [机器人产业发展规划（2016-2020年）]*. https://www.ndrc.gov.cn/xxgk/zcfb/ghwb/201604/t20160427_962181_ext.html

NDRC, & MOF. (2025). *Measures for Large-scale Equipment Renewal and Consumer Good Trade-In Policies [2026年实施大规模设备更新和消费品以旧换新政策]* (No. 发改环资〔2025〕1745号). https://www.ndrc.gov.cn/xxgk/zcfb/tz/202512/t20251230_1402851.html

NDRC, & NEA. (2016). *Energy Technology Revolution and Innovation Action Plan (2016–2030) [能源技术革命创新行动计划（2016–2030年）]* (No. 发改能源〔2016〕513号). https://www.ndrc.gov.cn/xxgk/zcfb/tz/201606/W020190905517012835441.pdf

NDRC, NEA, & MOF. (2021). *14th Five-Year Plan for Renewable Energy Development ["十四五"可再生能源发展规划]* (No. 发改能源〔2021〕1445号). https://www.ndrc.gov.cn/xwdt/tzgg/202206/P020220602315650388122.pdf

NDRC, & SAMR. (2025a). *4th Announcement of 2025 [公告2025年第4号]*. https://www.ndrc.gov.cn/xxgk/zcfb/gg/202510/t20251009_1400872.html

NDRC, & SAMR. (2025b). *Draft Amendments to the Price Law of the People's Republic of China (Public Consultation Draft) [中华人民共和国价格法修正草案（征求意见稿]*. https://yyglxxbsgw.ndrc.gov.cn/htmls/article/article.html?articleId=2c97d16c-9324f814-0198-39e9891b-0071

NEA, & Guoneng Hydrogen Innovation. (2025). *China Hydrogen Energy Development Report 2025 [中国氢能发展报告2025]*. https://www.nea.gov.cn/20250430/96022785b3a747248288ad1c57d3a025/20250430960227785b3a747248288ad1c57d3a025_35aeee443346424eb4a3da029cb007003c.pdf

NEA, & MOST. (2021). *14th Five-Year Plan of Energy Technology Innovation [十四五"能源领域科技创新规划]* (No. 国能发科技〔2021〕58号). https://zfxxgk.nea.gov.cn/1310540453_16488637054861n.pdf

Noland, M., & Pack, H. (2003). *Industrial policy in an era of globalization: Lessons from Asia*. Washington, D.C: Institute for International Economics.

Odenweller, A., & Ueckerdt, F. (2025). The green hydrogen ambition and implementation gap. *Nature Energy*, *10*, 110–123. https://doi.org/10.1038/s41560-024-01684-7

OECD. (2019). *Measuring distortions in international markets: The semiconductor value chain* (OECD Trade Policy Paper No. 234). OECD Publishing. https://doi.org/10.1787/8fe4491d-en

OECD. (2022). *Economic Surveys: China 2022*. OECD Publishing. https://doi.org/10.1787/b0e499cf-en

OECD. (2025a). *How Governments Back the Largest Manufacturing Firms: Insights from the OECD MAGIC Database* (289th ed.) [OECD Trade Policy Papers]. https://doi.org/10.1787/d93ed7db-en

OECD. (2025b). *Hydrogen in steel: Addressing emissions and dealing with overcapacity* (OECD Science, Technology and Industry Policy Paper No. 174). Organisation for Economic Co-Operation and Development (OECD). https://doi.org/10.1787/7e2edc69-en

OECD. (2025c). *Mapping the semiconductor value chain: Working towards identifying dependencies and vulnerabilities* (OECD Science, Technology and Industry Policy Paper No. 182). OECD Publishing. https://doi.org/10.1787/4154cdbf-en

Office of the United States Trade Representative. (2025). *Public Hearing Transcript Regarding Section 301 Investigation on China's Acts, Policies, and Practices Related to Targeting of the Semiconductor Industry for Dominance*. https://ustr.gov/trade-topics/enforcement/section-301-investigations/section-301-chinas-targeting-semiconductor-industry-dominance

Okimoto, D. I. (1989). *Between MITI and the Market: Japanese Industrial Policy for High Technology*. Stanford, CA: Stanford University Press.

People's Daily. (2017). MIIT: In the robotics sector, there is a risk that high-end industries will be pushed toward low-end output, along with overcapacity in low-end products [工信部：机器人领域有高端产业低端化和低端产品产能过剩风险]. *People's Daily*, 11.03.2017. https://finance.people.com.cn/n1/2017/0311/c1004-29138287.html

People's Government of Nanshan District, Shenzhen. (2024). *Measures for the Administration of the Nanshan District Special Funds for Promoting High-Quality Industrial Development [南山区促进产业高质量发展专项资金管理办法]* (No. 深南府规〔2024〕2号). https://www.szns.gov.cn/xxgk/qzfxxgkml/zcwj/qjzcwj/content/post_11217719.html

Pingan Securities Group. (2023). *Supply-Driven Market Structure: Identifying Opportunities in Offshore Wind, N-Type Solar Cells, and Green Hydrogen [证券研究报告重供给格局，掘金海风、N型电池及绿氢]* [Mid-2023 Strategic Outlook for the Power Equipment and New Energy Sector]. https://pdf.dfcfw.com/pdf/H3_AP202306151590958868_1.pdf

Polysilicon Energy Consumption will have a new National Standard: Approximately 30% of backward capacity will be eliminated [多晶硅能耗将有新国标，约30%落后产能将被淘汰？]. (2025). *Sina*, 23.09.2025. https://finance.sina.com.cn/money/future/wemedia/2025-09-23/doc-infrmpnr3980599.shtml

PwC. (2024). *State of the semiconductor industry*. https://www.pwc.com/gx/en/industries/technology/state-of-the-semicon-industry.html

Qiushi. (2024). Efforts to strengthen financial and policy guarantees for high-quality development [着力强化高质量发展的资金和政策保障]. *Qiushi,* 28.03.2024. https://www.qstheory.cn/laigao/ycjx/2024-03/28/c_1130096869.htm

Raj, G., & Wuttke, T. (2026). State subsidies, supply chain resilience and geopolitics influencing firm strategies in a post-2020 context: Case study of the German semiconductor industry. *Journal of Economic Geography*, *Forthcoming*.

Raza, W., & Tröster, B. (2025). *Moving beyond neoliberalism in EU Trade Policy*. FEPS (Foundation for European Progressive Studies). https://feps-europe.eu/wp-content/uploads/2025/12/Moving-beyond-neoliberalism-in-EU-trade-policy.pdf

Renner, C., Bahous, J., & Plotka, M. B. (2024). The Wind (Turbines) of Change: Ex officio Investigation and Dawn Raids are a New Double First under the FSR. *Lawflash (Morgan Lewis),* 24.04.2024. https://www.morganlewis.com/pubs/2024/04/the-wind-turbines-of-change-ex-officio-investigation-and-dawn-raids-are-a-new-double-first-under-the-fsr

Resources and Environment Research Division. (2025). *Letter Soliciting Public Comments on Three Mandatory National Standards, Including 'Energy Consumption Limits per Unit Product for Polycrystalline Silicon and Germanium' [《硅多晶和锗单位产品能源消耗限额》等3项强制性国家标准征求意见的函]*. https://www.cnis.ac.cn/bydt/bzyjzq/gbyjzq/202509/t20250917_61679.html

Rühlig, T. (2024). *China's growing legacy chip production – A challenge for Europe?* European Union Institute for Security Studies. https://www.iss.europa.eu/publications/commentary/chinas-growing-legacy-chip-production-challenge-europe

Rühlig, T. (2025). United Against Beijing—Or each other? In *Low Trust: Navigating transatlantic relations under Trump 2.0* (pp. 26–31). Belgium: Publications Office of the European Union. https://www.iss.europa.eu/sites/default/files/2025-10/CP_187_0.pdf

Scheiring, G., Serrano-Alarcón, M., Moise, A., McNamara, C., & Stuckler, D. (2024). The Populist Backlash Against Globalization: A Meta-Analysis of the Causal Evidence. *British Journal of Political Science*, *54*(3), 892–916. https://doi.org/10.1017/S0007123424000024

Schmitz, L., Seidl, T., & Wuttke, T. (2025). The costs of conditionality. IPCEIs and the constrained politics of EU industrial policy. *Competition & Change*, 10245294251320675. https://doi.org/10.1177/10245294251320675

Schumacher, A. (2024). *China's Mature Semiconductor Overcapacity: Does It Exist and Does It Matter?* Center for Strategic & International Studies. https://www.csis.org/analysis/chinas-mature-semiconductor-overcapacity-does-it-exist-and-does-it-matter

Sebastian, G., Barkin, N., & Kratz, A. (2024). *Ain't No Duty High Enough*. New York, NY: Rhodium Group. https://rhg.com/research/aint-no-duty-high-enough/

Sebastian, G., & Kratz, A. (2025). *Made in Europe 2025: Can Local Content Requirements Save Europe's Auto Industry?* New York, NY: Rhodium Group. https://rhg.com/research/made-in-europe-2025-can-local-content-requirements-save-europes-auto-industry/

Shanghai DRC, Shanghai STC, Shanghai EIC, & Shanghai PNR. (2022). *Mid- and Longterm Hydrogen Industry Development Plan of Shanghai Municipality (2022-2035) [上海市氢能产业发展中长期规划]*. https://fgw.sh.gov.cn/cmsres/89/890a5d14a42a497993546b558f525cb0/331d8ca143e43fd6548afed86b594190.pdf

Shanghai REFIRE Group Limited. (2024). *REFIRE: IPO [上海重塑能源集團股份有限公司: 全球發售]* (HKEX Stock Code: 257). https://www1.hkexnews.hk/listedco/listconews/sehk/2024/1128/2024112800012_c.pdf

Shen, G., & Chen, B. (2017). Zombie firms and over-capacity in Chinese manufacturing. *China Economic Review*, *44*, 327–342. https://doi.org/10.1016/j.chieco.2017.05.008

Shi, Dan. (2009). Analysis of China's Renewable Energy Development under the Current Economic and Technical Circumstances. *China & World Economy*, *17*(2), 94–109. https://doi.org/10.1111/j.1749-124X.2009.01144.x

Shi, Dinghuan. (1996). *Report: National New and Renewable Energy Development Plan (1996-2010) [国家新能源和可再生能源发展纲要(1996-2010)]*. https://kjpj.bit.edu.cn/docs/20150119212311371518.pdf

Shi, S., Li, H., Tang, H., & Liu, Y. (2023). A study of the impact of de-capacity policies on industry capacity utilization paths: Evidence from the Chinese steel industry. *PLOS ONE*, *18*(12), e0295613. https://doi.org/10.1371/journal.pone.0295613

SIASUN. (2025). *Annual report 2024 [2024 年年度报告]*. https://file.finance.sina.com.cn/211.154.219.97:9494/MRGG/CNSESZ_STOCK/2025/2025-4/2025-04-19/10906858.PDF

Sina Tech Digital. (2025). *The Ministry of Industry and Information Technology (MIIT) convened a symposium with manufacturing enterprises in the power and energy storage battery industry, demanding a firm resistance to irrational competitive practices. [工信部召开动力和储能电池行业制造业企业座谈会，要求坚决抵制非理性竞争行为]*. *Sina*, 28.11.2025. https://finance.sina.com.cn/tech/digi/2025-11-28/doc-infyyape7311569.shtml?froms=ggmp

Sinolink Securities. (2025). *Hydrogen and fuel cell industry research [氢能与燃料电车行业研究]* [Monthly Industry Report]. https://pdf.dfcfw.com/pdf/H3_AP202503221646091911_1.pdf

Sirma, G., & Wicke, T. (2025). *Competitive market for battery materials: Market leaders, technologies and cost analysis*. Fraunhofer Institute for Systems and Innovation Research (ISI). https://www.isi.fraunhofer.de/en/blog/themen/batterie-update/batterie-materialien-kosten-technologien-markt-fuehrer.html

Snyder, F. (2010). *The EU, the WTO and China: Legal Pluralism and International Trade Regulation*. London: Hart Publishing. https://doi.org/10.5040/9781472565204

Solheim, E. (2025). China's green "overcapacity" is the climate solution the world asked for. *Dialogue Earth,* 30.05.2025. https://dialogue.earth/en/business/opinion-chinas-green-overcapacity-is-the-climate-solution-the-world-asked-for/

Soong, C., & Gunter, J. (2024). *It's not us, it's you: China's surging overcapacities and distortive exports are pressuring many developing countries too*. MERICS. https://merics.org/en/tracker/its-not-us-its-you-chinas-surging-overcapacities-and-distortive-exports-are-pressuring-many

STA. (2026). *Export VAT Rebate Rate Inquiry [出口退税率查询]*, 14.01.2026. https://hd.chinatax.gov.cn/nszx2023/cktslcx2023.html

State Council. (2015). *Made in China 2025 [中国制造2025]* (No. 国发〔2015〕28号). https://www.ndrc.gov.cn/xxgk/zcfb/qt/201505/t20150520_967388.html

State Council. (2024). *Action Plan for Promoting Large-scale Equipment Renewals and Consumer Goods Trade-ins [推动大规模设备更新和消费品以旧换新行动方案]* (No. 国发〔2024〕7号). https://www.mee.gov.cn/zcwj/gwywj/202403/t20240318_1068594.shtml

State Council of the People's Republic of China. (2023). *China extends preferential purchase tax policy for NEVs*, 21.06.2023. https://english.www.gov.cn/news/202306/21/content_WS64929394c6d0868f4e8dd11c.html

State Council of the PRC. (2021). *Outline of the 14th Five-Year Plan for National Economic and Social Development and the Long-Range Objectives Through 2035 [中华人民共和国国民经济和社会发展 第十四个五年规划和2035 年远景目标纲要]*. https://www.ndrc.gov.cn/xxgk/zcfb/ghwb/202103/P020210323538797779059.pdf

Steel, fisheries subsidies take centre stage at committee meeting. (2018). *WTO News*, 24.04.2018. https://www.wto.org/english/news_e/news18_e/scm_24apr18_e.htm?utm.

Steindl, J. (1952). *Maturity and Stagnation in American Capitalism*. Oxford: Blackwell.

Sutton, T., & Williams, E. (2025). How China's New Trade Assertiveness Is Linked to Its Clean Energy Dominance. *Center on Global Energy Policy,* 31.10.2025. https://www.energypolicy.columbia.edu/how-chinas-new-trade-assertiveness-is-linked-to-its-clean-energy-dominance/

Suzhou MG. Several Measures of Suzhou City for Supporting the Innovative Development of the Embodied Intelligent Robotics Industry [苏州市支持具身智能机器人产业创新发展的若干措施]., 苏府规字〔2025〕7号 Regulation § (2025).

TechInsights. (2024). Automotive Semiconductor Demand Forecast 2021 to 2030. *TechInsights*. https://www.techinsights.com/blog/automotive-semiconductor-demand-forecast-2021-2030-january-2024

The Research Insights. (2025). *Automotive Semiconductor Market Share worth $77.76 billion by 2030*. The Research Insights. https://www.prnewswire.com/news-releases/automotive-semiconductor-market-share-worth-77-76-billion-by-2030--exclusive-report-by-the-research-insights-302462927.html

Topstar. (2025a). *2024 Annual Report*. https://file.finance.sina.com.cn/211.154.219.97:9494/MRGG/CNSESZ_STOCK/2025/2025-4/2025-04-22/10925474.PDF

Topstar. (2025b). *2025 Third Quarterly Report [拓斯达：2025年三季度报告新浪财经]*. https://file.finance.sina.com.cn/211.154.219.97:9494/MRGG/CNSESZ_STOCK/2025/2025-10/2025-10-28/11544855.PDF

Trachtenberg, D. M. (2025). *Section 301 of the Trade Act of 1974*. Congressional Research Service. https://www.congress.gov/crs-product/IF11346

TransitionAsia (2024). *Decoding China's Steel Capacity Replacement Policies: A Decade-Long Journey, Pausing to Forge Ahead*. https://transitionasia.org/wp-content/uploads/2024/09/Decoding_China_Steel_Capacity_240919.pdf

Transparency remains core priority in work of subsidies committee. (2025, October 28). *World Trade Organization News,* 28.10.2025. https://www.wto.org/english/news_e/news25_e/scm_28oct25_186_e.htm

Triolo, P. (2024). Legacy Chip Overcapacity in China: Myth and Reality. *Center for Strategic & International Studies*. https://www.csis.org/blogs/trustee-china-hand/legacy-chip-overcapacity-china-myth-and-reality

Tschiesner, A., Venus, A., Deubener, H., Schaufuss, P., Heuss, R., & Boers, M. (2025). A new 'ERA': Action plan for the European automotive industry. *McKinsey & Co.* https://www.mckinsey.com/features/mckinsey-center-for-future-mobility/our-insights/a-new-era-an-action-plan-for-the-european-automotive-industry

Update on the Foreign Subsidies Regulation ('FSR'): European Commission launches first ex officio investigation into wind turbine sector. (2024). *Noerr,* 11.04.2024. https://www.noerr.com/en/insights/update-on-the-foreign-subsidies-regulation-european-commission-launches-first-ex-officio-investigation-into-wind-turbine-sector

Urlesberger, F. W. (2020). EU law: News at one glance [Europarecht: Das Neueste auf einen Blick]. *Wirtschaftsrechtliche Blätter*, (4), 202–205.

U.S. Department of Energy. (2022). *Water Electrolyzers and Fuel Cell Supply Chain*. https://www.energy.gov/sites/default/files/2024-12/Fuel%2520Cells%2520%26%2520Electrolyzers%2520Supply%2520Chain%2520Report%2520-%2520Final%5B1%5D.pdf

van den Bossche, P., & Akpofure, S. (2020). *The Use and Abuse of the National Security Exception under Article XXI(b)(iii) of the GATT 1994* (WTI Working Paper No. 03/2020). world trade institute. https://www.wti.org/media/filer_public/50/57/5057fb22-f949-4920-8bd1-e8ad352d22b2/wti_working_paper_03_2020.pdf

van den Bossche, P., & Zdouc, W. (2022). *The Law and Policy of the World Trade Organization* (5th ed.). Cambridge: Cambridge University Press.

Vincent, S. (2026). Chinese PV Industry Brief: Polysilicon prices jump over 9% as China supply tightens. *Pv magazine International,* 09.01.2026. https://www.pv-magazine.com/2026/01/09/polysilicon-prices-jump-over-9-as-china-supply-tightens/

Wade, R. (1990). *Governing the Market: Economic Theory and the Role of Government in East Asian Industrialization*. Princeton University Press. https://doi.org/10.1515/9780691187181

Wade, R., & Veneroso, F. (1998). The Asian Crisis: The High Debt Model Versus the Wall Street–Treasury–IMF Complex. *New Left Review*, *I*(228), 3–23.

Wang, K. (2025). Trade-in initiative to be expanded. *Qiushi*, 20.02.2025. https://en.qstheory.cn/2025-02/20/c_1071544.htm

Wang, Y., Luo, G., & Guo, Y. (2014). Why is there overcapacity in China's PV industry in its early growth stage? *Renewable Energy*, *72*, 188–194. https://doi.org/10.1016/j.renene.2014.07.008

Watanabe, M. (2025). China's industrial policy a recipe of overcapacity. *East Asia Forum*, 23.12.2025. https://eastasiaforum.org/2025/12/23/chinas-industrial-policy-a-recipe-of-overcapacity/

Wei, W. (2025). *China's Hydrogen Energy Development Trends and Strategic Choices.* China-CEE Institute. https://china-cee.eu/working_papers/chinas-hydrogen-energy-development-trends-and-strategic-choices/

Whitfield, L., & Wuttke, T. (2025a). China's technological catch-up and leapfrogging in electric vehicles: A firm-level study of BYD and CATL. *Progress in Economic Geography*, *100054*(In Press). https://doi.org/10.1016/j.peg.2025.100054

Whitfield, L., & Wuttke, T. (2025b). Opportunities for Latecomer Technological Catch-up in the Era of Renewables Capitalism: Possible Pathways out of the Periphery. *Development and Change*, dech.70016. https://doi.org/10.1111/dech.70016

Wiegand, G. (2025). Diplomacy in an Age of Disruption: The EU-China Summit. *German Marshall Fund of the United States (GMF),* 12.07.2025. https://www.gmfus.org/news/diplomacy-age-disruption-eu-china-summit

Wood Mackenzie. (2025). Solar and storage costs are set to increase 9% in Q4 2025 as Chinese policy shifts end an era of low equipment prices. *Wood Mackenzie*, 02.10.2025. https://www.woodmac.com/press-releases/solar-and-storage-costs-are-set-to-increase-9-in-q4-2025-as-chinese-policy-shifts-end-an-era-of-low-equipment-prices/

World Bank. (2007). Statement from World Bank China Country Director on 'Cost of Pollution in China' Report. *World Bank News,* 11.07.2007. https://www.worldbank.org/en/news/press-release/2007/07/11/statement-world-bank-china-country-director-cost-pollution-china-report

World Bank. (undated). Exports of goods and services (% of GDP)—China. *World Bank Open Data.* https://data.worldbank.org/indicator/NE.EXP.GNFS.ZS?locations=CN

WTO. (2024). *GATT 1994—Article XXVIII (Practice)*. WTO. https://www.wto.org/english/res_e/publications_e/ai17_e/gatt1994_art28_oth.pdf

Wu, D. (2024). What is neijuan, and why is it worrying China's economic policymakers? *South China Morning Post*, 17.12.2024. https://www.scmp.com/economy/china-economy/article/3291163/what-neijuan-and-why-it-worrying-chinas-economic-policymakers

Wu, M. (2016). The "China, Inc." Challenge to Global Trade Governance. *Harvard International Law Journal*, *57*, 261–324.

Wu, W., & Lin, B. (2020). Reducing Overcapacity in China's Coal Industry: A Real Option Approach. *Computational Economics*, *55*(4), 1073–1093. https://doi.org/10.1007/s10614-018-9872-z

Wuttke, T. (2023). Global Value Chains and Local Inter-Industry Linkages: South Africa's Participation in the Automotive GVC. *The Journal of Development Studies*, *59*(2), 153–169. https://doi.org/10.1080/00220388.2022.2110491

Xi, J. (2025). To push forward the construction of a unified national market in depth [纵深推进全国统一大市场建设]. *Qiushi*, *18*. https://www.qstheory.cn/20250914/e5d3fb14f33c4771ba73977af642e99e/c.html

Xiang, S., Yin, X., & Zi, Y. (2025). *Two Decades of Chinese Industry Subsidies* (CEPR Discussion Paper No. 20878). Paris & London: CEPR Press.

Xiao, T. (2024). China Lowers the Export Tax Rebate Rate for Certain Products. *China Briefing,* 04.12.2024. https://www.china-briefing.com/news/navigating-chinas-latest-export-tax-rebate-adjustments-implications/

Xiao, Z., Sun, K., & Wang, X. (2025). *Industrial Policy and the Rise of China's Strategic Emerging Industries* [SSRN Scholarly Paper]. Rochester, NY: Social Science Research Network. https://doi.org/10.2139/ssrn.5478030

Xinhua. (2022). Full text of the report to the 20th National Congress of the Communist Party of China—People's Daily Online. *People's Daily,* 26.10.2022. https://en.people.cn/n3/2022/1026/c90000-10163397.html

Xinhua. (2026). Ministry of Finance: Cancelling export tax rebates for photovoltaic and related products is conducive to comprehensively addressing 'involution-style' competition [财政部：取消光伏等产品出口退税有利于综合整治"内卷式"竞争], 20.01.2026. https://www.news.cn/fortune/20260120/37f7da37727f4749a571ced2f1c5c8a0/c.html

Xinhua Finance and Economics. (2026). China Photovoltaic Industry Association: Timely reductions or the cancellation of export tax rebates for photovoltaic products help to promote a return to more rational pricing in overseas markets [中国光伏行业协会：适时降低或取消光伏产品的出口退税，有助于推动国外市场价格理性回归], 09.01.2026. https://m.cnfin.com/dz-lb//zixun/20260109/4363374_1.html

Xiong, Y. (2025). *Understanding China's 'Anti-involution' Drive*. Deutsche Bank Research Institute. https://www.dbresearch.com/PROD/RI-PROD/PDFVIEWER.calias?pdfViewerPdfUrl=PROD0000000000603307&rwnode=REPORT

Xu, D., & Liu, Y. (2018). Evaluation of Overcapacity with an Innovative Definition. In D. Xu & Y. Liu, *Understanding China's Overcapacity* (pp. 1–31). Singapore: Springer Singapore. https://doi.org/10.1007/978-981-13-0881-9_1

Xu, J. (2025). How is innovation competition exacerbating global overcapacity? *Bruegel,* 18.08.2025. https://www.bruegel.org/analysis/how-innovation-competition-exacerbating-global-overcapacity

Xu, Y. (2024). Harnessing venture capital in China. *Socio-Economic Review*, *22*(1), 349–371. https://doi.org/10.1093/ser/mwad037

Yamamura, K. (1986). Caveat Emptor: The Industrial Policy of Japan. In P. Krugman (Ed.), *Strategic Trade Policy and the New International Economics* (pp. 169–209). Cambridge: MIT Press.

Yang, H., & Zhao, D. (2015). Performance Legitimacy, State Autonomy and China's Economic Miracle. *Journal of Contemporary China*, *24*(91), 64–82. https://doi.org/10.1080/10670564.2014.918403

Yeung, H. W.-C. (2022). *Interconnected Worlds: Global Electronics and Production Networks in East Asia*. Stanford University Press.

Yicai. (2025). [Fact Check] Concerns About Overcapacity in Robot Industry Are Premature, Industry Insiders Say. *Yicai,* 12.11.2025. https://www.yicaiglobal.com/news/fact-check-concerns-about-overcapacity-in-robot-industry-are-premature-industry-insiders-say

Zeng, X., Huang, Q., Liu, Y., & Qin, Q. (2025). Decoding China's hydrogen development: A content analysis of policy objectives and instruments. *Renewable and Sustainable Energy Reviews*, *217*, 115729. https://doi.org/10.1016/j.rser.2025.115729

Zhang, J. (2024). Even After Going Overseas, the Competitors Are Still Domestic Peers: How Can the Robotics Industry Break Out of 'Involution'?[出海后竞争对手仍是国内同行，机器人"内卷"该如何破局？]. *Sina,* 27.12.2024. https://finance.sina.com.cn/jjxw/2024-12-27/doc-ineawtrv2616791.shtml

Zhang, Y., Zhang, M., Liu, Y., & Nie, R. (2017). Enterprise investment, local government intervention and coal overcapacity: The case of China. *Energy Policy*, *101*, 162–169. https://doi.org/10.1016/j.enpol.2016.11.036

Zhongda Leader. (2025). *2024 annual report*. https://www.100est.com/res/financial-report/r2024/SZ002896_202504171657998165.pdf

Zhou, W., Crochet, V., & Gao, H. (2025). Overcoming the geopolitics of overcapacity through cooperation. *East Asia Forum,* 09.06.2025. https://eastasiaforum.org/2025/06/09/overcoming-the-geopolitics-of-overcapacity-through-cooperation/.

ZVEI. (2025). *From chips to chances: The importance of and the economic case for supporting microelectronics*. German Electro and Digital Industry Association (ZVEI e.V.). https://www.zvei.org/en/press-media/publications/zvei-study-from-chips-to-chances

Annex: List of interviews

Interview	Case study	Interviewee	Date
Interview 1	Semiconductor	European semiconductor firm	November 2025
Interview 2	Semiconductor	SEMI Europe	November 2025
Interview 3	Semiconductor	European semiconductor firm	November 2025
Interview 4	EV/Batteries	International battery maker	November 2025
Interview 5	EV/Batteries	European battery maker	November 2025
Interview 6	EV/Batteries	European Automobile Manufacturers' Association	December 2025
Interview 7	EV/Batteries	International battery maker	November 2025
Interview 8	Hydrogen	European hydrogen industry alliance	December 2025
Interview 9	Hydrogen	European Trade Union Institute	December 2025
Interview 10	Robotics	IndustriAll	December 2025

The interviews were conducted with the informed consent of the participants, with full anonymity guaranteed.

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